

Running title: Theory of Ordinary Differential Equation

Theory of Ordinary Differential Equation

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1 Introduction

This manuscript is a lecture note for the theory of ordinary differential equations in the liberal arts educations of Tohoku University in Japan. Ordinary differential equations appear in many subjects for scientific problems. In the stand point of liberal arts educational views, it is interpreted that many scientific phenomena are often organized by some differential equations. Although we have some solvable differential equations, such solvable cases of differential equations are restricted and are often regarded to only toy models. The actual purpose of the present mathematical class for the liberal arts educations is for the students to understand that there exists theories to guarantee the existence and the uniqueness for some ordinary differential equations even though the equations cannot be given as a closed expression of the differential equation. In many recent developments of the information and communication technologies, we have a lot of computational resources and we can realize some mathematical inference and prediction modelings as massive differential equations as well as massive statistical models, although such mathematical models are often intractable in the stand point of solvable models.

In the present manuscript, some fundamental frameworks of the theory of ordinary differential equations, for example, first order ordinary differential equations, second order linear differential equations as well as simultaneous ordinary differential equations based on the mathematical frameworks of function sequences and power series in the calculus. In section 2, some examples of applications for problems in the fundamental physics including motions of simple oscillation and motions of charged particles under Lorentz force. In section 3, some definitions of limit, continuous and uniformly continuous of functions in the calculus. Moreover, some theorems of function sequence and function series as well as power series are introduced. They play some important roles in the frameworks of existence and uniqueness of each solution in ordinary differential equations with initial conditions. In section 4, the existence and the uniqueness of the solution in the first order ordinary differential equations with an initial condition are explained and the framework for the cases in separations of variables and variation of constants for first order linear ordinary differential equations. In section 5, the existence and the uniqueness of the solution in the second order linear ordinary differential equations with an initial condition are explained and the frameworks for both cases of homogeneous second order linear constant coefficient ordinary differential equations and homogeneous second order linear variable coefficient ordinary differential equations with initial conditions. Moreover, methods in variation of constants are also shown for inhomogeneous second order linear ordinary differential equations explicitly. In section 6, the existence and the uniqueness of the solution in the simultaneous ordinary differential equations with an initial condition are explained and the frameworks of simultaneous exact differential equations and simultaneous linear ordinary differential equations. Moreover, some examples of simultaneous linear ordinary differential equations are given. For homogeneous simultaneous linear ordinary differential equations, Frobenius method is also shown. In section 7, orthonormal systems and orthonormal expansions are explored. these are basic frameworks for ordinary differential equations with boundary conditions. In section 8, some frameworks of Sturm-Liouville equations are reviewed as one of familiar second order linear ordinary differential equations with boundary conditions. In section 9, we give some concluding remarks.

2 Some Elementary Examples of Ordinary Differential Equations in Fundamental Physics

Many scheme in fundamental theory of ordinary differential equations are produced in classical mechanics as well as electromagnetism. In the present section, It is clarified that some relationship to some examples in the fundamental physics and the fundamental theory of ordinary differential equations.

2.1 Motion of Free Fall

Let us suppose that a mass point with a weight $m[\text{kg}]$ falls vertically from a point at height $h[\text{m}]$ above the ground at time $t = 0$. The gravitational acceleration is denoted by $g[\text{m/sec}^2]$. The height at any time $t[\text{sec}]$ is denoted by $x(t)[\text{m}]$ and the velocity the mass point at $t[\text{seconds}]$ is given by $\frac{d}{dt}x(t)[\text{m/sec}]$. The energy at $t = 0$ is only the potential energy mgh and the one at any time t is given by the sum of the kinetic energy $\frac{1}{2}m\left(\frac{d}{dt}x(t)\right)^2$ and the potential energy $mgx(t)$ and then the law of conservation of energy is given as follows:

$$\frac{1}{2}m\left(\frac{d}{dt}x(t)\right)^2 + mgx(t) = mgh, \quad (2.1)$$

where

$$x(0) = h \text{ and } 0 \leq x(t) \leq h. \quad (2.2)$$

Eq.(2.1) can be reduced to

$$\frac{d}{dt}x(t) = -\sqrt{2g(h-x(t))}, \quad (2.3)$$

$$\frac{1}{\sqrt{h-x(t)}} \frac{d}{dt} x(t) = -\sqrt{2g}, \quad (2.4)$$

$$\int \frac{1}{\sqrt{h-x(t)}} dx(t) = -\int \sqrt{2g} dt, \quad (2.5)$$

$$-2\sqrt{h-x(t)} = -\sqrt{2gt} + \text{Constant}. \quad (2.6)$$

Here we remark that $\frac{d}{dt}x(t)$ should be non-positive real number for $0 \leq x(t) \leq h$. From Eq.(2.2), the constant of integration is determined as follows:

$$t = 0 \Rightarrow -2\sqrt{h-x(0)} = \text{Constant} \Rightarrow \text{Constant} = 0, \quad (2.7)$$

and then the closed expression of $x(t)$ is obtained as follows:

$$x(t) = h - \frac{1}{2}gt^2. \quad (2.8)$$

2.2 Motion of Simple Oscillation

The left end of the spring with the spring constant $k[\text{N/m}]$ is fixed, and a mass point with a weight $m[\text{kg}]$ is attached to the right end of the spring. The position of the mass point when the spring is in the natural length state is expressed as $x = 0$ along the horizontal direction and the position coordinate of the mass point is chosen so as to take a positive sign horizontally along the spring and to the right direction. Let us consider to pull the spring to the position of $x = a[\text{m}]$ and to release it at time $t = 0$. In this case, it is known that the mass point starts a simple oscillation. First of all, the kinetic energy is 0 and the potential energy is $\frac{1}{2}ka^2$ at time $t = 0$. The potential energy can be derived from the Hooke's theory that the pulling force of the spring when extended by $a[\text{m}]$ from the natural length can be expressed as $-ka[\text{N}]$. In this simple oscillation, the position of the mass point at any time $t[\text{sec}]$ is denoted by $x(t)[\text{m}]$. At any time $t[\text{sec}]$, the potential energy is $-\frac{1}{2}kx(t)^2$ and the kinetic energy is expressed as $\frac{1}{2}m\left(\frac{d}{dt}x(t)\right)^2$ and then the law of conservation of energy for the present example is given as follows:

$$\frac{1}{2}m\left(\frac{d}{dt}x(t)\right)^2 + \frac{1}{2}kx(t)^2 = \frac{1}{2}ka^2, \quad (2.9)$$

where

$$x(0) = 0 \quad \text{and} \quad \left[\frac{d}{dt}x(t)\right]_{t=0} = 0. \quad (2.10)$$

Eq.(2.9) can be reduced to

$$\frac{d}{dt}x(t) = \pm\sqrt{\left(\frac{k}{m}\right)(a^2 - x(t)^2)}, \quad (2.11)$$

$$\frac{1}{\sqrt{a^2 - x(t)^2}} \frac{d}{dt}x(t) = \pm\sqrt{\frac{k}{m}}, \quad (2.12)$$

Now we introduce the following transformation:

$$x(t) = a \times \sin(\theta(t)) \quad (t \geq 0), \quad (2.13)$$

where $\theta(t) \geq 0$. Eq.(2.12) can be reduced to

$$\frac{1}{a\sqrt{1 - \sin^2(\theta(t))}} \frac{d}{dt}(a\sin(\theta(t))) = \pm\sqrt{\frac{k}{m}}, \quad (2.14)$$

$$\frac{\cos(\theta(t))}{a\sqrt{1 - \sin^2(\theta(t))}} \frac{d}{dt}\theta(t) = \pm\sqrt{\frac{k}{m}}, \quad (2.15)$$

$$\frac{d}{dt}\theta(t) = \pm\sqrt{\frac{k}{m}}, \quad (2.16)$$

and then we obtain

$$\theta(t) = \sqrt{\frac{k}{m}} \times t + \text{Constant} \quad (t \geq 0), \quad (2.17)$$

where we remark to choose $\theta(t) \geq 0$ for $t \geq 0$. By substituting Eq.(2.17) to Eq.(2.13) and by using the initial condition $x(0) = 0$, the closed expression of $x(t)$ is given as follows:

$$x(t) = a \sin\left(\sqrt{\frac{k}{m}}t + \frac{\pi}{2}\right), \quad (t \geq 0). \quad (2.18)$$

The above arguments are referred to as an **energy integral**. On the other hand, by taking the first derivative of the both sides in the law of conservation of energy in Eq.(2.9) with respect to time t , the corresponding equations of motions are derived for each other. Particularly, for Eqs.(2.9), we can derive the equation of motion as follows:

$$m \frac{d^2}{dx^2} x(t) + kx(t) = 0. \quad (2.19)$$

We remark that Eq.(2.19) is one of homogeneous second order constant coefficient linear ordinary differential equations. In the class of the theory of ordinary differential equations, the general treatment of such homogeneous second order constant coefficient linear ordinary differential equations is one of main topics.

2.3 Motions of Charged Particles under Lorentz Force

In the electromagnetism, we have motions of charged particles under electric fields and magnetic fields as one of important topics. In the three space under the xyz coordinate system, we consider motions of a particle with a mass

$m[\text{kg}]$ and a charge $q[\text{C}]$ under a electric field $\vec{E}(x) = \begin{pmatrix} E_1(x) \\ E_2(y) \\ E_3(z) \end{pmatrix} [\text{N/C}]$ and a magnetic field (magnetic flux density) $\vec{B} = \begin{pmatrix} B_1(x) \\ B_2(y) \\ B_3(z) \end{pmatrix} [\text{T}]$ at each position vector $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} [\text{m}]$. It is usually assumed that the force on a particle with mass $m[\text{kg}]$ due to gravity, $mg[\text{N}]$, is negligible with respect to the force on a particle with charge $q[\text{C}]$ due to electric fields and magnetic fields, where the gravitational acceleration is denoted by $g[\text{m/sec}^2]$. The force $\begin{pmatrix} F_1(x(t)) \\ F_2(y(t)) \\ F_3(z(t)) \end{pmatrix} [\text{N}]$ acting on this charged particle at a position $\vec{r}(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} [\text{m}]$ at time $t[\text{sec}]$ is given as follows:

$$\begin{aligned} \begin{pmatrix} F_1(x(t)) \\ F_2(y(t)) \\ F_3(z(t)) \end{pmatrix} &= q \begin{pmatrix} E_1(x(t)) \\ E_2(y(t)) \\ E_3(z(t)) \end{pmatrix} + q \begin{pmatrix} \frac{d}{dt}x(t) \\ \frac{d}{dt}y(t) \\ \frac{d}{dt}z(t) \end{pmatrix} \times \begin{pmatrix} B_1(x(t)) \\ B_2(y(t)) \\ B_3(z(t)) \end{pmatrix} \\ &= q \begin{pmatrix} E_1(x(t)) \\ E_2(y(t)) \\ E_3(z(t)) \end{pmatrix} + q \begin{pmatrix} (\frac{d}{dt}y(t))B_3(z(t)) - (\frac{d}{dt}z(t))B_2(y(t)) \\ (\frac{d}{dt}z(t))B_1(x(t)) - (\frac{d}{dt}x(t))B_3(z(t)) \\ (\frac{d}{dt}x(t))B_2(y(t)) - (\frac{d}{dt}y(t))B_1(x(t)) \end{pmatrix}. \end{aligned} \quad (2.20)$$

The force expressed by Eq.(2.20) is referred to as a **Lorentz force**. Here $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ is referred to as a **vector product** defined by

$$\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \times \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \equiv \begin{pmatrix} a_2b_3 - b_2a_3 \\ a_3b_1 - b_3a_1 \\ a_1b_2 - b_1a_2 \end{pmatrix} = \det \begin{pmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{pmatrix}, \quad (2.21)$$

where

$$\vec{e}_1 \equiv \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \vec{e}_2 \equiv \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \vec{e}_3 \equiv \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (2.22)$$

Let us suppose that a mass point with a weight $m[\text{kg}]$ and $q[\text{C}]$ on a position $x(t)$ at time $t[\text{sec}]$ under a spatially

uniform electric field $\begin{pmatrix} 0 \\ 0 \\ E \end{pmatrix}$ [N/C] and a spatially uniform magnetic flux density $\vec{B} = \begin{pmatrix} 0 \\ 0 \\ B \end{pmatrix}$ [T]. The force

$\begin{pmatrix} F_1(x(t)) \\ F_2(y(t)) \\ F_3(z(t)) \end{pmatrix}$ [N] acting on this charged particle at a position $\vec{r}(t) = \begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix}$ [m] at time t [sec] is given as follows:

$$\begin{pmatrix} F_1(x(t)) \\ F_2(y(t)) \\ F_3(z(t)) \end{pmatrix} = q \begin{pmatrix} 0 \\ 0 \\ E \end{pmatrix} + q \begin{pmatrix} \left(\frac{d}{dt}y(t)\right)B \\ -\left(\frac{d}{dt}x(t)\right)B \\ 0 \end{pmatrix} \quad (2.23)$$

and then the equation of motion for the charged particle can be written down to

$$m \begin{pmatrix} \frac{d^2}{dt^2}x(t) \\ \frac{d^2}{dt^2}y(t) \\ \frac{d^2}{dt^2}z(t) \end{pmatrix} = q \begin{pmatrix} 0 \\ 0 \\ E \end{pmatrix} + q \begin{pmatrix} \left(\frac{d}{dt}y(t)\right)B \\ -\left(\frac{d}{dt}x(t)\right)B \\ 0 \end{pmatrix} - m \begin{pmatrix} 0 \\ 0 \\ g \end{pmatrix}, \quad (2.24)$$

$$\begin{cases} \frac{d}{dt} \left(\frac{d}{dt}x(t) \right) = \frac{qB}{m} \frac{d}{dt}y(t), \\ \frac{d}{dt} \left(\frac{d}{dt}y(t) \right) = -\frac{qB}{m} \frac{d}{dt}x(t), \\ \frac{d}{dt} \left(\frac{d}{dt}z(t) \right) = \frac{qE}{m}. \end{cases} \quad (2.25)$$

The third equation in Eq.(2.25) can be reduced to

$$\frac{d}{dt}z(t) = \left(\frac{qE}{m} \right)t + \gamma_1, \quad (2.26)$$

where γ_1 is an arbitrary constant. On the other hand, the first and the second equation of Eq.(2.25) can be regarded as homogeneous simultaneous constant coefficient linear first order ordinary differential equations of $\frac{d}{dt}x(t)$ and $\frac{d}{dt}y(t)$. In the first and the second equation in Eq.(2.25), we assume

$$\begin{cases} \frac{d}{dt}x(t) = A_1 e^{\lambda t}, \\ \frac{d}{dt}y(t) = A_2 e^{\lambda t}, \end{cases} \quad (2.27)$$

as one of the pairs of solutions. By substituting Eq.(2.27) to the first and the second equation of Eq.(2.25), we have

$$e^{\lambda t} \begin{pmatrix} -\lambda & \frac{qB}{m} \\ -\frac{qB}{m} & -\lambda \end{pmatrix} \begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (2.28)$$

Because it is known that $e^{\lambda x}$ is not zero at all for any finite complex number λx in the complex analysis, the necessary and sufficient condition for Eq.(2.28) to have non-trivial solutions $\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is given by

$$\det \begin{pmatrix} -\lambda & \frac{qB}{m} \\ -\frac{qB}{m} & -\lambda \end{pmatrix} = 0, \quad (2.29)$$

and then we can derive the expression of λ as follows:

$$\lambda = \pm i \frac{qB}{m}. \quad (2.30)$$

By substituting Eq.(2.30) to Eq.(2.28), the corresponding simultaneous linear equations of $\begin{pmatrix} A_1^\pm \\ A_2^\pm \end{pmatrix}$ for each case of $\lambda = \pm i \frac{qB}{m}$ can be written down to

$$\begin{pmatrix} \mp i \frac{qB}{m} & \frac{qB}{m} \\ -\frac{qB}{m} & \mp i \frac{qB}{m} \end{pmatrix} \begin{pmatrix} A_1^\pm \\ A_2^\pm \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad (2.31)$$

such that

$$\begin{cases} \mp i \frac{qB}{m} A_1^\pm + \frac{qB}{m} A_2^\pm = 0, \\ -\frac{qB}{m} A_1^\pm \mp i \frac{qB}{m} A_2^\pm = 0, \end{cases} \quad (2.32)$$

and then $A_1^\pm$ and $A_2^\pm$ are linearly dependent on each other. The coefficients A_2^+ and A_2^- can be expressed in terms of $A_1^\pm$ as follows:

$$A_2^\pm = \pm i A_1^\pm, \quad (2.33)$$

respectively. We obtain the solutions of the simultaneous linear ordinary differential equations consisting of the first and the second equation in Eq.(2.25) as follows:

$$\begin{aligned} A_1^\pm \begin{pmatrix} 1 \\ \pm i \end{pmatrix} e^{\pm \frac{qB}{m} t} &= A_1^\pm \begin{pmatrix} 1 \\ \pm i \end{pmatrix} \left(\cos\left(\frac{qB}{m} t\right) \pm i \sin\left(\frac{qB}{m} t\right) \right) \\ &= A_1^\pm \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ \pm i \end{pmatrix} \right) \left(\cos\left(\frac{qB}{m} t\right) \pm i \sin\left(\frac{qB}{m} t\right) \right) \\ &= A_1^\pm \left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} \cos\left(\frac{qB}{m} t\right) + \begin{pmatrix} 0 \\ \pm i \end{pmatrix} (\pm i) \sin\left(\frac{qB}{m} t\right) \right. \\ &\quad \left. + \begin{pmatrix} 1 \\ 0 \end{pmatrix} (\pm i) \sin\left(\frac{qB}{m} t\right) + \begin{pmatrix} 0 \\ \pm i \end{pmatrix} \cos\left(\frac{qB}{m} t\right) \right) \\ &= A_1^\pm \left(\begin{pmatrix} \cos\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) \end{pmatrix} \pm i \begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \end{pmatrix} \right) \end{aligned} \quad (2.34)$$

Eq.(2.26) and Eq.(2.34) mean that both $\begin{pmatrix} \frac{d}{dt}x(t) \\ \frac{d}{dt}y(t) \\ \frac{d}{dt}z(t) \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) \\ 0 \end{pmatrix}$, $\begin{pmatrix} \frac{d}{dt}x(t) \\ \frac{d}{dt}y(t) \\ \frac{d}{dt}z(t) \end{pmatrix} = \begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \\ 0 \end{pmatrix}$ and

$\begin{pmatrix} \frac{d}{dt}x(t) \\ \frac{d}{dt}y(t) \\ \frac{d}{dt}z(t) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \left(\frac{qE}{m}\right)t + \gamma_1 \end{pmatrix}$ are solutions of Eq.(2.30). Moreover we have

$$\det \left(\begin{pmatrix} \cos\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) \end{pmatrix}, \begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \end{pmatrix} \right) = \det \begin{pmatrix} \cos\left(\frac{qB}{m} t\right) & \sin\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) & -\cos\left(\frac{qB}{m} t\right) \end{pmatrix} = -1 \neq 0, \quad (2.35)$$

and then the simultaneous linear equations

$$C_1 \begin{pmatrix} \cos\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) \end{pmatrix} + C_2 \begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \end{pmatrix} = \begin{pmatrix} \cos\left(\frac{qB}{m} t\right) & \sin\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) & -\cos\left(\frac{qB}{m} t\right) \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (2.36)$$

have only the trivial solution $\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. It mean that $\begin{pmatrix} \cos\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) \end{pmatrix}$ and $\begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \end{pmatrix}$ are linearly

independent of each other and they are the fundamental system of solutions of the first and the second equation of Eq.(2.30). The general solution of Eq.(2.30) is given by

$$\begin{pmatrix} \frac{d}{dt}x(t) \\ \frac{d}{dt}y(t) \\ \frac{d}{dt}z(t) \end{pmatrix} = \alpha \begin{pmatrix} \cos\left(\frac{qB}{m} t\right) \\ \sin\left(\frac{qB}{m} t\right) \\ 0 \end{pmatrix} + \beta \begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{qE}{m} t + \gamma_1 \end{pmatrix}, \quad (2.37)$$

and

$$\begin{pmatrix} x(t) \\ y(t) \\ z(t) \end{pmatrix} = \alpha \frac{m}{qB} \begin{pmatrix} \sin\left(\frac{qB}{m} t\right) \\ -\cos\left(\frac{qB}{m} t\right) \\ 0 \end{pmatrix} + \beta \frac{m}{qB} \begin{pmatrix} -\cos\left(\frac{qB}{m} t\right) \\ -\sin\left(\frac{qB}{m} t\right) \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \frac{1}{2} \left(\frac{qE}{m}\right) t^2 + \gamma_1 t + \gamma_0 \end{pmatrix}, \quad (2.38)$$

where α , β , γ_0 and γ_1 are arbitrary constants. We remark that the charged particle is in the circle motion because it is always valid that $x(t)^2 + y(t)^2 = (\alpha^2 + \beta^2) \left(\frac{m}{qB} \right)^2$.

2.4 Resonant Circuit of Capacitor and Coil

As another example, we consider a circuit consisting of the direct connection of a capacitor with capacitance C [F] and a coil with self inductance L [H]. The capacitor has been charged. Let us suppose that the capacitor has the electric charge q [C] as an initial condition and that the switch is on at time $t = 0$ [sec]. At any time $t \geq 0$, the electric charge $Q(t)$ of the capacitor and the current $I(t)$ [A] have the following relation:

$$I(t) = \frac{d}{dt}Q(t). \quad (2.39)$$

and the energies of capacitor and the coil are given by $\frac{1}{2}LI(t)^2$ and $\frac{Q(t)^2}{2C}$, respectively. The law of conservation of energy can be written down to

$$\frac{1}{2}LI(t)^2 + \frac{Q(t)^2}{2C} = \frac{q^2}{2C}, \quad (2.40)$$

such that

$$\frac{1}{2}L \left(\frac{d}{dt}Q(t) \right)^2 + \frac{Q(t)^2}{2C} = \frac{q^2}{2C}, \quad (2.41)$$

where $0 \leq Q(t) \leq q$. Eq.(2.41) can be rewritten to

$$\frac{d}{dt}Q(t) = \sqrt{\frac{1}{LC}(q^2 - Q(t)^2)}, \quad (2.42)$$

and by introducing the transformation

$$Q(t) = q \sin(\theta(t)), \quad (2.43)$$

Eq.(2.42) is reduced to

$$\cos(\theta(t)) \frac{d}{dt}\theta(t) = \frac{q}{\sqrt{LC}}(1 - \sin^2(\theta(t))), \quad (2.44)$$

such that

$$\frac{d}{dt}\theta(t) = \frac{q}{\sqrt{LC}}, \quad (2.45)$$

and then we obtain

$$\theta(t) = \frac{q}{\sqrt{LC}}t + \text{Constant}. \quad (2.46)$$

Because of $Q(0) = q$ as an initial condition, the constant of integration should be equal to zero and the present solution is obtained by

$$Q(t) = q \sin\left(\frac{q}{\sqrt{LC}}t\right) \quad (t \geq 0). \quad (2.47)$$

3 Continuous Function, Function Series and Power Series

3.1 Limit, Continuous and Uniformly Continuous of Functions

It is rigorously defined that the limit value of a function $f(x)$ in the limit that x approaches to α is A , $\lim_{x \rightarrow \alpha} f(x) = A$ such that $\lim_{x \rightarrow \alpha} |f(x) - A| = 0$, as follows:

$$\forall \varepsilon > 0, \exists \delta(\varepsilon, \alpha) > 0 : |x - \alpha| < \delta(\varepsilon, \alpha) \Rightarrow |f(x) - A| < \varepsilon. \quad (3.1)$$

In the similar arguments, the right limit $\lim_{x \rightarrow \alpha+0} f(x) = A$ and the left limit $\lim_{x \rightarrow \alpha-0} f(x) = A$ are defined as follows:

$$\begin{cases} \lim_{h \rightarrow \alpha+0} f(x) = A \Rightarrow \forall \varepsilon > 0, \exists \delta(\varepsilon, \alpha) > 0 : x - \alpha < \delta(\varepsilon, \alpha) \Rightarrow |f(x) - A| < \varepsilon, \\ \lim_{h \rightarrow \alpha-0} f(x) = A \Rightarrow \forall \varepsilon > 0, \exists \delta(\varepsilon, \alpha) > 0 : \alpha - x < \delta(\varepsilon, \alpha) \Rightarrow |f(x) - A| < \varepsilon, \end{cases} \quad (3.2)$$

respectively. If I is a bounded and closed interval, there exists both $\max(I)$ and $\min(I)$ (**Weirstrass Axiom**), and the definition of $\lim_{x \rightarrow \alpha} f(x)$ for $\alpha = \max(I)$ and $\alpha = \min(I)$ are replaced to $\lim_{x \rightarrow \alpha-0} f(x)$ and $\lim_{x \rightarrow \alpha+0} f(x)$, respectively.

The definition that $f(x)$ is **continuous** at any point $x = \alpha$ in the interval I , such that $\lim_{x \rightarrow \alpha} |f(x) - f(\alpha)| = 0$ ($\forall \alpha \in I$), is given as follows:

$$\forall \varepsilon > 0, \forall \alpha \in I, \exists \delta(\varepsilon, \alpha) > 0 : |x - \alpha| < \delta(\varepsilon, \alpha) \Rightarrow |f(x) - f(\alpha)| < \varepsilon. \quad (3.3)$$

for any real number x in the interval I . Moreover, if $\delta(\varepsilon, \alpha)$ can be chosen so as not to depend on α in Eq.(3.73), then $f(x)$ is **uniformly continuous** in the interval I . If it is confirmed, for a function $f(x)$ in a interval I , that,

$$\forall \delta > 0, \forall \alpha \in I, \exists M(\alpha, \delta) > 0, \exists \phi(\alpha, \delta) \in I, \exists \psi(\alpha, \delta) \in I :$$

$$\left(\lim_{\delta \rightarrow +0} \phi(\alpha, \delta) = \alpha \right) \wedge \left(\lim_{\delta \rightarrow +0} \psi(\alpha, \delta) = \alpha \right) \wedge \left(|\phi(\alpha, \delta) - \psi(\alpha, \delta)| < \delta \Rightarrow |f(\phi(\alpha, \delta)) - f(\psi(\alpha, \delta))| \geq M(\alpha, \delta) \right) \quad (3.4)$$

$f(x)$ is not uniformly continuous in the interval I .

If there always exists $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ in the interval I , the function $f(x)$ is differentiable and the derivative $\frac{d}{dx}f(x)$ is defined by

$$\frac{d}{dx}f(x) \equiv \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \quad (\forall x \in I). \quad (3.5)$$

Theorem 3.1 If there always exists $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ in the interval I , the function $f(x)$ is continuous in the interval I .

(Proof) Because there always exists $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ in the interval I , we have

$$\forall x \in I, \forall \varepsilon > 0, \exists \delta(\varepsilon, x) \geq 0 : |h| < \delta(\varepsilon, x) \Rightarrow \left| \frac{f(x+h) - f(x)}{h} - \frac{d}{dx}f(x) \right| < \varepsilon. \quad (3.6)$$

The above inequality $\left| \frac{f(x+h) - f(x)}{h} - \frac{d}{dx}f(x) \right| < \varepsilon$ can be reduced to

$$-\varepsilon < \frac{f(x+h) - f(x)}{h} - \frac{d}{dx}f(x) < \varepsilon \quad (3.7)$$

$$\frac{d}{dx}f(x) - \varepsilon < \frac{f(x+h) - f(x)}{h} < \frac{d}{dx}f(x) + \varepsilon \quad (3.8)$$

$$|f(x+h) - f(x)| < |h| \times \min \left\{ \frac{d}{dx}f(x) + \varepsilon, \frac{d}{dx}f(x) - \varepsilon \right\}, \quad (3.9)$$

and then we derive

$$|f(x+h) - f(x)| \rightarrow 0 \quad (h \rightarrow 0). \quad (3.10)$$

for any positive real number ε and for any real number $x \in I$. It means that $f(x)$ is always continuous in the interval I .

If $f(x)$ is differentiable in an interval I , the second derivative $\frac{d^2}{dx^2}f(x)$ is defined by

$$\frac{d^2}{dx^2}f(x) \equiv \frac{d}{dx} \left(\frac{d}{dx}f(x) \right). \quad (3.11)$$

In the similar argument, the n -th order derivative $\frac{d^n}{dx^n}f(x)$ is defined by

$$\frac{d^n}{dx^n}f(x) \equiv \frac{d}{dx} \left(\frac{d^{n-1}}{dx^{n-1}}f(x) \right), \quad (3.12)$$

for any positive integer n . If the n -th order derivative $\frac{d^n}{dx^n}f(x)$ in an interval I , $f(x)$ is a **C^n -class function**. We remark that, for any positive integer n , the necessary condition that there exists the n -th order derivative $\frac{d^n}{dx^n}f(x)$ in an interval I

is that all of $f(x), \frac{d}{dx}f(x), \frac{d^2}{dx^2}f(x), \dots, \frac{d^{n-1}}{dx^{n-1}}f(x)$ are continuous and differentiable in the interval I . If all of $f(x), \frac{d}{dx}f(x), \frac{d^2}{dx^2}f(x), \dots$ are continuous, such that we can take n to infinity in the above definition of the C^n -class function, $f(x)$ is referred to as a $C^{+\infty}$ -**class function**.

Example 3.1. Let us consider a function $f(x) = \frac{1}{x}$. Answer the following questions.

- (1) Show that $\lim_{x \rightarrow 1} f(x) = 1$ by using (ε, δ) -definitions of limits.
- (2) Show that $f(x)$ is continuous at any point $x = \alpha$ in the interval $[1, +\infty)$ by using (ε, δ) -definitions of limits. Moreover, confirm that $f(x)$ uniformly continuous in the interval $[1, +\infty)$.
- (3) Show that $f(x)$ is continuous at any point $x = \alpha$ in the interval $(0, +1]$ by using (ε, δ) -definitions of limits.
- (4) Show that $f(x)$ is uniformly continuous at any point $x = \alpha$ in the interval $(\rho, +\infty)$ for any positive real number $\rho \in (0, +\infty)$ by using (ε, δ) -definitions of limits.
- (5) Show that $f(x)$ is not uniformly continuous in the interval $(0, +\infty)$ by using (ε, δ) -definitions of limits.

(Solution) (1) All we have to do are to find a neighbouring interval at any point $x = 1$ that satisfies $|\frac{1}{x} - 1| < \varepsilon$ in the interval $(0, +\infty)$, such that

$$\left| \frac{1}{x} - 1 \right| = \left| \frac{1-x}{x-1+1} \right| \leq \frac{|x-1|}{||x-1|-1|} < \varepsilon. \quad (3.13)$$

With restricted to $0 \leq |x-1| < 1$, we solve the inequality

$$\frac{|x-1|}{||x-1|-1|} < \varepsilon, \quad (3.14)$$

with respect to $|x-1|$ as follows:

$$|x-1| < \varepsilon ||x-1|-1|, \quad (3.15)$$

$$\varepsilon (|x-1|-1) < |x-1| < \varepsilon (1-|x-1|), \quad (3.16)$$

$$\begin{cases} 0 < \varepsilon < 1 & \Rightarrow -\frac{\varepsilon}{1-\varepsilon} < |x-1| < \frac{\varepsilon}{1+\varepsilon} & \Rightarrow |x-1| < \frac{\varepsilon}{1+\varepsilon} \left(\because -\frac{\varepsilon}{1-\varepsilon} < 0 < \frac{\varepsilon}{1+\varepsilon} \right), \\ \varepsilon > 1 & \Rightarrow |x-1| < \min\left\{ \frac{\varepsilon}{1+\varepsilon}, \frac{\varepsilon}{\varepsilon-1} \right\} & \Rightarrow |x-1| < \frac{\varepsilon}{1+\varepsilon} \left(\because -\frac{\varepsilon}{1-\varepsilon} > \frac{\varepsilon}{1+\varepsilon} > 0 \right), \end{cases} \quad (3.17)$$

then so that we have $|x-1| < \frac{\varepsilon}{1+\varepsilon}$. This fact means that

$$\forall \varepsilon > 0 : |x-1| < \frac{\varepsilon}{1+\varepsilon} \Rightarrow |f(x) - 1| < \varepsilon. \quad (3.18)$$

This means that $\delta(\varepsilon, 1)$ in the definition of Eq.(3.1) has been found as $\delta(\varepsilon, 1) = \frac{\varepsilon}{1+\varepsilon}$.

- (2) All we have to do are to find a neighbouring interval at any point $x = \alpha \in [1, +\infty)$ that satisfies $|\frac{1}{x} - \frac{1}{\alpha}| < \varepsilon$ in the interval $[1, +\infty)$. such that

$$\left| \frac{1}{x} - \frac{1}{\alpha} \right| = \left| \frac{x-\alpha}{\alpha x} \right| = \frac{|x-\alpha|}{|\alpha x|} \leq |x-\alpha| < \varepsilon, \quad (3.19)$$

then so that we have $|x-\alpha| < \varepsilon$. This fact means that

$$\forall \varepsilon > 0, \forall x, \alpha \in [1, +\infty) : |x-\alpha| < \varepsilon \Rightarrow |f(x) - f(\alpha)| < \varepsilon. \quad (3.20)$$

This means that $\delta(\varepsilon, \alpha)$ in the definition of Eq.(3.73) has been found as $\delta(\varepsilon, \alpha) = \varepsilon$ for any point $x = \alpha$ in the interval $(0, +\infty)$. This means that the function $f(x)$ is not only continuous but also uniformly continuous in the interval $[1, +\infty)$.

- (3) All we have to do are to find a neighbouring interval at any point $x = \alpha \in [\rho, +\infty)$ that satisfies $|\frac{1}{x} - \frac{1}{\alpha}| < \varepsilon$ in the interval $[\rho, +\infty)$. such that

$$\left| \frac{1}{x} - \frac{1}{\alpha} \right| = \left| \frac{x-\alpha}{\alpha x} \right| = \frac{|x-\alpha|}{|\alpha x|} \leq \rho^{-2} |x-\alpha| < \varepsilon, \quad (3.21)$$

then so that we have $|x-\alpha| < \rho^{-2} \varepsilon$. This fact means that

$$\forall \varepsilon > 0, \forall x, \alpha \in [1, +\infty) : |x-\alpha| < \rho^{-2} \varepsilon \Rightarrow |f(x) - f(\alpha)| < \varepsilon. \quad (3.22)$$

This means that $\delta(\varepsilon, \alpha)$ in the definition of Eq.(3.73) has been found as $\delta(\varepsilon, \alpha) = \rho^{-2} \varepsilon$ for any point $x = \alpha$ in

the interval $(0, +\infty)$. This means that the function $f(x)$ is not only continuous but also uniformly continuous in the interval $[\rho, +\infty)$ for any positive real number $\rho \in (0, \infty)$.

- (4) All we have to do are to find a neighbouring interval at any point $x = \alpha$ that satisfies $|\frac{1}{x} - \frac{1}{\alpha}| < \varepsilon$ in the interval $(0, 1]$, such that

$$\left| \frac{1}{x} - \frac{1}{\alpha} \right| = \left| \frac{\alpha - x}{\alpha(x - \alpha + \alpha)} \right| \leq \frac{|x - \alpha|}{\alpha|x - \alpha| - \alpha} < \varepsilon. \quad (3.23)$$

With restricted to $0 \leq |x - \alpha| < \alpha$, we solve the inequality

$$\frac{|x - \alpha|}{|x - \alpha| - \alpha} < \alpha\varepsilon, \quad (3.24)$$

with respect to $|x - \alpha|$ as follows:

$$|x - \alpha| < \varepsilon\alpha|x - \alpha| - \alpha, \quad (3.25)$$

$$\varepsilon\alpha(|x - \alpha| - \alpha) < |x - \alpha| < \varepsilon\alpha(\alpha - |x - \alpha|), \quad (3.26)$$

$$\begin{cases} 0 < \alpha\varepsilon < 1 & \Rightarrow -\frac{\varepsilon\alpha^2}{1-\varepsilon\alpha} < |x - \alpha| < \frac{\varepsilon\alpha^2}{1+\varepsilon\alpha} & \Rightarrow |x - \alpha| < \frac{\varepsilon\alpha^2}{1+\varepsilon\alpha} \left(\because -\frac{\varepsilon\alpha^2}{1-\varepsilon\alpha} < 0 < \frac{\varepsilon\alpha^2}{1+\varepsilon\alpha} \right), \\ \alpha\varepsilon > 1 & \Rightarrow |x - \alpha| < \min\left\{ \frac{\alpha^2\varepsilon}{1+\alpha\varepsilon}, \frac{\alpha^2\varepsilon}{\alpha\varepsilon-1} \right\} & \Rightarrow |x - \alpha| < \frac{\alpha^2\varepsilon}{1+\alpha\varepsilon} \left(\because \frac{\alpha^2\varepsilon}{\alpha\varepsilon-1} > \frac{\alpha^2\varepsilon}{1+\alpha\varepsilon} > 0 \right), \end{cases} \quad (3.27)$$

then so that we have $|x - \alpha| < \frac{\alpha^2\varepsilon}{1+\alpha\varepsilon}$. This fact means that

$$\forall \varepsilon > 0 : |x - \alpha| < \frac{\alpha^2\varepsilon}{1+\alpha\varepsilon} \Rightarrow |f(x) - f(\alpha)| < \varepsilon. \quad (3.28)$$

This means that $\delta(\varepsilon, \alpha)$ in the definition of Eq.(3.1) has been found as $\delta(\varepsilon, \alpha) = \frac{\alpha^2\varepsilon}{1+\alpha\varepsilon}$.

- (5) We introduce two functions $\phi(\alpha, \delta)$ and $\psi(\alpha, \delta)$ defined by

$$\begin{cases} \phi(\alpha, \delta) \equiv \alpha + \frac{1}{a(\delta)}, \\ \psi(\alpha, \delta) \equiv \alpha + \frac{1}{a(\delta)+1}, \end{cases} \quad (\forall \alpha \in (0, +\infty)), \quad (3.29)$$

where $a(\delta) \equiv \max\left\{1, \frac{1}{\sqrt{\delta}}\right\}$ for any positive real number δ . We can confirm that

$$\begin{cases} \lim_{\delta \rightarrow +0} \phi(\alpha, \delta) = \alpha, \\ \lim_{\delta \rightarrow +0} \psi(\alpha, \delta) = \alpha, \end{cases} \quad (\forall \alpha \in (0, +\infty)), \quad (3.30)$$

and

$$|f(\phi(\alpha, \delta)) - f(\psi(\alpha, \delta))| = \frac{1}{(\alpha a(\delta) + 1)(\alpha(a(\delta) + 1) + 1)} \quad (\forall \alpha \in (0, +\infty)). \quad (3.31)$$

It means that $M(\alpha, \delta)$ in Eq.(3.4) has been found as $M(\alpha, \delta) = \frac{1}{(\alpha a(\delta) + 1)(\alpha(a(\delta) + 1) + 1)}$ and then

$f(x)$ is not uniformly continuous in the interval $(0, +\infty)$.

Theorem 3.2 If a function $f(x)$ is continuous in a bounded and closed interval I , the function $f(x)$ is uniformly continuous in the interval I .

(Proof) Let us suppose that there exist $M(\alpha, \delta) > 0$, $\phi(\alpha, \delta) \in I$ and $\psi(\alpha, \delta) \in I$ satisfying

$$\begin{cases} \lim_{\delta \rightarrow +0} \phi(\alpha, \delta) = \alpha, \\ \lim_{\delta \rightarrow +0} \psi(\alpha, \delta) = \alpha, \end{cases} \quad (\forall \alpha \in I), \quad (3.32)$$

$$|\phi(\alpha, \delta) - \psi(\alpha, \delta)| < \delta \quad (\forall \alpha \in I), \quad (3.33)$$

and

$$|f(\phi(\alpha, \delta)) - f(\psi(\alpha, \delta))| \geq M(\alpha, \delta) \quad (\forall \alpha \in I), \quad (3.34)$$

for any positive real number δ . In the calculus, it is known that a continuous function $f(x)$ in a bounded and closed interval I is bounded, such that there exist both $\max_{x \in I} f(x)$ and $\min_{x \in I} f(x)$, and that

$$\lim_{\delta \rightarrow +0} |f(\phi(\alpha, \delta)) - f(\psi(\alpha, \delta))| = \left| f\left(\lim_{\delta \rightarrow +0} \phi(\alpha, \delta)\right) - f\left(\lim_{\delta \rightarrow +0} \psi(\alpha, \delta)\right) \right| = 0 \quad (\forall \alpha \in I). \quad (3.35)$$

Eq.(3.34) and Eq.(3.35) contradict each other, because $M(\alpha, \delta)$ should be a positive real number. The present proposition has been valid from *reductio ad absurdum*. We remark that some arguments in the present proof might be enough in the stand point of rigorous frameworks [10].

3.2 Function Sequence and Function Series

For a sequence $\{a_n | n = 0, 1, 2, \dots\}$, if it is valid that

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n \geq N(\varepsilon) \Rightarrow |a_n - a| < \varepsilon, \quad (3.36)$$

then **the sequence converges to a in the limit $n \rightarrow +\infty$** , such that we can describe $\lim_{n \rightarrow +\infty} |a_n - a| = 0$ and $\lim_{n \rightarrow +\infty} a_n = a$. If all the terms of sequence $\{a_n\}$ are always bounded, the sequence is referred to as a **bounded sequence**. A sequence in which an infinite number of terms are partially extracted from the sequence $\{a_n | n = 0, 1, 2, \dots\}$ and arranged in order, $\{a_{n_0}, a_{n_1}, a_{n_2}, \dots\}$ ($0 \leq n_0 < n_1 < n_2 < \dots$) is called a **subsequence** of the sequence $\{a_n | n = 0, 1, 2, \dots\}$.

Theorem 3.3 (Bolzano-Weierstrass Theorem). Every bounded sequence includes a subsequence.

(Proof) We consider the bounded and closed interval

$$I_0 \equiv \left[\min_{n \in \{0, 1, 2, 3, \dots\}} a_n, \max_{n \in \{0, 1, 2, 3, \dots\}} a_n \right], \quad (3.37)$$

and divide two bounded and closed intervals

$$\left[\min_{n \in \{0, 1, 2, 3, \dots\}} a_n, \frac{1}{2} \left(\min_{n \in \{0, 1, 2, 3, \dots\}} a_n + \max_{n \in \{0, 1, 2, 3, \dots\}} a_n \right) \right], \quad (3.38)$$

and

$$\left[\frac{1}{2} \left(\min_{n \in \{0, 1, 2, 3, \dots\}} a_n + \max_{n \in \{0, 1, 2, 3, \dots\}} a_n \right), \max_{n \in \{0, 1, 2, 3, \dots\}} a_n \right]. \quad (3.39)$$

The bounded and closed interval containing an infinite number of terms in the sequence $\{a_n | n = 0, 1, 2, \dots\}$ is chosen from the above two intervals and is denoted by I_1 . The interval I_1 is divided to two bounded and closed intervals with the midpoint of I_2 as a boundary, and the bounded and closed interval containing an infinite number of terms in the sequence $\{a_n | n = 0, 1, 2, \dots\}$ is chosen from the two intervals as I_2 . By iterating the similar procedure, the sequence of the intervals $\{I_0, I_1, I_2, I_3, \dots\}$ is generated. The length $|I_n|$ of the interval I_n is denoted by $|I_n|$ converges to 0 in the limit $n \rightarrow +\infty$. Moreover, there exists a real number a which belongs to all the intervals $I_0, I_1, I_2, I_3, \dots$, such that

$$\begin{cases} |I_n| \rightarrow 0 \quad (n \rightarrow +\infty), \\ \exists a \in I_n \quad (\forall n \in \{0, 1, 2, \dots\}). \end{cases} \quad (3.40)$$

From each of the bounded and closed intervals $I_0, I_1, I_2, I_3, \dots$, we extract the terms one by one from I_n as $\tilde{a}_n$ so as not to overlap each other and construct a subsequence $\{\tilde{a}_n | n = 0, 1, 2, \dots\}$. It is valid that

$$|\tilde{a}_n - a| \leq |I_n| \rightarrow 0 \quad (n \rightarrow +\infty). \quad (3.41)$$

If it is valid that

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n, m \geq N(\varepsilon) \Rightarrow |a_n - a_m| < \varepsilon, \quad (3.42)$$

the sequence $\{a_n | n = 0, 1, 2, \dots\}$ is referred to as a **Cauchy sequence**. It is known that every Cauchy sequence always converges in the calculus. In the calculus, that the following fundamental theorem is known:

Theorem 3.4. Every sequence to converge is a Cauchy sequence.

(Proof) Let us suppose that $\{a_n | n = 0, 1, 2, \dots\}$ converges to a , such that $\lim_{n \rightarrow +\infty} a_n = a$, and then we have

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : |a_n - a_m| \leq |a_n - a| + |a_m - a| < 2\varepsilon \quad (n > m \geq N(\varepsilon)). \quad (3.43)$$

from Eq.(3.36). It means that the sequence $\{a_n | n = 0, 1, 2, \dots\}$ is a Cauchy sequence.

Theorem 3.5. Every Cauchy sequence always converges.

(Proof) Let us suppose that $\{a_n | n = 0, 1, 2, \dots\}$ is a Cauchy sequence, such that we have

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > \forall m \geq N(\varepsilon) \Rightarrow |a_n - a_m| < \varepsilon. \quad (3.44)$$

By setting $\varepsilon = 1$ and $m = N(1)$ and then we have

$$\forall n > N(1) \Rightarrow |a_n - a_{N(1)}| < 1, \quad (3.45)$$

such that

$$\forall n > N(1) \Rightarrow a_{N(1)} - 1 < a_n < a_{N(1)} + 1. \quad (3.46)$$

Eq.(3.46) means that every Cauchy sequence $\{a_n | n = 0, 1, 2, \dots\}$ is bounded. By using the Bolzano-Weierstrass theorem, the bounded sequence $\{a_n | n = 0, 1, 2, \dots\}$ includes a subsequence $\{\tilde{a}_0, \tilde{a}_1, \tilde{a}_2, \tilde{a}_3, \dots\} = \{a_{n_0}, a_{n_1}, a_{n_2}, a_{n_3}, \dots\}$ where $0 \leq n_0 < n_1 < n_2 < n_3 < \dots$. By setting $a \equiv \lim_{n \rightarrow +\infty}$ such that

$$\exists N(\varepsilon) \geq 0 : |\tilde{a}_m - a| = |\tilde{a}_m - a| < \varepsilon \quad (m > N(\varepsilon), m \in \{n_0, n_1, n_2, \dots\}), \quad (3.47)$$

we have

$$|a_n - a| \leq |a_n - \tilde{a}_m| + |\tilde{a}_m - a| = |a_n - a_m| + |\tilde{a}_m - a| < 2\varepsilon \quad (n > m > N(\varepsilon), m \in \{n_0, n_1, n_2, \dots\}). \quad (3.48)$$

It means that the sequence $\{a_n | n = 0, 1, 2, \dots\}$ always converges.

For a sequence $\{a_n | n = 0, 1, 2, \dots\}$, $\sum_{n=0}^{+\infty} a_n \equiv \lim_{N \rightarrow +\infty} \sum_{n=0}^N a_n$ is referred to as a **series**. In the calculus, we have the following fundamental theorems:

Theorem 3.6. If a series $\sum_{n=0}^{+\infty} a_n \equiv \lim_{N \rightarrow +\infty} \sum_{n=0}^N a_n$ converges, then the sequence $\{a_n | n = 0, 1, 2, \dots\}$ always converges to 0 in the limit $n \rightarrow +\infty$, such that $\lim_{n \rightarrow +\infty} a_n = 0$. Moreover, the sequence is bounded and there exists a finite value M that satisfies $|a_n| < M$ for any non-negative integer n .

(Proof) Because $\lim_{N \rightarrow +\infty} \sum_{n=0}^N a_n$ converges, then the sequence $\left\{ \sum_{k=0}^n a_k | n = 0, 1, 2, \dots \right\}$ is a Cauchy sequence and then we have

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > m \geq N(\varepsilon) \Rightarrow \left| \sum_{k=0}^n a_k - \sum_{k=0}^m a_k \right| < \varepsilon. \quad (3.49)$$

By setting $m = n - 1$, we have

$$\forall n \geq N(\varepsilon) + 1 \Rightarrow \left| \sum_{k=0}^n a_k - \sum_{k=0}^{n-1} a_k \right| = |a_n| < \varepsilon, \quad (3.50)$$

and then

$$\lim_{n \rightarrow +\infty} a_n = 0. \quad (3.51)$$

Moreover, the sequence $\{a_n | n = 0, 1, 2, \dots\}$ to converge to 0 is a Cauchy sequence and then

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > m \geq N(\varepsilon) \Rightarrow |a_n - a_m| < \varepsilon. \quad (3.52)$$

By setting $\varepsilon = 1$ and $m = N(1)$ in Eq.(3.52),

$$\forall n > N(1) \Rightarrow |a_n - a_{N(1)}| < 1, \quad (3.53)$$

and then

$$\forall n > N(1) \Rightarrow a_{N(1)} - 1 < a_n < a_{N(1)} + 1 \text{ such that } |a_n| < \max\{|a_{N(1)} - 1|, |a_{N(1)} + 1|\}. \quad (3.54)$$

By adding $\{a_0, a_1, \dots, a_{N(1)-1}\}$, we derive

$$\forall n \geq 0 \Rightarrow |a_n| \leq \max\{|a_{N(1)} - 1|, |a_{N(1)} + 1|, |a_0|, |a_1|, \dots, |a_{N(1)-1}|\}. \quad (3.55)$$

It means that $M \equiv \max\{|a_{N(1)} - 1|, |a_{N(1)} + 1|, |a_0|, |a_1|, \dots, |a_{N(1)-1}|\}$ has been found.

Theorem 3.7. If a series $\sum_{n=0}^{+\infty} |a_n| \equiv \lim_{N \rightarrow +\infty} \sum_{n=0}^N |a_n|$ converges, then $\sum_{n=0}^{+\infty} a_n \equiv \lim_{N \rightarrow +\infty} \sum_{n=0}^N a_n$ always converges. Every series to absolutely converge always converges.

(Proof) We introduce the following sequences from $\{a_n | n = 0, 1, 2, \dots\}$:

$$a_n^+ \equiv \begin{cases} +a_n & (a_n \geq 0), \\ 0 & (a_n \leq 0), \end{cases} \quad (3.56)$$

$$a_n^- \equiv \begin{cases} -a_n & (a_n \leq 0), \\ 0 & (a_n \geq 0). \end{cases} \quad (3.57)$$

Because $\sum_{n=1}^{+\infty} a_n$ absolutely converges, such that the sequence $\left\{ \sum_{k=1}^n |a_k| \mid n = 0, 1, 2, \dots \right\}$ converges and is a Cauchy sequence. It is valid that

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > \forall m \geq N(\varepsilon) \Rightarrow \left| \sum_{k=1}^n a_k^+ - \sum_{k=1}^m a_k^+ \right| \leq \left| \sum_{k=1}^n |a_k| - \sum_{k=1}^m |a_k| \right| < \varepsilon \quad (3.58)$$

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > \forall m \geq N(\varepsilon) \Rightarrow \left| \sum_{k=1}^n a_k^- - \sum_{k=1}^m a_k^- \right| \leq \left| \sum_{k=1}^n |a_k| - \sum_{k=1}^m |a_k| \right| < \varepsilon \quad (3.59)$$

It means that both sequences $\left\{ \sum_{k=1}^n a_k^+ \mid n = 0, 1, 2, \dots \right\}$ and $\left\{ \sum_{k=1}^n a_k^- \mid n = 0, 1, 2, \dots \right\}$ are Cauchy sequences and then converge. Hence the series $\sum_{n=1}^{+\infty} a_n = \lim_{n \rightarrow +\infty} \sum_{k=1}^n a_k^+ - \lim_{n \rightarrow +\infty} \sum_{k=1}^n a_k^-$ also converges.

For a bounded and closed interval I in a set of all the real numbers on $(-\infty, +\infty)$, we consider a sequence of functions $f_n(x)$ ($n = 0, 1, 2, 3, \dots, x \in I$), which are referred to as a **function series**. When the function sequence satisfies

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n \geq N(\varepsilon) \Rightarrow |f_n(x) - f(x)| < \varepsilon \quad (\forall x \in I), \quad (3.60)$$

where the positive integer number $N(\varepsilon)$ is not depend on any real number x in the interval I , the function sequence $\{f_n(x) \mid n = 0, 1, 2, 3, \dots\}$ **uniformly converges** in the interval I .

Example 3.2. Let us consider a function sequence $f_n(x) = x^n$ for any non-negative integer n . Answer the following questions by means of (ε, δ) -definitions of limits.

- (1) Show that $f_n(x)$ always converges to $f(x) = 0$ in the interval $|x| < 1$.
- (2) Show that $f_n(x)$ always uniformly converges to $f(x) = 0$ in the interval $|x| < \rho$ for any real number $\rho \in (0, 1)$.
- (3) Show that $f_n(x)$ does not uniformly converge to $f(x) = 0$ in the interval $|x| < 1$.

(Solution) (1) All we have to do are to find the condition for n that satisfies the following inequality:

$$|f_n(x) - f(x)| = |x^n - 0| = |x|^n < \varepsilon, \quad (3.61)$$

such that

$$n \times \ln(|x|) < \ln(\varepsilon), \quad (3.62)$$

$$n > \frac{\ln(\varepsilon)}{\ln(|x|)}, \quad (3.63)$$

$$\forall \varepsilon > 0 : n > \frac{\ln(\varepsilon)}{\ln(|x|)} \Rightarrow |f_n(x) - f(x)| < \varepsilon. \quad (3.64)$$

This means that $N(\varepsilon, x)$ in the definition of Eq.(3.36) has been found as $N(\varepsilon, x) = \lfloor \frac{\ln(\varepsilon)}{\ln(|x|)} \rfloor + 1$, where $\lfloor a \rfloor$ is a floor function which outputs the integer part of a given real number a .

- (2) All we have to do are to find the condition for n that satisfies the following inequality:

$$|f_n(x) - f(x)| = |x^n - 0| = |x|^n \leq \rho^n < \varepsilon \quad (\forall \rho \in (0, 1)), \quad (3.65)$$

such that

$$n \times \ln(\rho) < \ln(\varepsilon), \quad (3.66)$$

$$n > \frac{\ln(\varepsilon)}{\ln(\rho)}, \quad (3.67)$$

and then we have

$$\forall \varepsilon > 0, \forall x \in [-\rho, +\rho] : n > \frac{\ln(\varepsilon)}{\ln(\rho)} \Rightarrow |f_n(x) - f(x)| < \varepsilon \quad (\forall \rho \in (0, 1)). \quad (3.68)$$

This means that $N(\varepsilon, x)$ in the definition of Eq.(3.60) has been found as $N(\varepsilon) = \lfloor \frac{\ln(\varepsilon)}{\ln(\rho)} \rfloor + 1$ and $N(\varepsilon)$ does

not depend on x at all for any real number $x \in [-\rho, +\rho]$.

- (3) In (1), we have already find out $N(\varepsilon)$ in the definition of Eq.(3.36) as $N(\varepsilon) = \lfloor \frac{\ln(\varepsilon)}{\ln(|x|)} \rfloor + 1$. This means that $N(\varepsilon)$ monotonically increases to infinity as $x \rightarrow 1 - 0$. This means that the function sequence $f_n(x)$ does not uniformly converge in the interval $|x| < 1$.

We confirm the following theorems for the uniformly convergence of function sequences:

Theorem 3.8. The necessary and sufficient condition that a function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ uniformly converges in the bounded and closed interval I is given as follows:

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > \forall m \geq N(\varepsilon) \Rightarrow |f_n(x) - f_m(x)| < \varepsilon \quad (\forall x \in I), \quad (3.69)$$

where the non-negative integer number $N(\varepsilon)$ does not depend on any real number x in the interval I .

(Proof for Necessity) If a function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ uniformly converges in the bounded and closed interval I

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n, m \geq N(\varepsilon) \Rightarrow |f_n(x) - f(x)| < \varepsilon \text{ and } |f_m(x) - f(x)| < \varepsilon \quad (\forall x \in I), \quad (3.70)$$

and then we have

$$|f_n(x) - f_m(x)| < |f_n(x) - f(x)| + |f_m(x) - f(x)| < 2\varepsilon \quad (\forall x \in I), \quad (3.71)$$

where $N(\varepsilon)$ does not depend on any real number x in the interval I .

(Proof of Sufficiency) Eq.(3.69) means that the function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ is a **Cauchy sequence**. It is known that every Cauchy sequence always converges to a function in the calculus. By setting $\lim_{n \rightarrow +\infty} f_n(x) = f(x)$ in Eq.(3.69), Eq.(3.69) can be reduced to

$$\forall \varepsilon > 0, \exists N(\varepsilon) > 0 : \forall m \geq N(\varepsilon) \Rightarrow |f(x) - f_m(x)| = |f_m(x) - f(x)| < \varepsilon \quad (\forall x \in I). \quad (3.72)$$

Here we remark again that $N(\varepsilon)$ does not depend on x in the interval I . Eq.(3.72) means that the function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ uniformly converge in the bounded and closed interval I .

The definition that $f(x)$ is continuous in the interval I , such that $\lim_{\xi \rightarrow x} |f(\xi) - f(x)| = 0 \quad (\forall x \in I)$, is given as follows:

$$\forall \varepsilon > 0, \exists \delta(\varepsilon, x) > 0 : |x - \xi| < \delta(\varepsilon, x) \quad (\xi \in I) \Rightarrow |f(x) - f(\xi)| < \varepsilon. \quad (3.73)$$

for any real number x in the interval I .

Theorem 3.9. If a continuous function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ uniformly converges to a function $f(x)$ in a bounded and closed interval I , $f(x)$ is a continuous function in the interval I .

(Proof) For any real number $x \in I$, we have

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n \geq N(\varepsilon) \Rightarrow |f_n(x) - f(x)| < \varepsilon, \quad (3.74)$$

where $N(\varepsilon)$ does not depend on any real number $x \in I$. On the other hand, $f_n(x)$ is continuous in the interval I for any finite non-negative integer n , such that

$$\forall \varepsilon > 0, \exists \delta(\varepsilon, x) > 0 : |x - \xi| < \delta(\varepsilon, x) \quad (\xi \in I) \Rightarrow |f_n(x) - f_n(\xi)| < \varepsilon, \quad (3.75)$$

for any real number x in the interval I . From Eqs.(3.74) and (3.75), we have

$$\begin{aligned} & \forall \varepsilon > 0, \exists \delta(\varepsilon, x) > 0, \exists N(\varepsilon) \geq 0 : \\ & |x - \xi| < \delta(\varepsilon, x) \quad (\xi \in I), \quad \forall n \geq N(\varepsilon) \\ & \Rightarrow |f(x) - f(\xi)| < |f_n(x) - f(x)| + |f_n(x) - f_n(\xi)| + |f_n(\xi) - f(\xi)| < 3\varepsilon \quad (\forall x \in I). \end{aligned} \quad (3.76)$$

Eq.(3.76) means that a function $f(x)$ to which the function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ converges should be continuous.

Theorem 3.10 If a function sequence $\{f_n(x)|n=0, 1, 2, \dots\}$ uniformly converges and $f_n(x)$ is always a continuous function in the bounded and closed interval I , it is valid that

$$\lim_{n \rightarrow +\infty} \int_{\alpha}^{\beta} f_n(x) dx = \int_{\alpha}^{\beta} \left(\lim_{n \rightarrow +\infty} f_n(x) \right) dx \quad (\forall [\alpha, \beta] \subseteq I). \quad (3.77)$$

(Proof) From theorem 3.9, $f(x) \equiv \lim_{n \rightarrow +\infty} f_n(x)$ is continuous in the bounded and closed interval I because the continuous function sequence $f_n(x)$ uniformly converges to $f(x)$ in the interval I . Here we remark that

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0, \forall n \geq N(\varepsilon) \Rightarrow |f_n(x) - f(x)| < \varepsilon \quad (\forall x \in I). \quad (3.78)$$

We obtain

$$\left| \int_{\alpha}^{\beta} f_n(x) dx - \int_{\alpha}^{\beta} f(x) dx \right| \leq \int_{\alpha}^{\beta} |f_n(x) - f(x)| dx < |I| \varepsilon \quad (\forall [\alpha, \beta] \subseteq I), \quad (3.79)$$

and then Eq.(3.77) has been confirmed.

Theorem 3.11 If both function sequences $\{f_n(x) | n = 0, 1, 2, \dots\}$ and $\{\frac{d}{dx}f_n(x) | n = 0, 1, 2, \dots\}$ uniformly converge and $f_n(x)$ is always C^1 -class function in the bounded and closed interval I , it is valid that

$$\lim_{n \rightarrow +\infty} \frac{d}{dx} f_n(x) = \frac{d}{dx} \left(\lim_{n \rightarrow +\infty} f_n(x) \right) \quad (x \in I). \quad (3.80)$$

(Proof) Because both function sequences $\{f_n(x) | n = 0, 1, 2, \dots\}$ and $\{\frac{d}{dx}f_n(x) | n = 0, 1, 2, \dots\}$ are uniformly convergence in the bounded and closed interval I , we introduce two functions $f(x)$ and $g(x)$:

$$\begin{cases} f(x) \equiv \lim_{n \rightarrow +\infty} f_n(x), \\ g(x) \equiv \lim_{n \rightarrow +\infty} \frac{d}{dx} f_n(x), \end{cases} \quad (x \in I). \quad (3.81)$$

By using Theorem 3.9, $f(x)$ and $g(x)$ are always continuous and integrable in the interval I , because $f_n(x)$ is always C^1 -class function in the bounded and closed interval I . By considering the integral of the second equation of Eq.(3.81), we have

$$\varepsilon > 0, \exists N(\varepsilon) \geq 0 :$$

$$n \leq N(\varepsilon) \Rightarrow \left| \int_{x_0}^x \left(\frac{d}{d\xi} f_n(\xi) \right) d\xi - \int_{x_0}^x (g(\xi)) d\xi \right| \leq \int_{x_0}^x \left| \frac{d}{d\xi} f_n(\xi) - g(\xi) \right| d\xi < |x - x_0| \varepsilon < |I| \varepsilon \quad (x \in I, x_0 \in I), \quad (3.82)$$

where the non-negative integer $N(\varepsilon)$ does not depend on x in the interval I and $|I|$ is the length of the interval I . Eq.(3.82) can be reduced to

$$\lim_{n \rightarrow +\infty} \int_{x_0}^x \left(\frac{d}{d\xi} f_n(\xi) \right) d\xi = \int_{x_0}^x (g(\xi)) d\xi. \quad (3.83)$$

On the other hand, we have

$$\int_{x_0}^x \left(\frac{d}{d\xi} f_n(\xi) d\xi \right) = f_n(x) - f_n(x_0) \quad (x \in I, x_0 \in I), \quad (3.84)$$

and

$$\lim_{n \rightarrow +\infty} \int_{x_0}^x \left(\frac{d}{d\xi} f_n(\xi) d\xi \right) = \lim_{n \rightarrow +\infty} (f_n(x) - f_n(x_0)) = f(x) - f(x_0) \quad (x \in I, x_0 \in I). \quad (3.85)$$

From Eqs.(3.82) and Eq.(3.85), we obtain

$$f(x) = \int_{x_0}^x g(\xi) d\xi + f(x_0) \quad (x \in I, x_0 \in I). \quad (3.86)$$

By differentiating both sides of Eq.(3.86) with respect to x , we derive

$$\frac{d}{dx} f(x) = g(x) \quad (x \in I), \quad (3.87)$$

and we can see that Eq.(3.87) is equivalent to Eq.(3.80).

For the bounded and closed interval I , we introduce the following sum of the function sequence:

$$S_N(x) \equiv \sum_{n=0}^N f_n(x) \quad (x \in I). \quad (3.88)$$

and consider the limit $N \rightarrow +\infty$ as follows:

$$\sum_{n=0}^{+\infty} f_n(x) \equiv \lim_{N \rightarrow +\infty} S_N(x) = \lim_{N \rightarrow +\infty} \sum_{n=0}^N f_n(x) \quad (x \in I). \quad (3.89)$$

Such a limit of the sum of function series, $\sum_{n=0}^{+\infty} f_n(x)$, is referred to as a **function series**. If $\sum_{n=0}^{+\infty} |f_n(x)|$ converges, then the

function series $\sum_{n=0}^{+\infty} f_n(x)$ **absolutely converges**.

Example 3.3. Let us consider a sum of function sequences, $S_n(x) = \sum_{k=0}^n x^k$, for any non-negative integer n . Here we remark that $x^0 = 1$ for any real number x and $x^n = 0$ at $x = 0$ for any positive integer n .

- (1) Show that the function series $\lim_{n \rightarrow +\infty} S_n(x)$ always converges to $\frac{1}{1-x}$ in the interval $|x| < 1$.
- (2) Show that the function series $\lim_{n \rightarrow +\infty} S_n(x)$ always uniformly converges to $\frac{1}{1-x}$ in the interval $|x| < \rho$ for any real number $\rho \in (0, 1)$.
- (3) Show that the function series $\lim_{n \rightarrow +\infty} S_n(x)$ always absolutely converge in the interval $|x| < 1$.
- (4) Show that the function series $\lim_{n \rightarrow +\infty} S_n(x)$ does not absolutely converge in the interval $|x| > 1$ at all.
- (5) Show that $\frac{d}{dx} \left(\lim_{n \rightarrow +\infty} \sum_{k=0}^n x^k \right) = \lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} x^k \right)$ in the interval $(-1, +1)$.

(Solution) (1) For any real number $x \in (-1, +1)$, we have

$$S_n(x) = \sum_{k=0}^n x^k = \frac{1-x^{n+1}}{1-x}, \quad (3.90)$$

and then

$$\left| S_n(x) - \frac{1}{1-x} \right| = \frac{|x|^{n+1}}{|1-x|} \rightarrow 0 \quad (n \rightarrow +\infty, \forall x \in (-1, +1)). \quad (3.91)$$

Rigorously we can show that

$$\left| S_n(x) - \frac{1}{1-x} \right| = \frac{|x|^{n+1}}{|1-x|} < \varepsilon, \quad (3.92)$$

$$|x|^{n+1} < \varepsilon |1-x|, \quad (3.93)$$

$$(n+1)\ln(|x|) < \ln(\varepsilon|1-x|), \quad (3.94)$$

$$n+1 > \frac{\ln(\varepsilon|1-x|)}{\ln(|x|)}, \quad (3.95)$$

and then

$$\forall \varepsilon > 0 : n > \frac{\ln(\varepsilon|1-x|)}{\ln(|x|)} - 1 \Rightarrow \left| S_n(x) - \frac{1}{1-x} \right| < \varepsilon. \quad (3.96)$$

- (2) In the bounded and closed interval $x \in [-\rho, +\rho]$ for any real number $\rho \in (-1, +1)$, we have

$$S_n(x) = \sum_{k=0}^n x^k = \frac{1-x^{n+1}}{1-x}, \quad (3.97)$$

and then

$$\left| S_n(x) - \frac{1}{1-x} \right| = \frac{|x|^{n+1}}{|1-x|} = \frac{|x|^{n+1}}{|1-|x||} \leq \frac{\rho^{n+1}}{|1-\rho|} \rightarrow 0 \quad (n \rightarrow +\infty, \forall x \in (-\rho, +\rho)). \quad (3.98)$$

Rigorously we can show that

$$\forall \varepsilon > 0 : n > \frac{\ln(\varepsilon|1-\rho|)}{\ln(\rho)} - 1 \Rightarrow \left| S_n(x) - \frac{1}{1-x} \right| < \varepsilon \quad (\forall x \in (-\rho, +\rho)). \quad (3.99)$$

- (3) For any real number $x \in (-1, +1)$, we have

$$\sum_{k=0}^n |x^k| = \frac{1-|x|^{n+1}}{1-|x|}, \quad (3.100)$$

and then

$$\left| \sum_{k=0}^n |x^k| - \frac{1}{1-|x|} \right| = \frac{|x|^{n+1}}{1-|x|} \rightarrow 0 \quad (n \rightarrow +\infty, \forall x \in (-1, +1)). \quad (3.101)$$

Rigorously we can show that

$$\forall \varepsilon > 0 : n > \frac{\ln(\varepsilon|1-|x||)}{\ln(|x|)} - 1 \Rightarrow \left| \sum_{k=0}^n |x^k| - \frac{1}{1-|x|} \right| < \varepsilon \quad (\forall x \in (-1, +1)). \quad (3.102)$$

(4) For any real number $x \in (-1, +1)$, we have

$$\sum_{k=0}^n |x^k| = \frac{1-|x|^{n+1}}{1-|x|}, \quad (3.103)$$

and then

$$\left| \sum_{k=0}^n |x^k| \right| = \left| \frac{1-|x|^{n+1}}{1-|x|} \right| = \frac{|1-|x|^{n+1}|}{|1-|x||} \rightarrow +\infty \quad (n \rightarrow +\infty, \forall x \in (-1, +1)). \quad (3.104)$$

Rigorously we can show that

$$\forall M > 0 : n > \frac{\ln(M|1-|x||+1)}{\ln(|x|)} - 1 \Rightarrow \left| \sum_{k=0}^n |x^k| \right| > M \quad (\forall x \in (-\rho, +\rho)). \quad (3.105)$$

(5) By using the equality $\lim_{n \rightarrow +\infty} S_n(x) = \frac{1}{1-x}$ in the interval $(-1, +1)$ in the above question (2), the left side of the present equality can be derived as follows:

$$\frac{d}{dx} \left(\lim_{n \rightarrow +\infty} \sum_{k=0}^n x^k \right) = \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{1}{(1-x)^2} \quad (3.106)$$

The right side of the present equality is rewritten to

$$\begin{aligned} \lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} x^k \right) &= \lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} kx^{k-1} \right) \\ &= \lim_{n \rightarrow +\infty} \sum_{k=1}^n (kx^{k-1}) = \frac{1}{x} \lim_{n \rightarrow +\infty} \sum_{k=1}^n (kx^k) \\ &= \lim_{n \rightarrow +\infty} \frac{-(n+1)x^n(1-x) + 1 - x^{n+1}}{(1-x)^2} \\ &= \lim_{n \rightarrow +\infty} \frac{-(n+1)x^n + (n+1)x^{n+1} + 1 - x^{n+1}}{(1-x)^2} \\ &= \lim_{n \rightarrow +\infty} \left(\frac{1 - x^n - n(1-x)x^n}{(1-x)^2} \right) \\ &= \lim_{n \rightarrow +\infty} \left(\frac{1}{(1-x)^2} - \frac{x^n}{(1-x)^2} - \frac{nx^n}{1-x} \right). \end{aligned} \quad (3.107)$$

Now we can confirm that

$$|x^n| \rightarrow 0 \quad (n \rightarrow +\infty), \quad (3.108)$$

and

$$\begin{aligned} |nx^n| &= \frac{n}{\left| \left(\frac{1}{x} \right)^n \right|} = \frac{n}{\left| \left(1 + \left(\frac{1}{x} - 1 \right) \right)^n \right|} = \frac{n}{\left| \sum_{l=0}^n \frac{n!}{l!(n-l)!} \left(\frac{1}{x} - 1 \right)^l \right|} \\ &= \frac{n}{\left| \frac{n(n-1)}{2} \left(\frac{1}{x} - 1 \right)^2 \right|} = \frac{2}{(n-1) \left| \frac{1}{x} - 1 \right|^2} \rightarrow 0 \quad (n \rightarrow +\infty), \end{aligned} \quad (3.109)$$

and then, from Eq.(3.107)-(3.109), we have

$$\lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} x^k \right) = \frac{1}{(1-x)^2}. \quad (3.110)$$

More rigorously, we can confirm the following inequalities for any positive real number ε :

$$\left| \frac{x^n}{(1-x)^2} \right| = \frac{|x|^n}{|1-x|^2} < \frac{1}{2}\varepsilon \Rightarrow n > \frac{\ln(\frac{1}{2}|1-x|^2\varepsilon)}{\ln(|x|)}, \quad (3.111)$$

$$\begin{aligned} \left| \frac{nx^n}{1-x} \right| &= \frac{n|x|^n}{|1-x|} = \frac{n}{|1-x|\left(\frac{1}{x}\right)^n} = \frac{n}{|1-x|\left(1+\left(\frac{1}{x}-1\right)\right)^n} \\ &= \frac{n}{|1-x|\left|\sum_{l=0}^n \frac{n!}{l!(n-l)!} \left(\frac{1}{x}-1\right)^l\right|} \\ &= \frac{n}{|1-x|\left|\frac{n(n-1)}{2} \left(\frac{1}{x}-1\right)^2\right|} \\ &= \frac{2}{(n-1)|x|\left|\frac{1}{x}-1\right|^3} < \frac{1}{2}\varepsilon \Rightarrow n > \frac{4}{|x|\left|\frac{1}{x}-1\right|^3\varepsilon} + 1, \end{aligned} \quad (3.112)$$

and then we have

$$\forall \varepsilon > 0 : n \geq N(\varepsilon, x) \Rightarrow \left| \sum_{k=0}^n \left(\frac{d}{dx} x^k \right) - \frac{1}{(1-x)^2} \right| = \left| -\frac{x^n}{(1-x)^2} - \frac{nx^n}{1-x} \right| \leq \left| \frac{x^n}{(1-x)^2} \right| + \left| \frac{nx^n}{1-x} \right| < \varepsilon \quad (3.113)$$

where

$$N(\varepsilon, x) \equiv \max \left\{ \left\lfloor \frac{\ln(\frac{1}{2}|1-x|^2\varepsilon)}{\ln(|x|)} \right\rfloor, \left\lfloor \frac{4}{|x|\left|\frac{1}{x}-1\right|^3\varepsilon} \right\rfloor + 1, 1 \right\}, \quad (3.114)$$

and $\lfloor a \rfloor$ is a floor function which outputs the integer part of a given real number a .

The sum of function sequence $S_N(x)$ can be regarded as one of function sequence and then we can apply the above theorems for function sequences to $\{S_n(x) | n = 0, 1, 2, \dots\}$.

Theorem 3.12. The necessary and sufficient condition for the uniformly convergence of the function sequence $\sum_{n=0}^{+\infty} f_n(x)$ in the bounded and closed interval I is as follows:

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : \forall n > m \geq N(\varepsilon) \Rightarrow |f_{m+1}(x) + f_{m+2}(x) + \dots + f_n(x)| < \varepsilon, \quad (3.115)$$

where $N(\varepsilon)$ does not depend on x in the interval I .

(Proof) In the theorem 3.8, by replacing $f_n(x)$ with $S_n(x) = \sum_{k=0}^n f_k(x)$, we have

$$S_n(x) - S_m(x) = f_{m+1}(x) + f_{m+2}(x) + \dots + f_n(x). \quad (3.116)$$

It is obvious that Eq.(3.69) is reduced to Eq.(3.115).

Theorem 3.13. The function sequence $\{f_n(x) | n = 0, 1, 2, \dots\}$ is always continuous and the function series $\sum_{n=0}^{+\infty} f_n(x)$

uniformly converges in the bounded and closed interval I , the function series $\sum_{n=0}^{+\infty} f_n(x)$ is always continuous in the interval I .

(Proof) The sum of function sequence, $S_n(x) \equiv \sum_{k=0}^n f_k(x)$, is always continuous and the function sequence $\{S_n(x) | n = 0, 1, 2, \dots\}$ uniformly converges. By using theorem 3.9, we can see that the function $\lim_{n \rightarrow +\infty} S_n(x)$ is always continuous in the interval I .

Theorem 3.14. Let us suppose that there exists a sequence $\{a_n > 0 | n = 0, 1, 2, 3, \dots\}$ that $\sum_{n=0}^{+\infty} a_n$ converges. If it is valid that

$$|f_n(x)| \leq a_n \quad (n = 0, 1, 2, \dots, x \in I), \quad (3.117)$$

the function series $\sum_{n=0}^{+\infty} f_n(x)$ always uniformly converges in the interval I .

(Proof) Because $\sum_{n=0}^{+\infty} a_n$ converges, the sequence $\left\{ \sum_{k=0}^n a_k \mid n = 0, 1, 2, \dots \right\}$ is a Cauchy sequence and then we have

$$\forall \varepsilon > 0, \exists N(\varepsilon) \geq 0 : n > m \geq N(\varepsilon) \Rightarrow \left| \sum_{k=0}^n a_k - \sum_{k=0}^m a_k \right| = |a_{m+1} + a_{m+2} + \dots + a_n| < \varepsilon. \quad (3.118)$$

By using Eqs.(3.117) and (3.118), we have

$$\begin{aligned} |f_{m+1}(x) + f_{m+2}(x) + \dots + f_n(x)| &\leq |f_{m+1}(x)| + |f_{m+2}(x)| + \dots + |f_n(x)| \\ &\leq a_{m+1} + a_{m+2} + \dots + a_n < \varepsilon \quad (n > m \geq N(\varepsilon), x \in I), \end{aligned} \quad (3.119)$$

where $N(\varepsilon)$ does not depend on x in the interval I . From theorem 3.12, the function series $\sum_{n=0}^{+\infty} f_n(x)$ always uniformly converges in the interval I .

A function series defined in terms of a sequence $\{c_n\}$ and a real number x_0 :

$$\sum_{n=0}^{+\infty} c_n (x - x_0)^n \quad (3.120)$$

is referred to as a **power series** around $x = x_0$. By setting $x_0 = 0$, the power series is reduced to

$$\sum_{n=0}^{+\infty} c_n x^n. \quad (3.121)$$

If the power series (3.121) converges in the open interval $|x| < r$, the power series (3.120) converges in the open interval $|x - x_0| < r$. Moreover

$$\sum_{n=0}^{+\infty} c_n x^{2n} \quad (3.122)$$

converges in the open interval $|x^2| < r$ such that $|x| < \sqrt{r}$. For such power series, we have the following theorems:

Theorem 3.15 (Abel theorem). Ever power series $\sum_{n=0}^{+\infty} c_n x^n$ to converges at $x = +r$ or $x = -r$ ($r > 0$) always absolutely converges in the open interval $|x| < r$.

(Proof) Because $\sum_{n=0}^{+\infty} c_n r^n$ converges, there exists a finite constant value M satisfying $|c_n r^n| < M$ for any non-negative integer n from theorem 3.6, and then we have

$$|c_n x^n| = |c_n r^n| \left| \frac{x}{r} \right|^n \leq M \left| \frac{x}{r} \right|^n \quad (3.123)$$

such that

$$\sum_{n=1}^{+\infty} |c_n x^n| \leq M \sum_{n=1}^{+\infty} \left| \frac{x}{r} \right|^n. \quad (3.124)$$

For the open interval $|x| < r$, $\sum_{n=0}^{+\infty} |x/r|^n$ converges and then $\sum_{n=1}^{+\infty} c_n x^n$ absolutely converges.

Theorem 3.16. Every power series $\sum_{n=0}^{+\infty} c_n x^n$ not to converge absolutely at $x = +r$ or $x = -r$ ($r > 0$) always diverges in $|x| > r$.

(Proof) If the power series $\sum_{n=0}^{+\infty} c_n x^n$ converges at a point $x = a$ in $|x| > r$, both points $x = +r$ and $x = -r$ satisfy $|r| < |a|$

and the $\sum_{n=0}^{+\infty} c_n a^n$ should converges from theorem 3.15. This fact goes against the premise of the present theorem.

The present theorem has been confirmed from *reductio ad absurdum*,

From theorems 3.15 and 3.16 there exists a real number r satisfying that the power series (3.121) always absolutely converges in $|x| < r$ and always diverges in $|x| > r$. Such a real number r is referred to as a **convergence radius**. In the case of $r = +\infty$, the power series (3.121) always converges in $|x| < +\infty$. In the case of $r = 0$, the power series (3.121)

always diverges for any real number x except $x = 0$.

Theorem 3.17. Every power series $\sum_{n=0}^{+\infty} c_n x^n$ with a convergence radius $r(>0)$ always uniformly converges in the bounded and closed interval $|x| \leq \rho$ for any real value $\rho \in (0, r)$.

(Proof) For $|x| \leq \rho$ ($0 < \forall \rho < r$), we have

$$\left| \sum_{k=0}^n c_k x^k \right| \leq \sum_{k=0}^n |c_k x^k| \leq \sum_{k=0}^n |c_k| |x|^k \leq \sum_{k=0}^n |c_k| \rho^k. \quad (3.125)$$

From theorems 3.14, the sequence $\left\{ \sum_{k=0}^n c_k x^k \mid n = 0, 1, 2, \dots \right\}$, such that the power series $\sum_{n=0}^{+\infty} c_n x^n$, always uniformly converge in $|x| \leq \rho$ for any real number $\rho \in (0, r)$.

Theorem 3.18. Every power series $\sum_{n=0}^{+\infty} c_n x^n$ with a convergence radius $r(>0)$ is always a $C^{+\infty}$ -class function and

$$\frac{d^l}{dx^l} \left(\lim_{n \rightarrow +\infty} \sum_{k=0}^n c_k x^k \right) = \lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d^l}{dx^l} (c_k x^k) \right) \quad (l = 1, 2, 3, \dots) \quad (3.126)$$

for any real number x in the open interval $|x| \leq \rho$ ($\rho \in (0, r)$).

(Proof) The power series $\sum_{n=0}^{+\infty} c_n x^n$ with a convergence radius $r(>0)$ always converges and then it is always continuous

in the interval $|x| \leq \rho$ for any real number $\rho \in (0, r)$ by using theorem 3.13. It means that the power series $\sum_{n=0}^{+\infty} c_n x^n$ is continuous in the open interval $|x| < r$. Next we consider the following function sequence:

$$\left\{ \sum_{k=0}^n \left(\frac{d}{dx} (c_k x^k) \right) = \sum_{k=1}^n k c_k x^{k-1} \mid n = 1, 2, 3, \dots \right\} \quad (3.127)$$

in the open interval $|x| < r$. For a positive real number ρ satisfying $|x| < \rho < r$ for any real number x in $|x| < r$,

(a) $\lim_{n \rightarrow +\infty} \sum_{k=0}^n c_k \rho^k$ converges and then there exists M satisfying $|c_n \rho^n| \leq M$ for any nonnegative integer n .

(b) From the above statement (a), we have the following inequality:

$$|n c_n x^{n-1}| = \frac{1}{\rho} |c_n \rho^n| \times n \left| \frac{x}{\rho} \right|^{n-1} \leq \frac{1}{\rho} M \times n \left| \frac{x}{\rho} \right|^{n-1}. \quad (3.128)$$

The sum of sequence $\sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1}$ can be derived as follows:

$$\begin{aligned} \left(\left| \frac{x}{\rho} \right| - 1 \right) \left(\sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1} \right) &= \left| \frac{x}{\rho} \right| \left(\sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1} \right) - \left(\sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1} \right) \\ &= \left(\sum_{k=1}^{n-1} k \left| \frac{x}{\rho} \right|^k \right) + n \left| \frac{x}{\rho} \right|^n - \left(\sum_{k=1}^{n-1} (k+1) \left| \frac{x}{\rho} \right|^k \right) - 1 \\ &= n \left| \frac{x}{\rho} \right|^n - \left(\sum_{k=0}^{n-1} \left| \frac{x}{\rho} \right|^k \right) \\ &= n \left| \frac{x}{\rho} \right|^n - \frac{1 - \left| \frac{x}{\rho} \right|^n}{1 - \left| \frac{x}{\rho} \right|} \end{aligned} \quad (3.129)$$

$$\sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1} = \frac{1}{1 - \left| \frac{x}{\rho} \right|} \times \left(-n \left| \frac{x}{\rho} \right|^n + \frac{1 - \left| \frac{x}{\rho} \right|^n}{1 - \left| \frac{x}{\rho} \right|} \right). \quad (3.130)$$

We can confirm that $\lim_{n \rightarrow +\infty} n \left| \frac{x}{\rho} \right|^{n-1} = 0$ as follows:

$$\forall n > 1 + \frac{\ln(\varepsilon)}{1 + \ln\left(\left|\frac{x}{\rho}\right|\right)} : \left| n \left| \frac{x}{\rho} \right|^{n-1} \right| = \left| \exp\left(\ln(n) + (n-1) \left| \frac{x}{\rho} \right|\right) \right| \leq \left| \exp\left(n - 1 + (n-1) \left| \frac{x}{\rho} \right|\right) \right| < \varepsilon, \quad (3.131)$$

and then the series $\lim_{n \rightarrow +\infty} \sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1}$ always converges. Hence we have confirmed that the series $\lim_{n \rightarrow +\infty} \sum_{k=1}^n k x^{k-1}$ absolutely converges for any real number x in the bounded and closed interval $|x| \leq \rho$ with $\forall \rho \in (0, r)$. It means that the convergence radius of the series $\lim_{n \rightarrow +\infty} \sum_{k=1}^n k \left| \frac{x}{\rho} \right|^{k-1}$ is larger than or equal to r . From theorem 3.11 and theorem 3.17, it is valid that the series $\lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} (c_k x^k) \right)$ always uniformly converges and

$$\frac{d}{dx} \left(\lim_{n \rightarrow +\infty} \sum_{k=0}^n c_k x^k \right) = \lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} (c_k x^k) \right), \quad (3.132)$$

for the bounded and closed interval $|x| \leq \rho$ for any positive real number $\rho \in (0, r)$. By iterating the similar procedures, we can confirm the series $\lim_{n \rightarrow +\infty} \sum_{k=0}^n \left(\frac{d}{dx} (c_k x^k) \right)$ always uniformly converges and we have Eq.(3.126).

Theorem 3.19 (Cauchy-Hadamard Formula). The convergence radius r of the power series (3.121) is given as follows:

$$\frac{1}{r} = \lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}} |c_m|^{1/m} \quad (3.133)$$

(Proof) From the definition of the limit superior $\lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}}$, the following two statements are valid for any

postive number ρ :

- (I) If we have $\frac{1}{\rho} < \lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}} |c_m|^{1/m}$, then, there exist an infinite number of c_n satisfying $\frac{1}{\rho} < |c_n|^{1/n}$, such that $|c_n| > \left(\frac{1}{\rho}\right)^n$.
- (II) If we have $\frac{1}{\rho} > \lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}} |c_m|^{1/m}$, there are only a finite number c_n satisfying $\frac{1}{\rho} < |c_n|^{1/n}$, such that $|c_n| > \left(\frac{1}{\rho}\right)^n$. Conversely, there exist an infinite number of c_n satifying $\frac{1}{\rho} > |c_n|^{1/n}$, such that $|c_n| < \left(\frac{1}{\rho}\right)^n$.

For any positive number ρ in the open interval $(0, r)$ with

$$r \equiv \left(\lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}} |c_m|^{1/m} \right)^{-1}, \quad (3.134)$$

there exist an infinite number of c_n satisfying $|c_n| < \left(\frac{1}{\rho}\right)^{1/n}$. And then, in the open interval $|x| < \rho$, there exists a positive integer N so that $|c_n x^n| < \left(\frac{|x|}{\rho}\right)^n$ for any integer $n \geq N$ and the power series $\sum_{n=0}^{+\infty} c_n x^n$ absolutely converges. For any positive number ρ in the open interval $(r, +\infty)$, there exist an infinite number of c_n satisfying $|c_n| > \left(\frac{1}{\rho}\right)^{1/n}$, and there exists a positive integer N so that it is valid that $|c_n x^n| > \left(\frac{|x|}{\rho}\right)^n$ for any positive integer $n \geq N$ in the open interval $|x| > \rho$. The power series $\sum_{n=0}^{+\infty} c_n z^n$ diverges in the open interval $|x| > \rho$. From these arguments, we have confirmed that r in Eq.(3.134) is the convergence radius of the power series $\sum_{n=0}^{+\infty} c_n z^n$.

Theorem 3.20 (Corollary of Cauchy-Hadamard Formula). If there exists the limit $\lim_{n \rightarrow +\infty} \left| \frac{c_{n+1}}{c_n} \right| = l$ as a finite value,

the converge radius r of the power series $\sum_{n=0}^{+\infty} c_n z^n$ is given by $r = \frac{1}{l}$.

(Proof) Because of $\lim_{n \rightarrow +\infty} \left| \frac{c_{n+1}}{c_n} \right| = l$, we have

$$\forall \varepsilon > 0, \exists N(\varepsilon) > 0; \forall n > N(\varepsilon) \Rightarrow \left| \frac{c_{n+1}}{c_n} - l \right| < \varepsilon, \text{ such that } l - \varepsilon < \left| \frac{c_{n+1}}{c_n} \right| < l + \varepsilon. \quad (3.135)$$

For a positive real number ε , we have

$$\begin{aligned} & \exists N(\varepsilon) > 0; \forall n > \forall m > N(\varepsilon) \\ & \Rightarrow |c_n x^n| = \left| \frac{c_n}{c_{n-1}} x \right| \left| \frac{c_{n-1}}{c_{n-2}} x \right| \times \cdots \times \left| \frac{c_{m+1}}{c_m} x \right| |c_m x^m| < (l + \varepsilon)^{n-m} |x|^{n-m} |c_m x^m| \end{aligned} \quad (3.136)$$

in the open interval $|x| < \frac{1}{l + \varepsilon}$. The series $\sum_{n=m}^{+\infty} (l + \varepsilon)^{n-m} |x|^{n-m} |c_m x^m|$ converges and then $\sum_{n=m}^{+\infty} |c_n x^n|$ also converges.

It means that the power series $\sum_{n=0}^{+\infty} c_n x^n$ absolutely converges. On the other hand, for any positive number ε , it is valid that

$$\begin{aligned} & \exists N(\varepsilon) > 0; \forall n > \forall m > N(\varepsilon) \\ & \Rightarrow |c_n x^n| = \left| \frac{c_n}{c_{n-1}} x \right| \left| \frac{c_{n-1}}{c_{n-2}} x \right| \times \cdots \times \left| \frac{c_{m+1}}{c_m} x \right| |c_m x^m| > (l - \varepsilon)^{n-m} |x|^{n-m} |c_m x^m| \end{aligned} \quad (3.137)$$

in the open interval $|x| > \frac{1}{l - \varepsilon}$. We remark that $\sum_{n=m}^{+\infty} (l - \varepsilon)^{n-m} |z|^{n-m} |c_m z^m|$ diverges and then $\sum_{n=m}^{+\infty} |c_n z^n|$ has diverges.

Hence the power series $\sum_{n=0}^{+\infty} c_n z^n$ diverges. It means that $r = \frac{1}{l}$ is the convergence radius of the power series $\sum_{n=0}^{+\infty} c_n z^n$.

4 First Order Ordinary Differential Equation

4.1 Existence and Uniqueness of Solution in First Order Differential Equation

For a real-valued function $y = w(x)$ for a real variable x in a bounded and closed interval $|x - x_0| \leq \alpha$, we consider a differential equation as follows:

$$F\left(x, w(x), \frac{d}{dx} w(x)\right) = 0, \quad (4.1)$$

This equation is referred to as a **first order differential equation**. Here we impose an initial condition:

$$w(x_0) = y_0. \quad (4.2)$$

Let us consider the case in which Eq.(4.1) can be reduced to

$$\frac{d}{dx} w(x) = f(x, w(x)) \quad (|x - x_0| \leq \alpha). \quad (4.3)$$

Here we remark that $f(x, y)$ is defined in a closed region $\{(x, y) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta\}$ on the xy -plane and should be a one-valued and continuous function at every point (x, y) in the region. There should exist the maximum value M of $|f(x, y)|$ in the closed region $\{(x, y) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta\}$ such that

$$|f(x, y)| \leq M \quad (\forall (x, y) \in \{(x, y) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta\}). \quad (4.4)$$

For any real number x in the closed region $|x - x_0| \leq \alpha$, we take an integration in the both side of Eq.(4.3) as follows:

$$\int_{x_0}^x \frac{d}{dx} w(x) dx = \int_{x_0}^x f(x, w(x)) dx, \quad (4.5)$$

such that

$$w(x) = y_0 + \int_{x_0}^x f(x, w(x)) dx \quad (|x - x_0| \leq \alpha). \quad (4.6)$$

Let us consider the existence and the uniqueness of the solution of the first order ordinary differential equation (4.3) with the initial condition (4.2) under the following condition:

$$\forall (x, y_1) \in \{(x, y) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta\}, \forall (x, y_2) \in \{(x, y) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta\}, \exists K > 0;$$

$$|f(x, y_1) - f(x, y_2)| \leq K|y_1 - y_2|. \quad (4.7)$$

It is referred to as a **Lipschitz condition**. We introduce a sequence of functions, $\{w_n(x) \mid |x - x_0| \leq \alpha, n = 0, 1, 2, \dots\}$, which is defined by

$$w_n(x) = y_0 + \int_{x_0}^x f(x, w_{n-1}(x)) dx \quad (|x - x_0| \leq \alpha), \quad (4.8)$$

where

$$w_0(x) = y_0 \quad (|x - x_0| \leq \alpha). \quad (4.9)$$

First we show the function $w_n(x)$ is always a continuous function in the closed region $|x - x_0| \leq \alpha$ for every positive integer n . Because $f(x, y_0)$ is a one valued continuous and bounded function in the closed region $|x - x_0| \leq \alpha$ as show in Eq.(4.4), then

$$w_1(x) = y_0 + \int_{x_0}^x f(\xi, y_0) d\xi \quad (|x - x_0| \leq \alpha), \quad (4.10)$$

is always a continuous and bounded function. In the similar arguments, the following sequence of function

$$\begin{cases} w_2(x) = y_0 + \int_{x_0}^x f(\xi, w_1(\xi)) d\xi, \\ w_3(x) = y_0 + \int_{x_0}^x f(\xi, w_2(\xi)) d\xi, \\ \vdots \\ w_n(x) = y_0 + \int_{x_0}^x f(\xi, w_{n-1}(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha), \quad (4.11)$$

are also always continuous functions in the closed region $|x - x_0| \leq \alpha$. If there exists the limit of n infinity, $\lim_{n \rightarrow +\infty} w_n(x)$, we can conclude that Eq.(4.6) has $w(x) = \lim_{n \rightarrow +\infty} w_n(x)$ as one of continuous and bounded solutions.

Nest step we consider the existence of solutions in Eq.(4.6). The sequence of function $\{w_n(x)\}$ can be rewritten as a function series as follows:

$$\begin{aligned} w_n(x) &= y_0 + (w_1(x) - w_0(x)) + (w_2(x) - w_1(x)) + \dots + (w_n(x) - w_{n-1}(x)) \\ &= y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)). \end{aligned} \quad (4.12)$$

If the function series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$ absolutely converges, the limit $\lim_{n \rightarrow +\infty} w_n(x)$ exists in the interval $[\alpha, \beta]$. All we have to do for the proof in the existence of solutions in the ordinary differential equation (4.3)

with the initial condition (4.2) is to show the absolutely convergence of the series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$

for any real number x in the interval $[x_0 - \alpha, x_0 + \alpha]$.

$$\begin{aligned} |w_n(x) - w_{n-1}(x)| &= \left| \int_{x_0}^x f(\xi_{n-1}, w_{n-1}(\xi_{n-1})) - f(\xi_{n-1}, w_{n-2}(\xi_{n-1})) d\xi_{n-1} \right| \\ &\leq \left| \int_{x_0}^x |f(\xi_{n-1}, w_{n-1}(\xi_{n-1})) - f(\xi_{n-1}, w_{n-2}(\xi_{n-1}))| d\xi_{n-1} \right| \\ &\leq K \left| \int_{x_0}^x |w_{n-1}(\xi_{n-1}) - w_{n-2}(\xi_{n-1})| d\xi_{n-1} \right| \\ &\leq K \left| \int_{x_0}^x \left| \int_{x_0}^{\xi_{n-1}} f(\xi_{n-2}, w_{n-2}(\xi_{n-2})) - f(\xi_{n-2}, w_{n-3}(\xi_{n-2})) d\xi_{n-2} \right| d\xi_{n-1} \right| \\ &\leq K \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} |f(\xi_{n-2}, w_{n-2}(\xi_{n-2})) - f(\xi_{n-2}, w_{n-3}(\xi_{n-2}))| d\xi_{n-2} d\xi_{n-1} \right| \\ &\leq K^2 \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} |w_{n-2}(\xi_{n-2}) - w_{n-3}(\xi_{n-2})| d\xi_{n-2} d\xi_{n-1} \right| \\ &\leq K^{n-2} \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \dots \int_{x_0}^{\xi_3} |w_2(\xi_2) - w_1(\xi_2)| d\xi_2 \dots d\xi_{n-2} d\xi_{n-1} \right| \end{aligned}$$

$$\begin{aligned}
&\leq K^{n-1} \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \cdots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} |w_1(\xi_1) - w_0(\xi_1)| d\xi_1 d\xi_2 \cdots d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq K^{n-1} \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \cdots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} \int_{x_0}^{\xi_1} |f(\xi_0, y_0)| d\xi_0 d\xi_1 d\xi_2 \cdots d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq K^{n-1} M \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \cdots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} \int_{x_0}^{\xi_1} d\xi_0 d\xi_1 d\xi_2 \cdots d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq K^{n-1} M \times \frac{1}{n!} |x - x_0|^n.
\end{aligned} \tag{4.13}$$

By means of Eq.(4.13), the absolutely convergence of the series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$ can be obtained as follows:

$$\begin{aligned}
|w_n(x)| &\leq |y_0| + \sum_{k=1}^n |w_k(x) - w_{k-1}(x)| \\
&\leq |y_0| + \sum_{k=1}^n K^{k-1} \times \frac{M}{n!} |x - x_0|^k \\
&\leq |y_0| + \sum_{k=1}^n K^{k-1} \times \frac{M}{n!} \alpha^k \\
&\rightarrow |y_0| + \frac{M}{K} (\exp(K\alpha) - 1) \quad (n \rightarrow +\infty, |x - x_0| \leq \alpha).
\end{aligned} \tag{4.14}$$

In the final equality, we use the Taylor expansion of $\exp(x)$ such that

$$\exp(x) = 1 + \lim_{N \rightarrow +\infty} \sum_{n=1}^N \frac{1}{n!} x^n \quad (|x| < +\infty). \tag{4.15}$$

Then it has been shown that the existence of the first order ordinary differential equation (4.3) with the initial condition (4.2) under the Lipshitz condition (4.7). From theorem 3.13, the function sequence $\{w_n(x) | n = 0, 1, 2, \dots\}$ defined by the recursion formulas (4.8) with the initial condition (4.2) uniformly converges and theorem 3.14, we can see that the function sequence $\{w_n(x) | n = 0, 1, 2, \dots\}$ always converges to a continuous function $w(x)$ in the closed interval $|x - x_0| \leq \alpha$, such that

$$|w(x) - w(\xi)| \leq |w(x) - w_n(x)| + |w_n(x) - w_n(\xi)| + |w_n(\xi) - w(\xi)| \rightarrow 0 \quad (|x - x_0| \leq \alpha, \xi \rightarrow x, n \rightarrow +\infty). \tag{4.16}$$

Finally, we shown the uniqueness for solutions in Eq.(4.6). Let us suppose that Eq.(4.6) have both function $w(x) = \Phi(x)$ and $w(x) = \tilde{\Phi}(x)$ as solutions. We consider the defference between $\Phi(x)$ and $\tilde{\Phi}(x)$ as follows:

$$\Phi(x) - \tilde{\Phi}(x) = \int_{x_0}^x (f(x, \Phi(x)) - f(x, \tilde{\Phi}(x))) dx, \tag{4.17}$$

such that

$$\begin{aligned}
|\Phi(x) - \tilde{\Phi}(x)| &= \left| \int_{x_0}^x (f(x, \Phi(x)) - f(x, \tilde{\Phi}(x))) dx \right| \leq \left| \int_{x_0}^x |f(x, \Phi(x)) - f(x, \tilde{\Phi}(x))| dx \right| \\
&\leq K \left| \int_{x_0}^x |\Phi(x) - \tilde{\Phi}(x)| dx \right| \\
&\leq K \left| \int_{x_0}^x \max_{x \in [x_0, x]} |\Phi(x) - \tilde{\Phi}(x)| dx \right| \leq K \left(\max_{x \in [x_0, x]} |\Phi(x) - \tilde{\Phi}(x)| \right) \left| \int_{x_0}^x dx \right| \\
&= K \left(\max_{x \in [x_0, x]} |\Phi(x) - \tilde{\Phi}(x)| \right) |x - x_0| = 2M |x - x_0|,
\end{aligned} \tag{4.18}$$

Here we remark that $[x_0, x]$ should be replaced by $[x, x_0]$ in the above inequalities for $x < x_0$. From the above derivations of the inequalities (4.18), we notice that the following two inequalities are valid:

$$|\Phi(x) - \tilde{\Phi}(x)| \leq K \left| \int_{x_0}^x |\Phi(x) - \tilde{\Phi}(x)| dx \right|, \tag{4.19}$$

$$|\Phi(x) - \tilde{\Phi}(x)| \leq K \left(\max_{x \in [x_0, x]} |\Phi(x) - \tilde{\Phi}(x)| \right) |x - x_0|. \quad (4.20)$$

By applying Eq.(4.20) to the right side of Eq.(4.19) iteratively, we obtain

$$\begin{aligned} |\Phi(x) - \tilde{\Phi}(x)| &\leq K \left| \int_{x_0}^x |\Phi(\xi_1) - \tilde{\Phi}(\xi_1)| d\xi_1 \right| \leq K^2 \left| \int_{x_0}^x \int_{x_0}^{\xi_1} |\Phi(\xi_2) - \tilde{\Phi}(\xi_2)| d\xi_2 d\xi_1 \right| \\ &\leq \dots \leq K^n \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \dots \int_{x_0}^{\xi_{n-1}} |\Phi(\xi_n) - \tilde{\Phi}(\xi_n)| d\xi_n \dots d\xi_2 d\xi_1 \right| \\ &\leq \left(\max_{|\xi - x_0| \leq \alpha} |\Phi(\xi) - \tilde{\Phi}(\xi)| \right) K^n \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \dots \int_{x_0}^{\xi_{n-1}} |\xi_n - x_0| d\xi_n \dots d\xi_2 d\xi_1 \right| \\ &= \left(\max_{|\xi - x_0| \leq \alpha} |\Phi(\xi) - \tilde{\Phi}(\xi)| \right) K^n \times \frac{1}{n!} |x - x_0|^n \\ &= \left(\max_{|\xi - x_0| \leq \alpha} |\Phi(\xi) - \tilde{\Phi}(\xi)| \right) K^n \times \frac{1}{n!} \alpha^n \rightarrow 0 \quad (n \rightarrow +\infty, |x - x_0| \leq \alpha). \end{aligned} \quad (4.21)$$

Here we remark that, for every integer m satisfied $m > 2\alpha K$, it is valid that

$$\begin{aligned} \left| K^n \times \frac{1}{n!} \alpha^n \right| &= \frac{(\alpha K)^m}{m!} \prod_{k=1}^{n-m} \frac{\alpha K}{m+k} \leq \frac{(\alpha K)^m}{m!} \prod_{k=1}^{n-m} \frac{\alpha K}{2\alpha K + k} \leq \frac{(\alpha K)^m}{m!} \prod_{k=1}^{n-m} \frac{\alpha K}{2\alpha K} \\ &\leq \frac{(\alpha K)^m}{m!} \left(\frac{1}{2} \right)^{n-m} \rightarrow 0 \quad (n \rightarrow +\infty, n \geq m). \end{aligned} \quad (4.22)$$

From Eq.(4.21) with Eq.(4.22), $\Phi(x)$ and $\tilde{\Phi}(x)$ are equivalent of each other and the uniqueness for solutions in Eq.(4.6) has been confirmed.

4.2 Separation of Variables

When two functions $P(x)$ and $Q(y)$ are given for each different variable x and y , respectively, the first order ordinary differential equation for $y = w(x)$:

$$\frac{dy}{dx} = P(x)Q(y), \quad (4.23)$$

such that

$$\frac{d}{dx} w(x) = P(x)Q(w(x)), \quad (4.24)$$

is referred to as a **separation of variables**. The present differential equation can be rewritten as

$$\left(\frac{1}{Q(y)} \right) \frac{dy}{dx} = P(x), \quad (4.25)$$

and take an indefinite integral with respect to x as follows:

$$\int \left(\left(\frac{1}{Q(y)} \right) \frac{dy}{dx} \right) dx = \int P(x) dx, \quad (4.26)$$

$$\left[\int \left(\frac{1}{Q(y)} \right) dy \right]_{y=w(x)} = \int P(x) dx, \quad (4.27)$$

4.3 Homogeneous Form

Let us consider the following first order ordinary differential equation for $y = w(x)$:

$$\frac{dy}{dx} = f\left(\frac{y}{x}\right), \quad (4.28)$$

such that

$$\frac{d}{dx} w(x) = f\left(\frac{w(x)}{x}\right), \quad (4.29)$$

where the right side of Eq.(4.29) is given as a homogeneous form of x and $y = w(x)$. In the present case, by replacing $w(x)$ by $v = \frac{w(x)}{x}$, Eq.(4.29) can be rewritten as

$$\frac{d}{dx}(xv(x)) = f(v(x)), \quad (4.30)$$

$$v(x) + x \frac{d}{dx}v(x) = f(v(x)), \quad (4.31)$$

$$\frac{d}{dx}v(x) = \frac{f(v(x)) - v(x)}{x}, \quad (4.32)$$

$$\frac{1}{f(v(x)) - v(x)} \frac{d}{dx}v(x) = \frac{1}{x}. \quad (4.33)$$

By integrating the both side of Eq.(4.33), we obtain the solution $y = w(x)$ of Eq.(4.29) as follows:

$$\int \frac{1}{f(v(x)) - v(x)} dv = \int \frac{1}{x} dx, \quad (4.34)$$

$$\left[\int \frac{1}{f(\eta) - \eta} d\eta \right]_{\eta = \frac{w(x)}{x}} = \int \frac{1}{x} dx. \quad (4.35)$$

We extend the present framework to the scheme of solving Eq.(4.29) to

$$\frac{d}{dx}w(x) = f\left(\frac{Ax+Bw(x)+E}{Cx+Dw(x)+F}\right), \quad \left(\det \begin{pmatrix} A & B \\ C & D \end{pmatrix} \neq 0\right). \quad (4.36)$$

By considering the variable transformation:

$$\begin{cases} x = \xi + \alpha \\ y = \eta + \beta \end{cases}, \quad (4.37)$$

where

$$\begin{cases} A\alpha + B\beta + E = 0 \\ C\alpha + D\beta + F = 0 \end{cases}, \quad (4.38)$$

for $f\left(\frac{Ax+Bw(x)+E}{Cx+Dw(x)+F}\right)$. Eq.(4.36) can be reduced to the following homogeneous form:

$$\frac{d(\eta + \alpha)}{d(\xi + \beta)} = f\left(\frac{A(\xi + \alpha) + B(\eta + \beta) + E}{C(\xi + \alpha) + D(\eta + \beta) + F}\right) = f\left(\frac{A\xi + A\alpha + B\eta + B\beta + E}{C\xi + C\alpha + D\eta + D\beta + F}\right) = f\left(\frac{A\xi + B\eta}{C\xi + D\eta}\right), \quad (4.39)$$

such that

$$\frac{d\eta}{d\xi} = f\left(\frac{A + B\left(\frac{\eta}{\xi}\right)}{C + D\left(\frac{\eta}{\xi}\right)}\right), \quad (4.40)$$

and it can be solved in the similar arguments of Eqs. (4.29)-(4.42) by replacing η by $\eta = v\xi$, as follows:

$$\frac{d(\xi v)}{d\xi} = f\left(\frac{A + Bv}{C + Dv}\right), \quad (4.41)$$

such that

$$\left[\left[\int \frac{1}{f\left(\frac{A+B\zeta}{C+D\zeta}\right) - \zeta} d\zeta \right]_{\zeta = \frac{w(\xi)-\beta}{\xi}} \right]_{\xi = x-\alpha} = \left[\int \frac{1}{\xi} d\xi \right]_{\xi = x-\alpha}. \quad (4.42)$$

4.4 First Order Linear Differential Equation

For a given real valued function $P(x)$, we consider a **homogeneous first order linear differential equation** for a real valued function $y = w(x)$ as follows:

$$\frac{dy}{dx} + P(x)y = 0, \quad (4.43)$$

such that

$$\frac{d}{dx}w(x) + P(x)w(x) = 0. \quad (4.44)$$

By regarding Eq.(4.43) as a separation form of variables, it can be solve as follows:

$$\frac{1}{w(x)} \frac{d}{dx}w(x) = -P(x), \quad (4.45)$$

$$\int \frac{1}{w(x)} \left(\frac{d}{dx}w(x) \right) dx = - \int P(x) dx, \quad (4.46)$$

$$\ln(|w(x)|) + \text{Constant} = - \int P(x) dx, \quad (4.47)$$

such that

$$w(x) = \text{Constant} \times \exp\left(- \int P(x) dx\right). \quad (4.48)$$

Next we consider a following **inhomogeneous first order linear equation** for a real valued function $y = w(x)$:

$$\frac{d}{dx}w(x) + P(x)w(x) = Q(x), \quad (4.49)$$

under giving real valued functions $P(x)$ and $Q(x)$. Now we restrict a solution of Eq.(4.49) to a following form:

$$w(x) = v(x) \times \exp\left(- \int P(x) dx\right), \quad (4.50)$$

and consider to determine the unknown function $v(x)$ so as to satisfy Eq.(4.49). We remark that Eq.(4.50) corresponds to the one replaced Constant in Eq.(4.48) by the unknown function $v(x)$. This scheme is referred to as a **method of variation of constants**. By substituting Eq.(4.50) to Eq.(4.49), we obtain the unknown function $v(x)$ as follows:

$$\frac{d}{dx} \left(v(x) \exp\left(- \int P(x) dx\right) \right) + P(x) \left(v(x) \exp\left(- \int P(x) dx\right) \right) = Q(x), \quad (4.51)$$

$$\left(\frac{d}{dx}v(x) \right) \exp\left(- \int P(x) dx\right) - v(x) \left(P(x) \exp\left(- \int P(x) dx\right) \right) + P(x) \left(v(x) \exp\left(- \int P(x) dx\right) \right) = Q(x), \quad (4.52)$$

$$\left(\frac{d}{dx}v(x) \right) \exp\left(- \int P(x) dx\right) = Q(x), \quad (4.53)$$

$$\frac{d}{dx}v(x) = Q(x) \exp\left(\int P(x) dx\right), \quad (4.54)$$

$$v(x) = \int Q(x) \exp\left(\int P(x) dx\right) dx + \text{Constant}. \quad (4.55)$$

By substituting Eq.(4.55) to Eq.(4.49), the solution $y = w(x)$ can be derived as

$$w(x) = \left\{ \int Q(x) \exp\left(\int P(x) dx\right) dx + \text{Constant} \right\} \times \exp\left(- \int P(x) dx\right). \quad (4.56)$$

4.5 Exact Equation

First of all in the present subsection, we summarize some important items for partial derivative and total derivative in the calculus as follows:

(I) Boundedness of Continuous Function in Closed and Bounded Region: For every region D and its boundary ∂D

In the xy -plane, their union $D \cup \partial D$ is referred to as a **closed and bounded region**. Every continuous function in a closed and bounded region is always bounded, such that it always has both the maximum and the minimum value.

(II) Definition of Partial Derivative Every point (x, y) in the xy -plane,

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) \equiv \lim_{\xi \rightarrow 0} \frac{1}{\xi} (f(x + \xi, y) - f(x, y)) \quad (4.57)$$

$$\frac{\partial w}{\partial y} = \frac{\partial}{\partial y} f(x, y) \equiv \lim_{\eta \rightarrow 0} \frac{1}{\eta} (f(x, y + \eta) - f(x, y)) \quad (4.58)$$

are referred to as **partial differential functions**) with respect to x and y , respectively. If both partial differential functions $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist for every point in the region D , then $f(x, y)$ is **partial differentiable** in the region D .

(III) Definition and Fundamental Theorem of Total Differentiability: For any points (x, y) and $(x + \xi, y + \eta)$ in the region D on the xy -plane, if we take $\rho \equiv \sqrt{\xi^2 + \eta^2}$, there exists constants $A(x, y)$ and $B(x, y)$ for each point (x, y) , that do not depend on ξ and η . If it is valid that

$$\lim_{\rho \rightarrow 0} \varepsilon(x, y, x + \xi, y + \eta) = 0, \quad (4.59)$$

where

$$\varepsilon(x, y, x + \xi, y + \eta) \equiv \frac{1}{\rho} \left((f(x + \xi, y + \eta) - f(x, y)) - (\xi A(x, y) + \eta B(x, y)) \right), \quad (4.60)$$

then $f(x, y)$ is **totally differentiable** in the region D . By replacing ξ and η by dx and dy , a **total derivative** $df(x, y)$ of the function $f(x, y)$ can be introduced by

$$dw(x, y) \equiv A(x, y)dx + \eta B(x, y)dy. \quad (4.61)$$

In the calculus, it is known that if $f(x, y)$ is total differentiable in the region D , then the following two statements are valid:

(i) $f(x, y)$ is continuous in the region D .

(ii) $f(x, y)$ is partial differentiable with respect to both x and y in the region D . In Eq.(4.60), $A(x, y)$ and $B(x, y)$ are given as $A(x, y) = \frac{\partial}{\partial x} w(x, y)$, $B(x, y) = \frac{\partial}{\partial y} w(x, y)$.

as one of fundamental theorem for total differentiability. This fundamental theorem means that if $f(x, y)$ is total differentiable in the region D , then it is valid that

$$\lim_{\rho \rightarrow 0} \frac{1}{\rho} \left((f(x + \xi, y + \eta) - f(x, y)) - \left(\xi \frac{\partial}{\partial x} f(x, y) + \eta \frac{\partial}{\partial y} f(x, y) \right) \right) = 0 \quad (\forall (x, y) \in D). \quad (4.62)$$

Eqs.(4.60)-(4.62) can be formally denoted by

$$df(x, y) = \left(\frac{\partial}{\partial x} f(x, y) \right) dx + \left(\frac{\partial}{\partial y} f(x, y) \right) dy \quad (\forall (x, y) \in D). \quad (4.63)$$

(IV) Total Differentiability Theorem for C^1 -Function If $f(x, y)$ is partial differentiable and both $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are continuous in the region D , then $f(x, y)$ is a C^1 -function. If $w(x, y)$ is a C^1 -function in the region D , then $w(x, y)$ is a total differentiable in the region D , such that $w(x, y)$ is continuous and Eq.(4.62) holds for every point (x, y) in the region D .

(V) Formulas of Derivative for C^1 -Function: In the region D on the xy -plane, let us suppose that two real-valued functions $u(x, y)$, $v(x, y)$ are C^1 -functions. Both $u(x, y) \pm v(x, y)$ and $u(x, y)v(x, y)$ are also C^1 -function in the region D and it is valid that

$$\frac{\partial}{\partial x} (u(x, y) \pm v(x, y)) = \frac{\partial}{\partial x} u(x, y) \pm \frac{\partial}{\partial x} v(x, y), \quad (4.64)$$

$$\frac{\partial}{\partial y} (u(x, y) \pm v(x, y)) = \frac{\partial}{\partial y} u(x, y) \pm \frac{\partial}{\partial y} v(x, y), \quad (4.65)$$

$$\frac{\partial}{\partial x} (u(x, y)v(x, y)) = \left(\frac{\partial}{\partial x} u(x, y) \right) v(x, y) + u(x, y) \left(\frac{\partial}{\partial x} v(x, y) \right), \quad (4.66)$$

$$\frac{\partial}{\partial y} (u(x, y)v(x, y)) = \left(\frac{\partial}{\partial y} u(x, y) \right) v(x, y) + u(x, y) \left(\frac{\partial}{\partial y} v(x, y) \right). \quad (4.67)$$

(VI) Partial Differentiability of Composite Function Two mapping functions $u = \phi(x, y)$ and $v = \psi(x, y)$ from the region D on xy -plane to the region Δ on the uv -plane is C^1 functions in the region D , and $f(u, v)$ is a C^1 -function in the region Δ , the composite function $f(\phi(x, y), \psi(x, y))$ is a C^1 -function in the region D and the following equalities hold:

$$\frac{\partial}{\partial x} f(\phi(x, y), \psi(x, y)) = \left[\frac{\partial}{\partial u} f(u, v) \right]_{u=\phi(x, y), v=\psi(x, y)} \left(\frac{\partial}{\partial x} \phi(x, y) \right)$$

$$+ \left[\frac{\partial}{\partial v} f(u, v) \right]_{u=\phi(x,y), v=\psi(x,y)} \left(\frac{\partial}{\partial x} \psi(x, y) \right), \quad (4.68)$$

$$\begin{aligned} \frac{\partial}{\partial y} f(\phi(x, y), \psi(x, y)) &= \left[\frac{\partial}{\partial u} f(u, v) \right]_{u=\phi(x,y), v=\psi(x,y)} \left(\frac{\partial}{\partial y} \phi(x, y) \right) \\ &+ \left[\frac{\partial}{\partial v} f(u, v) \right]_{u=\phi(x,y), v=\psi(x,y)} \left(\frac{\partial}{\partial y} \psi(x, y) \right). \end{aligned} \quad (4.69)$$

(VI) Interchange of the Order of Integral and Derivative: Let us suppose that $w(x, y)$ is a C^1 -function in the region D and the bounded and closed region $\{(x, y) | a \leq x \leq b, c \leq y \leq d\}$ is included in the region D . It is valid that

$$\frac{d}{dy} \int_a^b f(x, y) dx = \int_a^b \left(\frac{\partial}{\partial y} f(x, y) \right) dx \quad (c \leq y \leq d), \quad (4.70)$$

$$\frac{d}{dx} \int_c^d f(x, y) dy = \int_c^d \left(\frac{\partial}{\partial x} f(x, y) \right) dy \quad (a \leq x \leq b). \quad (4.71)$$

(VII) Interchange of the Order of Partial Derivatives If the real-valued function $f(x, y)$ is a C^1 -function in the region D on the xy -plane, then it is valid that

$$\frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f(x, y) \right) = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} f(x, y) \right) \quad (\forall (x, y) \in D). \quad (4.72)$$

Let us consider the following differential equation with respect to $y = w(x)$:

$$P(x, w(x)) + Q(x, w(x)) \frac{d}{dx} w(x) = 0, \quad (4.73)$$

for two real-valued functions $P(x, y)$ and $Q(x, y)$ which are C^1 -functions in the region D on the xy -plane and satisfy the following relation:

$$\frac{\partial}{\partial y} P(x, y) = \frac{\partial}{\partial x} Q(x, y) \quad (\forall (x, y) \in D). \quad (4.74)$$

Eq.(4.73) can be rewritten as

$$P(x, y) dx + Q(x, y) dy = 0 \quad (\forall (x, y) \in D). \quad (4.75)$$

We introduce a function $f(x, y)$ whose total derivative $df(x, y) = \left(\frac{\partial}{\partial x} f(x, y) \right) dx + \left(\frac{\partial}{\partial y} f(x, y) \right) dy$ in Eq.(4.63) is given as

$$df(x, y) = P(x, y) dx + Q(x, y) dy, \quad (4.76)$$

for any point in the region D . From Eq.(4.75) and Eq.(4.76), the differential equation (4.73) can be reduced to

$$df(x, y) = 0, \quad (4.77)$$

such that

$$f(x, y) = \text{Constant}, \quad (4.78)$$

One of important points is to find the function $f(x, y)$ which satisfy Eq.(4.76). By comparing the right sides of Eqs.(4.63) and (4.76), we have

$$\begin{cases} \frac{\partial}{\partial x} f(x, y) = P(x, y) \\ \frac{\partial}{\partial y} f(x, y) = Q(x, y) \end{cases} \quad (\forall (x, y) \in D), \quad (4.79)$$

as a sufficient condition for Eq.(4.76). By differentiate the first and the second equation in Eq.(4.76) with respect to y and x , respectively, we have

$$\begin{cases} \frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} f(x, y) \right) = \frac{\partial}{\partial y} P(x, y) \\ \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f(x, y) \right) = \frac{\partial}{\partial x} Q(x, y) \end{cases} \quad (\forall (x, y) \in D). \quad (4.80)$$

The functions $P(x, y)$ and $Q(x, y)$ are C^1 -functions and Eq.(4.80) means that $\frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} f(x, y) \right)$ and $\frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f(x, y) \right)$ are

continuous functions and then we should have $\frac{\partial}{\partial y} \left(\frac{\partial}{\partial x} f(x, y) \right) = \frac{\partial}{\partial x} \left(\frac{\partial}{\partial y} f(x, y) \right)$ in the region D . Then Eq.(4.74) should be valid and is necessary condition for which $df(x, y)$ given in Eq.(4.76) is a total derivative.. Eqs.(4.73)-(4.75) are referred to as an **exact equation** or a **total differential equation**.

For a specified point (x_0, y_0) in the region D , we consider a closed bounded rectangular region $(x_0, x) \times (y_0, y)$ included by the region D on the xy -plane. By integrating the first equation in Eq.(4.79) with respect to x , the function $f(x, y)$ can be expressed in terms of the function P as follows:

$$\int_{x_0}^x \frac{\partial}{\partial x} f(x, y) dx = \int_{x_0}^x P(x, y) dx, \quad (4.81)$$

$$f(x, y) = f(x_0, y) + \int_{x_0}^x P(x, y) dx. \quad (4.82)$$

By substituting Eq.(4.82) to the second equation of Eq.(4.79) and integrating its both sides with respect to y , $f(x_0, y)$ can be derived as follows:

$$\frac{\partial}{\partial y} f(x_0, y) + \frac{\partial}{\partial y} \int_{x_0}^x P(x, y) dx = Q(x, y), \quad (4.83)$$

$$\int_{y_0}^y \left(\frac{\partial}{\partial y} f(x_0, y) \right) dy + \int_{y_0}^y \left(\frac{\partial}{\partial y} \int_{x_0}^x P(x, y) dx \right) dy = \int_{y_0}^y Q(x, y) dy, \quad (4.84)$$

$$f(x_0, y) = f(x_0, y_0) - \int_{y_0}^y \left(\frac{\partial}{\partial y} \int_{x_0}^x P(x, y) dx \right) dy + \int_{y_0}^y Q(x, y) dy. \quad (4.85)$$

By applying Eq.(4.85) to Eq.(4.82), $f(x, y)$ can be reduced to the following expression:

$$f(x, y) = f(x_0, y_0) + \int_{x_0}^x P(x, y) dx - \int_{y_0}^y \left(\frac{\partial}{\partial y} \int_{x_0}^x P(x, y) dx \right) dy + \int_{y_0}^y Q(x, y) dy, \quad (4.86)$$

and then the solution $y = w(x)$ is derived as an implicit function as follows:

$$\int_{x_0}^x P(x, y) dx + \int_{y_0}^y \left(Q(x, y) - \left(\frac{\partial}{\partial y} \int_{x_0}^x P(x, y) dx \right) \right) dy = \text{Constant}. \quad (4.87)$$

Next, we consider the case in which Eq.(4.74) does not hold while there exists a function $\lambda(x, y)$ which satisfy the following equality:

$$\frac{\partial}{\partial y} (\lambda(x, y) P(x, y)) = \frac{\partial}{\partial x} (\lambda(x, y) Q(x, y)), \quad (4.88)$$

in the region D . We remark that $\lambda(x, y) P(x, y) dx + \lambda(x, y) Q(x, y) dy$ is a total derivative, although $P(x, y) dx + Q(x, y) dy$ is not a total derivative, The function $\lambda(x, y)$ is referred to as an **integrating factor** of $\lambda(x, y) P(x, y) dx + \lambda(x, y) Q(x, y) dy$. Eq.(4.88) can be reduced to

$$\lambda(x, y) \frac{\partial}{\partial y} P(x, y) + P(x, y) \frac{\partial}{\partial y} \lambda(x, y) = \lambda(x, y) \frac{\partial}{\partial x} Q(x, y) + Q(x, y) \frac{\partial}{\partial x} \lambda(x, y), \quad (4.89)$$

$$P(x, y) \frac{\partial}{\partial y} \lambda(x, y) - Q(x, y) \frac{\partial}{\partial x} \lambda(x, y) = -\lambda(x, y) \left(\frac{\partial}{\partial y} P(x, y) - \frac{\partial}{\partial x} Q(x, y) \right). \quad (4.90)$$

Now we consider the differential equation (4.73) under there exists $\lambda(x, y)$ in which satisfy Eq.(4.88). Then the differential equation

$$\lambda(x, y) P(x, y) dx + \lambda(x, y) Q(x, y) dy = 0, \quad (4.91)$$

is an exact equation. One of the choices is that we restrict to the case in which $\lambda(x, y)$ does not depend on y and does depend on only x , such that $\lambda(x, y) = \lambda(x)$. In such a case, we have $\frac{\partial}{\partial y} \lambda(x, y) = 0$ and then Eq.(4.90) can be reduced to

$$-Q(x, y) \frac{d}{dx} \lambda(x) = -\lambda(x, y) \left(\frac{\partial}{\partial y} P(x, y) - \frac{\partial}{\partial x} Q(x, y) \right), \quad (4.92)$$

$$\frac{1}{\lambda(x)} \frac{d}{dx} \lambda(x) = \frac{1}{Q(x, y)} \left(\frac{\partial}{\partial y} P(x, y) - \frac{\partial}{\partial x} Q(x, y) \right). \quad (4.93)$$

$$\ln(\lambda(x)) = \int \frac{1}{Q(x,y)} \left(\frac{\partial}{\partial y} P(x,y) - \frac{\partial}{\partial x} Q(x,y) \right) dx + \text{Constant}, \quad (4.94)$$

$$\lambda(x) = \text{Constant} \times \exp \left(\int \frac{1}{Q(x,y)} \left(\frac{\partial}{\partial y} P(x,y) - \frac{\partial}{\partial x} Q(x,y) \right) dx \right). \quad (4.95)$$

Another choice is that we restrict to the case in which $\lambda(x,y)$ does not depend on x and does depend on only y , such that $\lambda(x,y) = \lambda(y)$. Also in this case, we find out the integrating factor $\lambda(y)$ in the similar arguments to the ones in Eq.(4.95) as follows:

$$\lambda(y) = \text{Constant} \times \exp \left(- \int \frac{1}{P(x,y)} \left(\frac{\partial}{\partial y} P(x,y) - \frac{\partial}{\partial x} Q(x,y) \right) dy \right). \quad (4.96)$$

5 Second Order Linear Ordinary Differential Equation

5.1 General Framework of Second Order Linear Ordinary Differential Equation

For two continuous functions $p(x)$ and $q(x)$ in the bounded and closed region $|x - x_0| \leq \alpha$

$$\frac{d^2}{dx^2} w(x) + p(x) \frac{d}{dx} w(x) + q(x) w(x) = 0 \quad (|x - x_0| \leq \alpha), \quad (5.1)$$

where

$$w(x_0) = y_0, \quad \left[\frac{d}{dx} w(x) \right]_{x=x_0} = y'_0. \quad (5.2)$$

By introducing a function $v(x)$ defined by

$$v(x) \equiv \frac{d}{dx} w(x), \quad (5.3)$$

Eq.(5.1) can be reduced to the following simultaneous first order linear differential equations:

$$\begin{cases} \frac{d}{dx} w(x) = v(x), \\ \frac{d}{dx} v(x) = -p(x)v(x) - q(x)w(x), \end{cases} \quad (|x - x_0| \leq \alpha). \quad (5.4)$$

First we show the functions $w(x)$ and $v(x)$ are always continuous functions in the closed region $|x - x_0| \leq \alpha$.

$$\begin{cases} w_0(x) = y_0, \\ v_0(x) = y'_0, \end{cases} \quad (|x - x_0| \leq \alpha). \quad (5.5)$$

Because $p(x)$ and $q(x)$ are one valued continuous and bounded function in the region $|x - x_0| \leq \alpha$, then

$$\begin{cases} w_1(x) = y_0 + \int_{x_0}^x v_0(\xi) d\xi, \\ v_1(x) = y'_0 - \int_{x_0}^x (p(\xi)v_0(\xi) - q(\xi)w_0(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha). \quad (5.6)$$

is always a continuous and bounded function. In the similar arguments, the following sequence of function

$$\begin{cases} w_2(x) = y_0 + \int_{x_0}^x v_1(\xi) d\xi, \\ v_2(x) = y'_0 - \int_{x_0}^x (p(\xi)v_1(\xi) + q(\xi)w_1(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha). \quad (5.7)$$

$\vdots$

$$\begin{cases} w_n(x) = y_0 + \int_{x_0}^x v_{n-1}(\xi) d\xi, \\ v_n(x) = y'_0 - \int_{x_0}^x (p(\xi)v_{n-1}(\xi) + q(\xi)w_{n-1}(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha). \quad (5.8)$$

are also always continuous functions in the bounded and closed region $|x - x_0| \leq \alpha$. If there exists the limit of n infinity, $\lim_{n \rightarrow +\infty} w_n(x)$ and $\lim_{n \rightarrow +\infty} v_n(x)$, we can conclude that Eq.(5.4) has $w(x) = \lim_{n \rightarrow +\infty} w_n(x)$ as one of continuous and bounded solutions.

Next step we consider the existence of solutions in Eq.(4.6). The sequence of function $\{w_n(x)\}$ can be rewritten as a function series as follows:

$$\begin{cases} w_n(x) = y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)), \\ v_n(x) = y_0 + \sum_{k=1}^n (v_k(x) - v_{k-1}(x)). \end{cases} \quad (|x - x_0| \leq \alpha). \quad (5.9)$$

If the function series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$ absolutely converges, the limit $\lim_{n \rightarrow +\infty} w_n(x)$ exists in the bounded and closed region $|x - x_0| \leq \alpha$. All we have to do for the proof in the existence of solutions in the ordinary differential equation (5.1) with the initial condition (5.2) is to show the absolutely convergence of the series

$\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$ for any real number x in the bounded and closed region $|x - x_0| \leq \alpha$. There exists the finite upper bound of $p(x)$ and $q(x)$ in the bounded closed interval $|x - x_0| \leq \alpha$ because $p(x)$ and $q(x)$ are continuous functions. By introducing M and m defined by

$$M \equiv \max_{|x - x_0| \leq \alpha} \{ |p(x)|, |q(x)|, 1 \}, \quad (5.10)$$

and

$$m \equiv \max \{ |y_0|, |y'_0| \}. \quad (5.11)$$

The upper bounds of $|w_n(x) - w_{n-1}(x)|$ and $|v_n(x) - v_{n-1}(x)|$ are estimated as follows:

$$|v_1(x) - v_0(x)| = \left| \int_{x_0}^x w_0(\xi) d\xi \right| \leq \left| \int_{x_0}^x |y_0| d\xi \right| \leq Mm \left| \int_{x_0}^x d\xi_0 \right| = Mm|x - x_0| \quad (5.12)$$

$$\begin{aligned} |w_1(x) - w_0(x)| &= \left| \int_{x_0}^x (p(\xi)v_0(\xi) + q(\xi)w_0(\xi)) d\xi \right| \leq \left| \int_{x_0}^x (|p(\xi)||y'_0| + |q(\xi)||y_0|) d\xi \right| \\ &\leq m \left| \int_{x_0}^x (|p(\xi)| + |q(\xi)|) d\xi_0 \right| \leq Mm \left| \int_{x_0}^x d\xi \right| = Mm|x - x_0| \end{aligned} \quad (5.13)$$

$$\begin{aligned} |v_2(x) - v_1(x)| &= \left| \int_{x_0}^x (w_1(\xi) - w_0(\xi)) d\xi \right| \leq \left| \int_{x_0}^x |w_1(\xi) - w_0(\xi)| d\xi \right| \\ &\leq M^2m \left| \int_{x_0}^x |\xi - x_0| d\xi_0 \right| = \frac{1}{2!} M^2m|x - x_0|^2, \end{aligned} \quad (5.14)$$

$$\begin{aligned} |w_2(x) - w_1(x)| &= \left| \int_{x_0}^x (p(\xi)(v_1(\xi) - v_0(\xi)) + q(\xi)(w_1(\xi) - w_0(\xi))) d\xi \right| \\ &\leq \left| \int_{x_0}^x (|p(\xi)||v_1(\xi) - v_0(\xi)| + |q(\xi)||w_1(\xi) - w_0(\xi)|) d\xi \right| \\ &\leq Mm \left| \int_{x_0}^x (|p(\xi)| + |q(\xi)|) |\xi - x_0| d\xi \right| \\ &\leq M^2m \left| \int_{x_0}^x |\xi - x_0| d\xi \right| = \frac{1}{2!} M^2m|x - x_0|^2, \end{aligned} \quad (5.15)$$

$$\begin{aligned} |v_3(x) - v_2(x)| &= \left| \int_{x_0}^x (w_2(\xi) - w_1(\xi)) d\xi \right| \leq \left| \int_{x_0}^x |w_2(\xi) - w_1(\xi)| d\xi \right| \\ &\leq M^3m \left| \int_{x_0}^x \left(\frac{1}{2!} |\xi - x_0|^2 \right) d\xi_0 \right| = \frac{1}{3!} M^3m|x - x_0|^3, \end{aligned} \quad (5.16)$$

$$\begin{aligned}
|w_3(x) - w_2(x)| &= \left| \int_{x_0}^x (p(\xi)(v_2(\xi) - v_1(\xi)) + q(\xi)(w_2(\xi) - w_1(\xi))) d\xi \right| \\
&\leq \left| \int_{x_0}^x (|p(\xi)||v_2(\xi) - v_1(\xi)| + |q(\xi)||w_2(\xi) - w_1(\xi)|) d\xi \right| \\
&\leq M^2 m \left| \int_{x_0}^x (|p(\xi)| + |q(\xi)|) \left(\frac{1}{2!} |\xi - x_0|^2 \right) d\xi \right| \\
&\leq M^3 m \left| \int_{x_0}^x \left(\frac{1}{2!} |\xi - x_0|^2 \right) d\xi \right| = \frac{1}{3!} M^3 m |x - x_0|^3
\end{aligned} \tag{5.17}$$

$$\begin{cases} |v_k(x) - v_{k-1}(x)| \leq \frac{1}{k!} M^k m |x - x_0|^k, \\ |w_k(x) - w_{k-1}(x)| \leq \frac{1}{k!} M^k m |x - x_0|^k, \end{cases} \quad (|x - x_0| \leq \alpha). \tag{5.18}$$

By means of Eq.(5.18), the absolutely convergence of the series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{n=1}^n (w_k(x) - w_{k-1}(x)) \right)$ can be obtained as follows:

$$\begin{aligned}
|w_n(x)| &\leq |y_0| + \sum_{k=1}^n |w_k(x) - w_{k-1}(x)| \\
&\leq |y_0| + m \sum_{k=1}^n \frac{1}{k!} M^k |x - x_0|^k \\
&\leq |y_0| + m \sum_{k=1}^n \frac{1}{k!} M^k \alpha^k \\
&\rightarrow |y_0| + m(\exp(M\alpha) - 1) \quad (n \rightarrow +\infty, |x - x_0| \leq \alpha).
\end{aligned} \tag{5.19}$$

In the final equality, we use the Taylor expansion of $\exp(x)$ in Eq.(4.15). Then it has been shown that the existence of the second order linear differential equation (5.1) with the initial condition in Eqs.(5.2). From theorem 3.13, the function sequence $\{w_n(x) | n = 0, 1, 2, \dots\}$ defined by the recursion formulas (5.5)-(5.8) under Eqs.(5.2)-(5.3) uniformly converges and theorem 3.14, we can see that the function sequence $\{w_n(x) | n = 0, 1, 2, \dots\}$ always converges to a continuous function $w(x)$ in the closed interval $|x - x_0| \leq \alpha$.

Finally, we show the uniqueness of the second order linear differential equation (5.1) with the initial condition (5.2). Let us suppose that Eq.(5.1) have both set of functions $(w(x), v(x)) = (\Phi(x), \Psi(x))$ and $(\tilde{w}(x), \tilde{v}(x)) = (\tilde{\Phi}(x), \tilde{\Psi}(x))$ as solutions. Because it has been shown that there exists solutions of the second order linear differential equation (5.1) with the initial condition (5.2) as continuous functions in the bounded and closed interval $[x_0 - \alpha, x_0 + \alpha]$, we can see that there exists the following maximum value defined by

$$L \equiv \max_{x \in [x_0 - \alpha, x_0 + \alpha]} (|\Phi(x) - \tilde{\Phi}(x)| + |\Psi(x) - \tilde{\Psi}(x)|). \tag{5.20}$$

We consider the difference between them as follows:

$$\begin{aligned}
|\Phi(x) - \tilde{\Phi}(x)| &= \left| \int_{x_0}^x (\Psi(\xi) - \tilde{\Psi}(\xi)) d\xi \right| \\
&\leq \left| \int_{x_0}^x |\Psi(\xi) - \tilde{\Psi}(\xi)| d\xi \right| \\
&\leq \left| \int_{x_0}^x (|\Psi(\xi) - \tilde{\Psi}(\xi)| + |\Psi(\xi) - \tilde{\Psi}(\xi)|) d\xi \right| \\
&\leq M \left| \int_{x_0}^x (|\Psi(\xi) - \tilde{\Psi}(\xi)| + |\Psi(\xi) - \tilde{\Psi}(\xi)|) d\xi \right| \\
|\Psi(x) - \tilde{\Psi}(x)| &= \left| \int_{x_0}^x (p(\xi)(\Psi(\xi) - \tilde{\Psi}(\xi)) + q(\xi)(\Phi(\xi) - \tilde{\Phi}(\xi))) d\xi \right|
\end{aligned} \tag{5.21}$$

$$\begin{aligned}
&\leq \left| \int_{x_0}^x \left(|p(\xi)| |\Psi(\xi) - \tilde{\Psi}(\xi)| + |q(\xi)| |\Phi(\xi) - \tilde{\Phi}(\xi)| \right) d\xi \right| \\
&\leq M \left| \int_{x_0}^x (|\Psi(\xi) - \tilde{\Psi}(\xi)| + |\Psi(\xi) - \tilde{\Psi}(\xi)|) d\xi \right|, \tag{5.22}
\end{aligned}$$

and then we obtain the following inequality:

$$\begin{aligned}
|\Phi(x) - \tilde{\Phi}(x)| + |\Psi(x) - \tilde{\Psi}(x)| &\leq 2M \left| \int_{x_0}^x (|\Psi(\xi) - \tilde{\Psi}(\xi_1)| + |\Psi(\xi) - \tilde{\Psi}(\xi_1)|) d\xi_1 \right| \\
&\leq 2^2 M^2 \left| \int_{x_0}^x \int_{x_0}^{\xi_1} (|\Psi(\xi_2) - \tilde{\Psi}(\xi_2)| + |\Psi(\xi_2) - \tilde{\Psi}(\xi_2)|) d\xi_2 d\xi_1 \right| \\
&\vdots \\
&\leq 2^n M^n \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \cdots \int_{x_0}^{\xi_{n-1}} (|\Psi(\xi_n) - \tilde{\Psi}(\xi_n)| + |\Psi(\xi_n) - \tilde{\Psi}(\xi_n)|) d\xi_n \cdots d\xi_2 d\xi_1 \right| \\
&\leq 2^n M^n L \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \cdots \int_{x_0}^{\xi_{n-1}} d\xi_n \cdots d\xi_2 d\xi_1 \right| \\
&\leq 2^n M^n L \times \frac{1}{n!} |x - x_0|^n \leq 2^n M^n L \times \frac{1}{n!} \alpha^n \quad (n \in \{1, 2, 3, \dots\}, |x - x_0| \leq \alpha). \tag{5.23}
\end{aligned}$$

Because we have

$$|\Phi(x) - \tilde{\Phi}(x)| + |\Psi(x) - \tilde{\Psi}(x)| \leq 2^n M^n L \times \frac{1}{n!} \alpha^n \rightarrow 0 \quad (n \rightarrow +\infty), \tag{5.24}$$

in the similar arguments to the one in Eq.(4.22), the uniqueness of solutions in the second order linear differential equation (5.1) with the initial condition in Eq.(5.2). has been established.

If C^1 -functions $f^{(1)}(x)$ and $f^{(2)}(x)$ in the interval I on the xy -plane satisfy

$$C_1 f^{(1)}(x) + C_2 f^{(2)}(x) = 0 \quad (\forall x \in I), \tag{5.25}$$

where $|c_1| + |c_2| > 0$ such that either c_1 or c_2 is nonzero, then $f^{(1)}(x)$ and $f^{(2)}(x)$ are linearly dependent on each other. If $f^{(1)}(x)$ and $f^{(2)}(x)$ are not linearly dependent, then $f^{(1)}(x)$ and $f^{(2)}(x)$ are linearly independent on each other. By differentiating both sides of Eq.(5.25) with respect to x , we have

$$C_1 \frac{d}{dx} f^{(1)}(x) + C_2 \frac{d}{dx} f^{(2)}(x) = 0 \quad (\forall x \in I). \tag{5.26}$$

For any real value x in the interval I , it is known that the necessary and sufficient condition for which the simultaneous linear equations (5.25) and (5.26) for c_1 and c_2 have a set (c_1, c_2) of trivial solutions with $c_1 = 0$ and $c_2 = 0$ as all the possible solutions and there exists no set (c_1, c_2) of non-trivial solutions with either $c_1 \neq 0$ or $c_2 \neq 0$ as the solutions can be reduced to the following equality:

$$\det \begin{pmatrix} f^{(1)}(x) & \frac{d}{dx} f^{(1)}(x) \\ f^{(2)}(x) & \frac{d}{dx} f^{(2)}(x) \end{pmatrix} = 0 \quad (\forall x \in I), \tag{5.27}$$

in the theory of linear algebra.

Let us consider that the second order linear ordinary differential equation (5.1) has two different solutions $w^{(1)}(x)$ and $w^{(2)}(x)$.

$$\frac{d^2}{dx^2} w^{(1)}(x) + p(x) \frac{d}{dx} w^{(1)}(x) + q(x) w^{(1)}(x) = 0 \quad (|x - x_0| \leq \alpha), \tag{5.28}$$

$$\frac{d^2}{dx^2} w^{(2)}(x) + p(x) \frac{d}{dx} w^{(2)}(x) + q(x) w^{(2)}(x) = 0 \quad (|x - x_0| \leq \alpha), \tag{5.29}$$

By multiplying both sides of Eqs.(5.28) and (5.29) by $w^{(1)}(x)$ and $w^{(2)}(x)$, respectively and by subtracting both sides of them, we have

$$\left(w^{(1)}(x) \left(\frac{d^2}{dx^2} w^{(2)}(x) \right) - \left(\frac{d^2}{dx^2} w^{(1)}(x) \right) w^{(2)}(x) \right) + p(x) \left(w^{(1)}(x) \left(\frac{d}{dx} w^{(2)}(x) \right) - \left(\frac{d}{dx} w^{(1)}(x) \right) w^{(2)}(x) \right) = 0 \tag{5.30}$$

$$\frac{d}{dx} \det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix} + p(x)e^{\lambda_1 x} = 0 \quad (5.31)$$

By solving Eq.(5.31) with respect to $\det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix}$, we have

$$\det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix} = \text{Constant} \times \exp \left(- \int_{x_0}^x p(\xi) d\xi \right) \quad (|x - x_0| \leq \alpha). \quad (5.32)$$

In Eq.(5.32), $\exp \left(- \int_{x_0}^x p(\xi) d\xi \right)$ is always nonzero for any real number x in $|x - x_0| \leq \alpha$ because $p(x)$ is a continuous function in the present section. We conclude the following two statements:

1. If $\det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix} = 0$ at a point $x = x_0$ in $|x - x_0| \leq \alpha$, then the constant factor in the right side of

Eq.(5.32) should be zero and the Wronskii determinant $\det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix}$ of the solutions $w^{(1)}(x)$ and

$w^{(2)}(x)$ in Eq.(5.1) is always zero at every point x in $|x - x_0| \leq \alpha$, such that $w^{(1)}(x)$ and $w^{(2)}(x)$ are linearly dependent on each other in $|x - x_0| \leq \alpha$.

2. If $\det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix} \neq 0$ at a point $x = x_0$ in $|x - x_0| \leq \alpha$, then the constant factor in the right side of

Eq.(5.32) should be nonzero and the Wronskii determinant $\det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} w^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} w^{(2)}(x) \end{pmatrix}$ of the solutions $w^{(1)}(x)$

and $w^{(2)}(x)$ in Eq.(5.1) is always nonzero at every point x in $|x - x_0| \leq \alpha$, such that $w^{(1)}(x)$ and $w^{(2)}(x)$ are linearly independent of each other in $|x - x_0| \leq \alpha$.

For the second order linear ordinary differential equation in Eq.(5.1), if two solutions $w^{(1)}(x)$ and $w^{(2)}(x)$ are linearly independent of each other in $|x - x_0| \leq \alpha$, then the solutions are referred to as **basic solutions** or a **fundamental system of solutions** of Eq.(5.1). Moreover, every solution $w(x)$ for Eq.(5.1) is expressed as

$$w(x) = c_1 w^{(1)}(x) + c_2 w^{(2)}(x) \quad (|x - x_0| \leq \alpha), \quad (5.33)$$

in terms of the fundamental system of solutions, $w^{(1)}(x)$ and $w^{(2)}(x)$. And by specifying both values of $w(x)$ and $\frac{d}{dx} w(x)$ at a point $x = x_0$ in $|x - x_0| \leq \alpha$ as shown in Eq.(5.2), the solution $w(x)$ in Eq.(5.33) are uniquely expressed because

$$\det \begin{pmatrix} w^{(1)}(x_0) & w^{(2)}(x_0) \\ \left[\frac{d}{dx} w^{(1)}(x) \right]_{x=x_0} & \left[\frac{d}{dx} w^{(2)}(x) \right]_{x=x_0} \end{pmatrix} \neq 0, \quad (5.34)$$

and the solutions c_1 and c_2 of the following simultaneous linear equations:

$$\begin{cases} c_1 w^{(1)}(x_0) + c_2 w^{(2)}(x_0) = y_0, \\ c_1 \left[\frac{d}{dx} w^{(1)}(x) \right]_{x=x_0} + c_2 \left[\frac{d}{dx} w^{(2)}(x) \right]_{x=x_0} = y'_0, \end{cases} \quad (5.35)$$

are uniquely determined.

If there exists the fundamental system of solution, $w^{(1)}(x)$ and $w^{(2)}(x)$, for Eq.(5.1),

$$\left\{ c_1 w^{(1)}(x) + c_2 w^{(2)}(x) \mid c_1 \in (-\infty, +\infty), c_2 \in (-\infty, +\infty) \right\} \quad (5.36)$$

is referred to as a **solution space** of the second order linear ordinary differential equation (5.1). In the solution space, it is valid that

$$(c_1 w^{(1)}(x) + c_2 w^{(2)}(x)) + (\tilde{c}_1 w^{(1)}(x) + \tilde{c}_2 w^{(2)}(x)) = (c_1 + \tilde{c}_1) w^{(1)}(x) + (c_2 + \tilde{c}_2) w^{(2)}(x). \quad (5.37)$$

It means that the above solution space for Eq.(5.1) is a linear space in which $w^{(1)}(x)$ and $w^{(2)}(x)$ are basis.

5.2 Second Order Homogeneous Constant Coefficient Linear Ordinary Differential Equation

For two coefficients A and B , we consider the following second order constant coefficient linear ordinary differential equation:

$$\frac{d^2}{dx^2}w(x) + A\frac{d}{dx}w(x) + Bw(x) = 0 \quad (|x - x_0| \leq \alpha), \quad (5.38)$$

with Eqs.(5.2) and (5.3). Now we introduce the following linear operator:

$$\left(\frac{d}{dx} - C\right)w(x) \equiv \left(\frac{d}{dx}w(x) - C\right)w(x), \quad (5.39)$$

for a constant C , and then we have

$$\begin{aligned} \left(\frac{d}{dx} - \lambda_1\right)\left(\frac{d}{dx} - \lambda_2\right)w(x) &= \left(\frac{d}{dx} - \lambda_1\right)\left(\frac{d}{dx}w(x) - \lambda_2w(x)\right) \\ &= \frac{d}{dx}\left(\frac{d}{dx}w(x) - \lambda_2w(x)\right) - \lambda_1\left(\frac{d}{dx}w(x) - \lambda_2w(x)\right) \\ &= \frac{d^2}{dx^2}w(x) - (\lambda_1 + \lambda_2)w(x) + \lambda_1\lambda_2w(x), \end{aligned} \quad (5.40)$$

for two constants λ_1 and λ_2 . By imposing the following relationships:

$$\lambda_1 + \lambda_2 = -A, \quad \lambda_1\lambda_2 = B, \quad (5.41)$$

Eq.(5.38) can be reduced to

$$\left(\frac{d}{dx} - \lambda_1\right)\left(\frac{d}{dx} - \lambda_2\right)w(x) = 0, \quad (5.42)$$

where λ_1 and λ_2 are solutions of the quadratic equation:

$$\lambda^2 + A\lambda + B = 0. \quad (5.43)$$

From the definition (5.39), the linear operators $\left(\frac{d}{dx} - \lambda_1\right)$ and $\left(\frac{d}{dx} - \lambda_2\right)$ are commutable of each other, and then Eq.(5.42) can be rewritten as follows:

$$\left(\frac{d}{dx} - \lambda_2\right)\left(\frac{d}{dx} - \lambda_1\right)w(x) = 0, \quad (5.44)$$

Eq.(5.43) is referred to as a **characteristic equation** for the second order constant coefficient linear ordinary differential equation (5.38). It is obvious that the solution $w^{(1)}(x) = e^{\lambda_1 x}$ of the first order ordinary differential equation:

$$\left(\frac{d}{dx} - \lambda_1\right)w^{(1)}(x) = 0, \quad (5.45)$$

is also the one in Eq.(5.44), such that Eq.(5.38). We introduce a function $v(x)$ defined by

$$v(x) \equiv \left(\frac{d}{dx} - \lambda_1\right)\tilde{w}(x), \quad (5.46)$$

for any function $\tilde{w}(x)$. From Eq.(5.44), the solution of

$$\left(\frac{d}{dx} - \lambda_2\right)v(x) = 0, \quad (5.47)$$

is also the one of (5.42) and is obtained by $v(x) = e^{\lambda_2 x}$. By substituting $v(x) = e^{\lambda_2 x}$ to Eq.(5.46), we have the following

$$\left(\frac{d}{dx} - \lambda_1\right)\tilde{w}(x) = e^{\lambda_2 x}, \quad (5.48)$$

By operating $\left(\frac{d}{dx} - \lambda_1\right)$ with the both side of Eq.(5.48), it can be confirmed that $\tilde{w}(x)$ is also the solution of Eq.(5.42). Because $u(x) = e^{\lambda_2 x}$ is the solution of the first order homogeneous linear equation:

$$\left(\frac{d}{dx} - \lambda_1\right)u(x) = 0, \quad (5.49)$$

we obtain the solution of Eq.(5.48) by setting

$$\tilde{w}(x) = v(x)u(x) = v(x)e^{\lambda_1 x}, \quad (5.50)$$

and by applying the method of variation of constants in Eqs.(4.49)-(4.56) to the inhomogeneous first order linear differential equation (5.48), the solution $\tilde{w}(x)$ is derived as

$$\begin{aligned} v(x) &= \int e^{\lambda_2 x} e^{-\lambda_1 x} dx + \text{Constant} \\ &= \begin{cases} (\lambda_2 - \lambda_1)^{-1} e^{(\lambda_2 - \lambda_1)x} + \text{Constant} & (\lambda_1 \neq \lambda_2), \\ x + \text{Constant} & (\lambda_1 = \lambda_2), \end{cases} \end{aligned} \quad (5.51)$$

$$\tilde{w}(x) = v(x)e^{\lambda_1 x} = \begin{cases} (\lambda_2 - \lambda_1)^{-1} \times e^{\lambda_2 x} + \text{Constant} \times e^{\lambda_1 x} & (\lambda_1 \neq \lambda_2), \\ x e^{\lambda_1 x} + \text{Constant} \times e^{\lambda_1 x} & (\lambda_1 = \lambda_2). \end{cases} \quad (5.52)$$

For $\lambda_1 \neq \lambda_2$, $w^{(1)}(x) = e^{\lambda_1 x}$ and $w^{(2)}(x) = e^{\lambda_2 x}$ are independent of each other and are the fundamental system of solutions for Eq.(5.38), because of $\det \begin{pmatrix} e^{\lambda_1 x} & \frac{d}{dx} e^{\lambda_1 x} \\ e^{\lambda_2 x} & \frac{d}{dx} e^{\lambda_2 x} \end{pmatrix} = (\lambda_2 - \lambda_1)e^{(\lambda_1 + \lambda_2)x} \neq 0$. For $\lambda_1 = \lambda_2$, $w^{(1)}(x) = e^{\lambda_1 x}$ and $w^{(2)}(x) = x e^{\lambda_1 x}$ are independent of each other and are the fundamental system of solutions for Eq.(5.38), because of $\det \begin{pmatrix} e^{\lambda_1 x} & \frac{d}{dx} e^{\lambda_1 x} \\ x e^{\lambda_1 x} & \frac{d}{dx} (x e^{\lambda_1 x}) \end{pmatrix} = e^{2\lambda_1 x} \neq 0$ for $\lambda_1 = \lambda_2$. Every solution for the second order constant coefficient linear ordinary differential equation (5.38) is given by

$$w(x) = \begin{cases} c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} & (\lambda_1 \neq \lambda_2), \\ c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x} & (\lambda_1 = \lambda_2), \end{cases} \quad (5.53)$$

where λ_1 and λ_2 are the solutions of the characteristic equation (5.43), and c_1 and c_2 are determined so as to satisfy Eqs.(5.2) and (5.3).

For the case that the solutions λ_1 and λ_2 of the characteristic equation (5.43) degenerate, such that $\lambda_1 = \lambda_2$, one of the solutions of the second order homogeneous constant coefficient ordinary differential equation (5.38) has been given as $e^{\lambda_1 x}$. Although we have to find out another solution of Eq.(5.38) that is linearly independent of $e^{\lambda_1 x}$. Another solution in such a case has been obtained as the second equation for $\lambda_1 = \lambda_2$ of Eq.(5.53). Now we introduce the other method to derive the other solution of Eq.(5.38), although the result is the same as the second equation of Eq.(5.53). First we consider a function $e^{\lambda x}$ for any real number $\lambda \in (-\infty, +\infty)$. We can confirm that $\left[\frac{d}{d\lambda} e^{\lambda x}\right]_{\lambda=\lambda_1} = e^{\lambda_1 x} + x e^{\lambda_1 x}$ is a solution of Eq.(5.38) by substituting it to the present differential equation. This scheme is referred to as a **Frobenius method**. And it has already been shown that $e^{\lambda_1 x}$ and $x e^{\lambda_1 x}$ are linearly independent of each other in the above statements.

For any real numbers x and y in the open interval $(-\infty, +\infty)$, we define

$$e^{x+iy} \equiv e^x (\cos(y) + i \sin(y)). \quad (5.54)$$

If the solutions λ_1 and λ_2 of Eq.(5.43) are complex numbers, they should be complex conjugate of each other and can be expressed as

$$\begin{cases} \lambda_1 = \kappa + i\omega, \\ \lambda_2 = \bar{\lambda}_1 = \kappa - i\omega, \end{cases} \quad (5.55)$$

where κ and ω are the real and the imaginary part of λ_1 , respectively. The first equality of Eq.(5.53) can be reduced as follows:

$$w(x) = (c_1 + c_2) e^{\kappa x} \cos(\omega x) + i(c_1 - c_2) e^{\kappa x} \sin(\omega x) \quad (\lambda_1 \neq \lambda_2 \text{ such that } \omega \neq 0) \quad (5.56)$$

For $\omega \neq 0$, $w^{(1)}(x) = e^{\kappa x} \cos(\omega x)$ and $w^{(2)}(x) = e^{\kappa x} \sin(\omega x)$ are independent of each other, because of

$$\begin{aligned} \det \begin{pmatrix} e^{\kappa x} \cos(\omega x) & \frac{d}{dx} e^{\kappa x} \cos(\omega x) \\ e^{\kappa x} \sin(\omega x) & \frac{d}{dx} e^{\kappa x} \sin(\omega x) \end{pmatrix} &= \det \begin{pmatrix} e^{\kappa x} \cos(\omega x) & \kappa e^{\kappa x} \cos(\omega x) - \omega e^{\kappa x} \sin(\omega x) \\ e^{\kappa x} \sin(\omega x) & \kappa e^{\kappa x} \sin(\omega x) + \omega e^{\kappa x} \cos(\omega x) \end{pmatrix} \\ &= e^{\kappa x} \cos(\omega x) (\kappa e^{\kappa x} \sin(\omega x) + \omega e^{\kappa x} \cos(\omega x)) \\ &\quad - e^{\kappa x} \sin(\omega x) (\kappa e^{\kappa x} \cos(\omega x) - \omega e^{\kappa x} \sin(\omega x)) \\ &= \omega e^{\kappa x} \neq 0. \end{aligned} \quad (5.57)$$

By substituting $e^{\kappa x} \cos(\omega x)$ and $e^{\kappa x} \sin(\omega x)$ to $w(x)$ in Eq.(5.38), it is confirmed that $w^{(1)}(x) = e^{\kappa x} \cos(\omega x)$ and

$w^{(2)}(x) = e^{\kappa x} \sin(\omega x)$ are the fundamental system of solutions for Eq.(5.38) with

$$\begin{cases} A = 2\kappa, \\ B = \kappa^2 + \omega^2. \end{cases} \quad (5.58)$$

5.3 Second Order Homogeneous Linear Ordinary Differential Equation around Analytic Point

In calculus, it is known that every C^∞ -class function $f(x)$ around a bounded and open interval $|x - x_0| < r$ ($r > 0$) can be expressed as the following N -th order polynomial of $x - x_0$:

$\forall N$ (N is any positive integer), $\exists \theta$ ($0 < \theta < 1$):

$$\begin{aligned} f(x) = f(x_0) + \sum_{n=1}^{N-1} \frac{1}{n!} \left[\frac{d^{n-1}}{d\xi^{n-1}} f(\xi) \right]_{\xi=a} (x - x_0)^n \\ + \frac{1}{N!} \left[\frac{d^N}{d\xi^N} f(\xi) \right]_{\xi=a+\theta(x-x_0)} (x - x_0)^N \quad (|x - x_0| < r), \end{aligned} \quad (5.59)$$

as the Taylor theorem. If we have

$$\lim_{N \rightarrow +\infty} \frac{1}{N!} \left[\frac{d^N}{d\xi^N} f(\xi) \right]_{\xi=x_0+\theta(x-x_0)} (x - x_0)^N = 0 \quad (|x - x_0| < r), \quad (5.60)$$

Eq.(5.59) can be reduced to the following power series:

$$\begin{aligned} f(x) &= \sum_{n=0}^{+\infty} \frac{1}{n!} \left[\frac{d^{n-1}}{d\xi^{n-1}} f(\xi) \right]_{\xi=x_0} (x - x_0)^n \\ &= f(x_0) + \lim_{N \rightarrow +\infty} \sum_{n=1}^N \frac{1}{n!} \left[\frac{d^{n-1}}{d\xi^{n-1}} f(\xi) \right]_{\xi=a} (x - x_0)^n \quad (|x - x_0| < r), \end{aligned} \quad (5.61)$$

and if there exists such a positive real number r , then $f(x)$ is analytic in the point $x = x_0$ and the radius of convergence for the power series of the function $f(x)$ around the point $x = x_0$ should be larger than or equal to r . In the case, we have also the following equalities for termwise derivatives:

$$\begin{cases} \frac{d}{dx} f(x) = \sum_{n=0}^{+\infty} \frac{d}{dx} \left(\frac{1}{n!} \left[\frac{d^{n-1}}{d\xi^{n-1}} f(\xi) \right]_{\xi=a} (x - x_0)^n \right), \\ \frac{d^2}{dx^2} f(x) = \sum_{n=0}^{+\infty} \frac{d^2}{dx^2} \left(\frac{1}{n!} \left[\frac{d^{n-1}}{d\xi^{n-1}} f(\xi) \right]_{\xi=a} (x - x_0)^n \right), \end{cases} \quad (|x - x_0| < r). \quad (5.62)$$

Moreover we have

$$\begin{aligned} \det \begin{pmatrix} 1 & \frac{d}{dx} 1 & \frac{d^2}{dx^2} 1 & \frac{d^3}{dx^3} 1 & \cdots & \frac{d^N}{dx^N} 1 \\ x - x_0 & \frac{d}{dx} (x - x_0) & \frac{d^2}{dx^2} (x - x_0) & \frac{d^3}{dx^3} (x - x_0) & \cdots & \frac{d^N}{dx^N} (x - x_0) \\ (x - x_0)^2 & \frac{d}{dx} (x - x_0)^2 & \frac{d^2}{dx^2} (x - x_0)^2 & \frac{d^3}{dx^3} (x - x_0)^2 & \cdots & \frac{d^N}{dx^N} (x - x_0)^2 \\ (x - x_0)^3 & \frac{d}{dx} (x - x_0)^3 & \frac{d^2}{dx^2} (x - x_0)^3 & \frac{d^3}{dx^3} (x - x_0)^3 & \cdots & \frac{d^N}{dx^N} (x - x_0)^3 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (x - x_0)^N & \frac{d}{dx} (x - x_0)^N & \frac{d^2}{dx^2} (x - x_0)^N & \frac{d^3}{dx^3} (x - x_0)^N & \cdots & \frac{d^N}{dx^N} (x - x_0)^N \end{pmatrix} \\ = \det \begin{pmatrix} 1 & 0 & 0 & 0 & \cdots & 0 \\ x - x_0 & 1! & 0 & 0 & \cdots & 0 \\ (x - x_0)^2 & 2(x - x_0) & 2! & 0 & \cdots & 0 \\ (x - x_0)^3 & 3(x - x_0)^2 & 3 \cdot 2 \cdot (x - x_0) & 3! & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ (x - x_0)^N & N(x - x_0)^{N-1} & N(N-1)(x - x_0)^{N-2} & N(N-1)(N-2)(x - x_0)^{N-3} & \cdots & N! \end{pmatrix} \\ = \prod_{n=0}^N n! \neq 0 \quad (-\infty < x < +\infty), \end{aligned} \quad (5.63)$$

and then the polynomial functions, $\{1, x - x_0, (x - x_0)^2, (x - x_0)^3, \dots, (x - x_0)^N\}$, are linearly independent of each other.

We consider the second order linear ordinary differential equation (5.1) in which both coefficient functions $p(x)$ and $q(x)$ are C^∞ -functions in any point of $|x - x_0| < r$ for a positive real number $r > \alpha$, and are analytic at $x = x_0$ and can be

expressed as the following power series a round $x = x_0^*$:

$$\begin{cases} p(x) = \sum_{n=0}^{+\infty} P_n(x-x_0)^n, \\ q(x) = \sum_{n=0}^{+\infty} Q_n(x-x_0)^n, \end{cases} \quad (|x-x_0| \leq \alpha), \quad (5.64)$$

where

$$\begin{cases} P_n \equiv \frac{1}{n!} \left[\frac{d^n}{d\xi^n} p(\xi) \right]_{\xi=x_0} \quad (n=0,1,2,\dots), \\ Q_n \equiv \frac{1}{n!} \left[\frac{d^n}{d\xi^n} q(\xi) \right]_{\xi=x_0} \quad (n=0,1,2,\dots). \end{cases} \quad (5.65)$$

Moreover, $\frac{d^n}{dx^n} p(x)$ and $\frac{d^n}{d\xi^n} q(\xi)$ are continuous functions and then we have

$$\begin{cases} P_n(x-x_0)^n \leq P_n \alpha^n \leq \frac{1}{n!} \alpha^n \max_{k \in \{0,1,2,\dots\}} \left[\frac{d^k}{d\xi^k} p(\xi) \right]_{\xi=x_0} \leq \left(\prod_{k < \alpha} \frac{1}{k} \alpha \right) \left(\max_{k \in \{0,1,2,\dots\}} \left[\frac{d^k}{d\xi^k} p(\xi) \right]_{\xi=x_0} \right), \\ Q_n(x-x_0)^n \leq Q_n \alpha^n \leq \frac{1}{n!} \alpha^n \max_{k \in \{0,1,2,\dots\}} \left[\frac{d^k}{d\xi^k} q(\xi) \right]_{\xi=x_0} \leq \left(\prod_{k < \alpha} \frac{1}{k} \alpha \right) \left(\max_{k \in \{0,1,2,\dots\}} \left[\frac{d^k}{d\xi^k} q(\xi) \right]_{\xi=x_0} \right), \end{cases} \quad (n=0,1,2,\dots), \quad (5.66)$$

in the bounded and closed interval $|x-x_0| \leq \alpha$ for any real number $\alpha \in (0, r)$ and any non-negative integer n .[†] From theorem 3.14, the function series in the right sides of Eq.(5.64) always uniformly converge in the bounded and closed interval $|x-x_0| \leq \alpha$ for any positive real number $\alpha \in (0, r)$. Now we restrict the solution $w(x)$ of Eq.(5.1) to a $C^{+\infty}$ -class function and is assume to be the following power series functional form:

$$w(x) = \sum_{n=0}^{+\infty} C_n(x-x_0)^n, \quad (|x-x_0| \leq \alpha), \quad (5.67)$$

where

$$C_n \equiv \frac{1}{n!} \left[\frac{d^{n-1}}{d\xi^{n-1}} p(\xi) \right]_{\xi=x_0} \quad (n=0,1,2,\dots). \quad (5.68)$$

The power series $\sum_{n=0}^{+\infty} C_n(x-x_0)^n$ always uniformly converges and is termwise differentiable in the bounded and closed interval $|x-x_0| \leq \alpha$ for any positive real number $\alpha \in (0, r)$. By using Eq.(5.64) and Eq.(5.67), Eq.(5.1) can be reduced as follows:

$$\begin{aligned} \frac{d^2}{dx^2} \left(\sum_{n=0}^{+\infty} C_n(x-x_0)^n \right) &+ \left(\sum_{m=0}^{+\infty} P_m(x-x_0)^m \right) \frac{d}{dx} \left(\sum_{n=0}^{+\infty} C_n(x-x_0)^n \right) \\ &+ \left(\sum_{m=0}^{+\infty} Q_m(x-x_0)^m \right) \left(\sum_{n=0}^{+\infty} C_n(x-x_0)^n \right) = 0, \end{aligned} \quad (5.69)$$

$$\begin{aligned} \left(\sum_{n=2}^{+\infty} n(n-1)C_n(x-x_0)^{n-2} \right) &+ \left(\sum_{m=0}^{+\infty} P_m(x-x_0)^m \right) \left(\sum_{n=1}^{+\infty} nC_n(x-x_0)^{n-1} \right) \\ &+ \left(\sum_{m=0}^{+\infty} Q_m(x-x_0)^m \right) \left(\sum_{n=0}^{+\infty} C_n(x-x_0)^n \right) = 0, \end{aligned} \quad (5.70)$$

$$\left(\sum_{n=0}^{+\infty} (n+2)(n+1)C_{n+2}(x-x_0)^n \right) + \left(\sum_{m=0}^{+\infty} P_m(x-x_0)^m \right) \left(\sum_{n=0}^{+\infty} (n+1)C_{n+1}(x-x_0)^n \right)$$

* The positive number $r(> \alpha)$ should be smaller than or equal to both convergence radiuses in the power series of $p(x)$ and $q(x)$ around $x = x_0$.

[†] $\frac{1}{n!} \alpha^n = \left(\prod_{k < \alpha} \frac{1}{k} \alpha \right) \left(\prod_{\alpha \leq k < n} \frac{1}{k} \alpha \right) \leq \left(\prod_{k < \alpha} \frac{1}{k} \alpha \right)$

$$+ \left(\sum_{m=0}^{+\infty} Q_m (x-x_0)^m \right) \left(\sum_{n=0}^{+\infty} C_n (x-x_0)^n \right) = 0, \quad (5.71)$$

$$\begin{aligned} & \left(\sum_{n=0}^{+\infty} (n+2)(n+1)C_{n+2}(x-x_0)^n \right) + \sum_{n=0}^{+\infty} \sum_{m=0}^{+\infty} (n+1)P_m C_{n+1}(x-x_0)^{m+n} \\ & + \sum_{n=0}^{+\infty} \sum_{m=0}^{+\infty} Q_m C_n (x-x_0)^{m+n} = 0, \end{aligned} \quad (5.72)$$

$$\begin{aligned} & \left(\sum_{n=0}^N (n+2)(n+1)C_{n+2}(x-x_0)^n \right) + \sum_{n=0}^{+\infty} \sum_{k=n}^{+\infty} (n+1)P_{k-n} C_{n+1}(x-x_0)^k \\ & + \sum_{n=0}^{+\infty} \sum_{k=n}^{+\infty} Q_{k-n} C_n (x-x_0)^k = 0, \end{aligned} \quad (5.73)$$

$$\begin{aligned} & \left(\sum_{k=0}^{+\infty} (k+2)(k+1)C_{k+2}(x-x_0)^k \right) + \sum_{k=0}^{+\infty} \sum_{n=0}^k (n+1)P_{k-n} C_{n+1}(x-x_0)^k \\ & + \sum_{k=0}^{+\infty} \sum_{n=0}^k Q_{k-n} C_n (x-x_0)^k = 0, \end{aligned} \quad (5.74)$$

$$\sum_{k=0}^{+\infty} \left((k+2)(k+1)C_{k+2} + \sum_{n=0}^k (n+1)P_{k-n} C_{n+1} + \sum_{n=0}^k Q_{k-n} C_n \right) (x-x_0)^k = 0 \quad (|x-x_0| \leq \alpha). \quad (5.75)$$

From the sufficient condition for Eq.(5.75), we have

$$(k+2)(k+1)C_{k+2} + \sum_{n=0}^k (n+1)P_{k-n} C_{n+1} + \sum_{n=0}^k Q_{k-n} C_n = 0 \quad (k=0, 1, 2, \dots). \quad (5.76)$$

such that

$$C_{k+2} = -\frac{1}{(k+1)(k+2)} \left(\sum_{n=0}^k (n+1)P_{k-n} C_{n+1} + \sum_{n=0}^k Q_{k-n} C_n \right) \quad (k=0, 1, 2, \dots). \quad (5.77)$$

By using the recursion formula (5.77) with $(C_0, C_1) = (1, 0)$ and $(C_0, C_1) = (0, 1)$, we can uniquely generate two sequences

$$\left\{ C_k^{(1)} \middle| C_0^{(1)} = 1, C_1^{(1)} = 0, C_{k+2}^{(1)} = -\frac{1}{(k+1)(k+2)} \left(\sum_{n=0}^k (n+1)P_{k-n} C_{n+1}^{(1)} + \sum_{n=0}^k Q_{k-n} C_n^{(1)} \right), k=0, 1, 2, \dots \right\}, \quad (5.78)$$

$$\left\{ C_k^{(2)} \middle| C_0^{(2)} = 0, C_1^{(2)} = 1, C_{k+2}^{(2)} = -\frac{1}{(k+1)(k+2)} \left(\sum_{n=0}^k (n+1)P_{k-n} C_{n+1}^{(2)} + \sum_{n=0}^k Q_{k-n} C_n^{(2)} \right), k=0, 1, 2, \dots \right\}, \quad (5.79)$$

respectively, and then the fundamental system of the solutions for the second order homogeneous linear ordinary differential equation around an analytic point in Eq.(5.1) can be given by

$$\begin{cases} w^{(1)}(x) = \sum_{n=0}^{+\infty} C_n^{(1)} (x-x_0)^n, \\ w^{(2)}(x) = \sum_{n=0}^{+\infty} C_n^{(2)} (x-x_0)^n. \end{cases} \quad (|x-x_0| \leq \alpha). \quad (5.80)$$

By using Eq.(5.66), the coefficients P_n and Q_n of the power series for $p(x)$ and $q(x)$ in Eq.(5.64) with Eq.(5.65) can be estimated as follows:

$$\begin{cases} P_n \mu^n \leq M, \\ Q_n \mu^n \leq M, \end{cases} \quad (n=0, 1, 2, \dots) \quad (5.81)$$

where

$$\begin{aligned}\mu &\equiv \min\{\alpha, 1\} \\ M &\equiv \max\left\{\left(\prod_{k<\alpha} \frac{1}{k}\alpha\right)\left(\max_{k\in\{0,1,2,\dots\}}\left[\frac{d^k}{d\xi^k}P(\xi)\right]_{\xi=x_0}\right), \left(\prod_{k<\alpha} \frac{1}{k}\alpha\right)\left(\max_{k\in\{0,1,2,\dots\}}\left[\frac{d^k}{d\xi^k}P(\xi)\right]_{\xi=x_0}\right), 1\right\},\end{aligned}\quad (5.82)$$

for any positive number α in the open $(0, r)$.

If it is valid that

$$|C_0| < 1, |C_1| < \frac{M}{\mu}, |C_2| < \frac{M^2}{\mu^2}, \dots, |C_{k+1}| < \frac{M^{k+1}}{\mu^{k+1}}, \quad (5.83)$$

the recursion formula (5.77) can be reduced to the following inequality:

$$\begin{aligned}|C_{k+2}^{(1)}| &= \frac{1}{|k+1||k+2|} \left| \sum_{n=0}^k (n+1)P_{k-n}C_{n+1} + \sum_{n=0}^k Q_{k-n}C_n \right| \\ &\leq \frac{1}{|k+1||k+2|} \left(\sum_{n=0}^k (n+1)|P_{k-n}||C_{n+1}| + \sum_{n=0}^k |Q_{k-n}||C_n| \right) \\ &\leq \frac{1}{|k+1||k+2|} \left(\sum_{n=0}^k (n+1)\mu^{k-n}M|C_{n+1}| + \sum_{n=0}^k \mu^{k-n}M|C_n| \right) \\ &\leq \frac{1}{|k+1||k+2|} \left(\sum_{n=0}^k (n+1)\mu^{-k+n}M\frac{M^{n+1}}{\mu^{n+1}} + \sum_{n=0}^k \mu^{-k+n}M\frac{M^n}{\mu^n} \right) \\ &\leq \frac{1}{|k+1||k+2|} \mu^{-k-1} \left(\sum_{n=0}^k (n+1)M^{n+2} + \sum_{n=0}^k M^n \right) \\ &\leq \frac{1}{|k+1||k+2|} \mu^{-k-1} M^{n+2} \left(\sum_{n=0}^k (n+1) + \sum_{n=0}^k \mu \right) \\ &\leq \frac{1}{|k+1||k+2|} \alpha^{-k-1} M^{n+2} (k(k+1) + (k+1)(\mu+1)) \\ &\leq \frac{1}{|k+1||k+2|} \mu^{-k-1} M^{n+2} (k+1)(\mu+1+k) \\ &\leq \mu^{-k-2} M^{n+2}.\end{aligned}\quad (5.84)$$

From mathematical inductions, it is valid that

$$|C_k| < \mu^{-k} M^k \quad (k = 0, 1, 2, 3, \dots), \quad (5.85)$$

and then we have

$$|C_k|(x-x_0)^k < \left(\frac{M}{\mu}\right)^k \left(\frac{\alpha}{M}\right)^k \quad (k = 0, 1, 2, 3, \dots). \quad (5.86)$$

in the bounded and closed interval $|x-x_0| \leq \frac{\alpha}{M}$. Moreover the series $\sum_{n=0}^{+\infty} \left(\frac{M}{\mu}\right)^n \left(\frac{\alpha}{M}\right)^n$ converges as follows:

$$\sum_{n=0}^{+\infty} \left(\frac{M}{\mu}\right)^n \left(\frac{\alpha}{M}\right)^n = \frac{1}{1 - \frac{\alpha}{\mu}} \quad (5.87)$$

because of $\left|\frac{\alpha}{\mu}\right| < 1$. Fome theorem 3.14, we have confirmed that $\sum_{n=0}^{+\infty} C_n(x-x_0)^n$ uniformly converges at least in the

bounded and closed interval $|x-x_0| \leq \frac{\alpha}{M}$.

The recursion formula (5.76) can be reduced to the following inequality:

$$(k+2)(k+1)|C_{k+2}| \leq \sum_{n=0}^k (n+1)|P_{k-n}||C_{n+1}| + \sum_{n=0}^k |Q_{k-n}||C_n|$$

$$\begin{aligned}
&\leq \sum_{n=0}^k (n+1)\alpha^{k-n}M|C_{n+1}| + \sum_{n=0}^k \alpha^{k-n}M|C_n| \\
&\leq \sum_{n=0}^k (n+1)\alpha^{k-n}M|C_{n+1}| + \sum_{n=0}^k \alpha^{k-n}M|C_n| + M\alpha|C_{k+1}| \quad (k=0,1,2,\dots). \quad (5.88)
\end{aligned}$$

Now we introduce a sequence defined by the following recursion formula:

$$(k+2)(k+1)A_{k+2} = \sum_{n=0}^k (n+1)\alpha^{k-n}MA_{n+1} + \sum_{n=0}^k \alpha^{k-n}MA_n + M\alpha A_{k+1} \quad (k=0,1,2,\dots), \quad (5.89)$$

where $A_0 \equiv |C_0|$ and $A_1 \equiv |C_1|$. If it is valid that $|C_n| \leq A_n$ ($n=1,2,\dots,k+1$), we have

$$\sum_{n=0}^k \alpha^{k-n}M|C_n| + M\alpha|C_{k+1}| \leq \sum_{n=0}^k (n+1)\alpha^{k-n}MA_{n+1} + \sum_{n=0}^k \alpha^{k-n}MA_n + M\alpha A_{k+1}, \quad (5.90)$$

and then

$$|C_{k+2}| \leq A_{k+2}. \quad (5.91)$$

From mathematical inductions, we have

$$|C_n| \leq A_n \quad (n=0,1,2,\dots). \quad (5.92)$$

$$\begin{aligned}
\alpha(k+1)kA_{k+1} - k(k-1)A_k &= \sum_{n=0}^{k-1} (n+1)\alpha^{k-n}MA_{n+1} + \sum_{n=0}^{k-1} \alpha^{k-n}MA_n + M\alpha A_k \\
&\quad - \sum_{n=0}^{k-1} (n+1)\alpha^{k-1-n}MA_{n+1} - \sum_{n=0}^{k-1} \alpha^{k-1-n}MA_n - M\alpha A_k \\
&= (Mk\alpha + m\alpha^2)A_n \quad (5.93)
\end{aligned}$$

and then we obtain

$$\frac{A_{n+1}}{A_n} = \frac{k(k-1) + Mk\alpha + m\alpha^2}{\alpha(k+1)k} \rightarrow \frac{1}{\alpha} \quad (k \rightarrow +\infty). \quad (5.94)$$

From theorem 3.18 and theorem 3.19, we have

$$\left(\lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}} |C_m|^{1/m} \right)^{-1} \geq \left(\lim_{n \rightarrow +\infty} \sup_{m \in \{n, n+1, n+2, \dots\}} |A_m|^{1/m} \right)^{-1}, \quad (5.95)$$

and then the convergence radius of the power series $w(x) = \sum_{n=0}^{+\infty} C_n(x-x_0)^n$ is larger than or equal to any positive number $\alpha \in (0, r)$.

We can confirm that

$$\det \begin{pmatrix} w^{(1)}(a) & \left[\frac{d}{dx} w^{(1)}(x) \right]_{x=x_0} \\ w^{(2)}(a) & \left[\frac{d}{dx} w^{(2)}(x) \right]_{x=x_0} \end{pmatrix} = \det \begin{pmatrix} C_0^{(1)} & C_1^{(1)} \\ C_0^{(2)} & C_1^{(2)} \end{pmatrix} = \det \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = 1 \neq 0, \quad (5.96)$$

and $w^{(1)}(x)$ and $w^{(2)}(x)$ are independent of each other in $|x-x_0| \leq \alpha$. The exact solution $w(x)$ of Eq.(5.1) can be expressed by

$$w(x) = c_1 w^{(1)}(x) + c_2 w^{(2)}(x) \quad (|x-x_0| \leq \alpha), \quad (5.97)$$

where c_1 and c_2 are constants determined so as to satisfy Eqs.(5.2) and (5.3).

5.4 Second Order Inhomogeneous Linear Ordinary Differential Equation

For two continuous functions $p(x)$, $q(x)$ and $h(x)$ in the bounded and closed region $|x-x_0| \leq \alpha$, we consider the following second order inhomogeneous linear ordinary differential equation:

$$\frac{d^2}{dx^2} w(x) + p(x) \frac{d}{dx} w(x) + q(x) w(x) = h(x) \quad (|x-x_0| \leq \alpha), \quad (5.98)$$

with Eq.(5.2). Let us suppose that we find out a function $\psi(x)$ as a special solution of Eq.(5.98). For the general solution $\phi(x)$ of Eq.(5.98), it is shown that the difference between $\phi(x)$ and $\psi(x)$ is a general solution of the second order homogeneous linear ordinary differential equation (5.1) as follows:

$$\frac{d^2}{dx^2}(\phi(x) - \psi(x)) + p(x)\frac{d}{dx}(\phi(x) - \psi(x)) + q(x)(\phi(x) - \psi(x)) = h(x) - h(x) = 0 \quad (|x - x_0| \leq \alpha). \quad (5.99)$$

It means that the difference $\phi(x) - \psi(x)$ can be expressed in terms of the fundamental system of solutions $w^{(1)}(x)$ and $w^{(2)}(x)$ for the second order homogeneous linear ordinary differential equation (5.1) as follows:

$$\phi(x) - \psi(x) = c_1 w^{(1)}(x) + c_2 w^{(2)}(x) \quad (|x - x_0| \leq \alpha), \quad (5.100)$$

such that the general solution $\phi(x)$ of Eq.(5.98) is expressed as

$$\phi(x) = c_1 w^{(1)}(x) + c_2 w^{(2)}(x) + \psi(x) \quad (|x - x_0| \leq \alpha), \quad (5.101)$$

where c_1 and c_2 are constant and are determined so as to satisfy Eq.(5.2).

One of the methods to obtain such a special solution $\psi(x)$ in the second order inhomogeneous linear ordinary differential equation (5.98) is a **variation of constants**. In the variation of constants, the special solution is restricted to the following functional form:

$$\psi(x) = u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x) \quad (|x - x_0| \leq \alpha), \quad (5.102)$$

where $u^{(1)}(x)$ and $u^{(2)}(x)$ satisfy

$$\left(\frac{d}{dx}u^{(1)}(x)\right)w^{(1)}(x) + \left(\frac{d}{dx}u^{(2)}(x)\right)w^{(2)}(x) = 0 \quad (|x - x_0| \leq \alpha). \quad (5.103)$$

We remark that Eq.(5.102) corresponds to the one in which the constants c_1 and c_2 in the general solution of Eq.(5.1) are replaced by functions of x , $u^{(1)}(x)$ and $u^{(2)}(x)$. By substituting Eq.(5.102) to

$$\frac{d^2}{dx^2}\psi(x) + p(x)\frac{d}{dx}\psi(x) + q(x)\psi(x) = h(x) \quad (|x - x_0| \leq \alpha), \quad (5.104)$$

we have

$$\begin{aligned} \frac{d^2}{dx^2}\left(u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x)\right) &+ p(x)\frac{d}{dx}\left(u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x)\right) \\ &+ q(x)\left(u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x)\right) = h(x), \end{aligned} \quad (5.105)$$

$$\begin{aligned} &\frac{d}{dx}\left(u^{(1)}(x)\left(\frac{d}{dx}w^{(1)}(x)\right) + u^{(2)}(x)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) + \left(\frac{d}{dx}u^{(1)}(x)\right)w^{(1)}(x) + \left(\frac{d}{dx}u^{(2)}(x)\right)w^{(2)}(x) \\ &+ p(x)\left(u^{(1)}(x)\left(\frac{d}{dx}w^{(1)}(x)\right) + u^{(2)}(x)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) + \left(\frac{d}{dx}u^{(1)}(x)\right)w^{(1)}(x) + \left(\frac{d}{dx}u^{(2)}(x)\right)w^{(2)}(x) \\ &+ q(x)\left(u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x)\right) = h(x), \end{aligned} \quad (5.106)$$

$$\begin{aligned} &\frac{d}{dx}\left(u^{(1)}(x)\left(\frac{d}{dx}w^{(1)}(x)\right) + u^{(2)}(x)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) + p(x)\left(u^{(1)}(x)\left(\frac{d}{dx}w^{(1)}(x)\right) + u^{(2)}(x)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) \\ &+ q(x)\left(u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x)\right) = h(x), \end{aligned} \quad (5.107)$$

$$\begin{aligned} &\left(\left(\frac{d}{dx}u^{(1)}(x)\right)\left(\frac{d}{dx}w^{(1)}(x)\right) + \left(\frac{d}{dx}u^{(2)}(x)\right)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) + \left(u^{(1)}(x)\left(\frac{d^2}{dx^2}w^{(1)}(x)\right) + u^{(2)}(x)\left(\frac{d^2}{dx^2}w^{(2)}(x)\right)\right) \\ &+ p(x)\left(u^{(1)}(x)\left(\frac{d}{dx}w^{(1)}(x)\right) + u^{(2)}(x)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) + q(x)\left(u^{(1)}(x)w^{(1)}(x) + u^{(2)}(x)w^{(2)}(x)\right) = h(x), \end{aligned} \quad (5.108)$$

$$\begin{aligned} &\left(\left(\frac{d}{dx}u^{(1)}(x)\right)\left(\frac{d}{dx}w^{(1)}(x)\right) + \left(\frac{d}{dx}u^{(2)}(x)\right)\left(\frac{d}{dx}w^{(2)}(x)\right)\right) \\ &+ u^{(1)}(x)\left(\left(\frac{d^2}{dx^2}w^{(1)}(x)\right) + p(x)\left(\frac{d}{dx}w^{(1)}(x)\right) + q(x)w^{(1)}(x)\right) \\ &+ u^{(2)}(x)\left(\left(\frac{d^2}{dx^2}w^{(2)}(x)\right) + p(x)\left(\frac{d}{dx}w^{(2)}(x)\right) + q(x)w^{(2)}(x)\right) = h(x), \end{aligned}$$

$$+u^{(2)}(x)\left(\left(\frac{d^2}{dx^2}w^{(2)}(x)\right)+p(x)\left(\frac{d}{dx}w^{(2)}(x)\right)+q(x)w^{(2)}(x)\right)=h(x), \quad (5.109)$$

By applying Eq.(5.103) to Eq.(5.109), we obtain

$$\left(\frac{d}{dx}u^{(1)}(x)\right)\left(\frac{d}{dx}w^{(1)}(x)\right)+\left(\frac{d}{dx}u^{(2)}(x)\right)\left(\frac{d}{dx}w^{(2)}(x)\right)=h(x), \quad (5.110)$$

From Eqs.(5.103) and (5.110), the simultaneous deterministic equations for $u^{(1)}(x)$ and $u^{(2)}(x)$ are reduced to

$$\begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ \frac{d}{dx}w^{(1)}(x) & \frac{d}{dx}w^{(1)}(x) \end{pmatrix} \begin{pmatrix} \frac{d}{dx}u^{(1)}(x) \\ \frac{d}{dx}u^{(2)}(x) \end{pmatrix} = \begin{pmatrix} 0 \\ h(x) \end{pmatrix}. \quad (5.111)$$

By solving Eq.(5.111) with respect to $\frac{d}{dx}u^{(1)}(x)$ and $\frac{d}{dx}u^{(2)}(x)$ and by integrating them, $u^{(1)}(x)$ and $u^{(2)}(x)$ are obtained as follows:

$$\begin{pmatrix} \frac{d}{dx}u^{(1)}(x) \\ \frac{d}{dx}u^{(2)}(x) \end{pmatrix} = \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ \frac{d}{dx}w^{(1)}(x) & \frac{d}{dx}w^{(1)}(x) \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ h(x) \end{pmatrix}, \quad (5.112)$$

$$\begin{pmatrix} u^{(1)}(x) \\ u^{(2)}(x) \end{pmatrix} = \int_{x_0}^x \begin{pmatrix} w^{(1)}(\xi) & w^{(2)}(\xi) \\ \frac{d}{dx}w^{(1)}(\xi) & \frac{d}{dx}w^{(1)}(\xi) \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ h(\xi) \end{pmatrix} d\xi. \quad (5.113)$$

The special solution $\psi(x)$ is derived as follows:

$$\psi(x) = (w^{(1)}(x), w^{(2)}(x)) \begin{pmatrix} u^{(1)}(x) \\ u^{(2)}(x) \end{pmatrix} = (w^{(1)}(x), w^{(2)}(x)) \int_{x_0}^x \begin{pmatrix} w^{(1)}(\xi) & w^{(2)}(\xi) \\ \frac{d}{dx}w^{(1)}(\xi) & \frac{d}{dx}w^{(1)}(\xi) \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ h(\xi) \end{pmatrix} d\xi. \quad (5.114)$$

6 Simultaneous Ordinary Differential Equations

6.1 Fundamental Theorem of Simultaneous Ordinary Differential Equations

Let us consider the following simultaneous ordinary differential equations:

$$\begin{cases} \frac{d}{dx}w(x) = f_1(x, w(x), v(x)) \quad (|x - x_0| \leq \alpha), \\ \frac{d}{dx}v(x) = f_2(x, w(x), v(x)) \quad (|x - x_0| \leq \alpha). \end{cases} \quad (6.1)$$

Here we remark that both $f_1(x, y, z)$ and $f_2(x, y, z)$ are defined in a closed region

$\{(x, y, z) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta, |z - z_0| \leq \gamma\}$ on the xyz -plane and should be a one-valued and continuous function at every point (x, y, z) in the region and that the following initial conditions are imposed:

$$\begin{cases} w(x_0) = y_0, \\ v(x_0) = z_0. \end{cases} \quad (6.2)$$

There should exist the maximum values M_1 and M_2 of $|f_1(x, y, z)|$ and $|f_2(x, y, z)|$, respectively, in the region

$\{(x, y, z) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta, |z - z_0| \leq \gamma\}$, such that

$$|f_1(x, y, z)| \leq M_1, \quad |f_2(x, y, z)| \leq M_2 \quad (\forall (x, y, z) \in \{(x, y, z) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta, |z - z_0| \leq \gamma\}). \quad (6.3)$$

For any real number x in the closed region $|x - x_0| \leq \alpha$, we take an integration in the both side of Eq.(6.1) as follows:

$$\begin{cases} \int_{x_0}^x \frac{d}{dx}w(x)dx = \int_{x_0}^x f_1(x, w(x), v(x))dx, \\ \int_{x_0}^x \frac{d}{dx}v(x)dx = \int_{x_0}^x f_2(x, w(x), v(x))dx, \end{cases} \quad (6.4)$$

such that

$$\begin{cases} w(x) = y_0 + \int_{x_0}^x f_1(x, w(x), v(x))dx \quad (|x - x_0| \leq \alpha), \\ v(x) = z_0 + \int_{x_0}^x f_2(x, w(x), v(x))dx \quad (|x - x_0| \leq \alpha). \end{cases} \quad (6.5)$$

Let us consider the existence and the uniqueness of the solution of the first order ordinary differential equation (6.1) with the initial condition (6.2) under the following condition:

$$\forall (x, y_1, z_1) \in \{(x, y, z) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta, |z - z_0| \leq \gamma\},$$

$$\begin{aligned}
& \forall (x, y_2, z_2) \in \{(x, y, z) \mid |x - x_0| \leq \alpha, |y - y_0| \leq \beta, |z - z_0| \leq \gamma\}, \\
& \exists K > 0; |f_1(x, y_1, z_1) - f_1(x, y_2, z_2)| \leq \frac{K}{2} (|y_1 - y_2| + |z_1 - z_2|), \\
& |f_2(x, y_1, z_1) - f_2(x, y_2, z_2)| \leq \frac{K}{2} (|y_1 - y_2| + |z_1 - z_2|).
\end{aligned} \tag{6.6}$$

It is referred to as a **Lipschitz condition**. We introduce a sequence of functions, $\{(w_n(x), v_n(x)) \mid |x - x_0| \leq \alpha, n = 0, 1, 2, \dots\}$, which is defined by

$$\begin{cases} w_n(x) = y_0 + \int_{x_0}^x f_1(\xi, w_{n-1}(\xi), v_{n-1}(\xi)) d\xi, \\ v_n(x) = z_0 + \int_{x_0}^x f_2(\xi, w_{n-1}(\xi), v_{n-1}(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha), \tag{6.7}$$

where

$$\begin{cases} w_0(x) = y_0, \\ v_0(x) = z_0, \end{cases} \quad (|x - x_0| \leq \alpha). \tag{6.8}$$

First we show the function $w_n(x)$ is always a continuous function in the closed region $|x - x_0| \leq \alpha$ for every positive integer n . Because $f(x, y_0)$ is a one valued continuous and bounded function in the closed region $|x - x_0| \leq \alpha$ as show in Eq.(6.3), then

$$\begin{cases} w_1(x) = y_0 + \int_{x_0}^x f_1(\xi, w_0(\xi), v_0(\xi)) d\xi, \\ v_1(x) = z_0 + \int_{x_0}^x f_2(\xi, w_0(\xi), v_0(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha), \tag{6.9}$$

is always a continuous and bounded function. In the similar arguments, the following sequence of function

$$\begin{cases} w_2(x) = y_0 + \int_{x_0}^x f_1(\xi, w_1(\xi), v_1(\xi)) d\xi, \\ v_2(x) = z_0 + \int_{x_0}^x f_2(\xi, w_1(\xi), v_1(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha), \tag{6.10}$$

$$\begin{cases} w_3(x) = y_0 + \int_{x_0}^x f_1(\xi, w_2(\xi), v_2(\xi)) d\xi, \\ v_3(x) = z_0 + \int_{x_0}^x f_2(\xi, w_2(\xi), v_2(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha), \tag{6.11}$$

$\vdots$

$$\begin{cases} w_n(x) = y_0 + \int_{x_0}^x f_1(\xi, w_{n-1}(\xi), v_{n-1}(\xi)) d\xi, \\ v_n(x) = z_0 + \int_{x_0}^x f_2(\xi, w_{n-1}(\xi), v_{n-1}(\xi)) d\xi, \end{cases} \quad (|x - x_0| \leq \alpha), \tag{6.12}$$

are also always continuous functions in the closed region $|x - x_0| \leq \alpha$. If there exists the limit of n infinity, $\lim_{n \rightarrow +\infty} w_n(x)$ and $\lim_{n \rightarrow +\infty} v_n(x)$, we can conclude that Eq.(6.5) has $w(x) = \lim_{n \rightarrow +\infty} w_n(x)$ and $v(x) = \lim_{n \rightarrow +\infty} v_n(x)$ as one of continuous and bounded solutions.

Nest step we consider the existence of solutions in Eq.(6.5). The sequence of function $\{w_n(x)\}$ can be rewritten as a function series as follows:

$$\begin{aligned} w_n(x) &= y_0 + (w_1(x) - w_0(x)) + (w_2(x) - w_1(x)) + \dots + (w_n(x) - w_{n-1}(x)) \\ &= y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)). \end{aligned} \tag{6.13}$$

$$\begin{aligned} v_n(x) &= z_0 + (v_1(x) - v_0(x)) + (v_2(x) - v_1(x)) + \dots + (v_n(x) - v_{n-1}(x)) \\ &= z_0 + \sum_{k=1}^n (v_k(x) - v_{k-1}(x)). \end{aligned} \tag{6.14}$$

If the function series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$ absolutely converges, the limit $\lim_{n \rightarrow +\infty} w_n(x)$ exists in the

interval $[\alpha, \beta]$. All we have to do for the proof in the existence of solutions in the ordinary differential equation (6.1) with the initial condition (6.2) is to show the absolutely convergence of the series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (w_k(x) - w_{k-1}(x)) \right)$ and $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{k=1}^n (v_k(x) - v_{k-1}(x)) \right)$ for any real number x in the interval $[x_0 - \alpha, x_0 + \alpha]$. By using Eqs.(6.3) and (6.6), upper bounds of $|w_n(x) - w_{n-1}(x)|$ and $|v_n(x) - v_{n-1}(x)|$ in the above series of functions can be estimated as follows:

$$\begin{aligned}
|w_n(x) - w_{n-1}(x)| &= \left| \int_{x_0}^x f_1(\xi_{n-1}, w_{n-1}(\xi_{n-1}), v_{n-1}(\xi_{n-1})) - f_1(\xi_{n-1}, w_{n-2}(\xi_{n-1}), v_{n-1}(\xi_{n-1})) d\xi_{n-1} \right| \\
&\leq \left| \int_{x_0}^x |f_1(\xi_{n-1}, w_{n-1}(\xi_{n-1}), v_{n-1}(\xi_{n-1})) - f_1(\xi_{n-1}, w_{n-2}(\xi_{n-1}), v_{n-1}(\xi_{n-1}))| d\xi_{n-1} \right| \\
&\leq K \left| \int_{x_0}^x (|w_{n-1}(\xi_{n-1}) - w_{n-2}(\xi_{n-1})| + |v_{n-1}(\xi_{n-1}) - v_{n-2}(\xi_{n-1})|) d\xi_{n-1} \right| \\
&\leq K \left| \int_{x_0}^x \left(\left| \int_{x_0}^{\xi_{n-1}} (f_1(\xi_{n-2}, w_{n-2}(\xi_{n-2}), v_{n-2}(\xi_{n-2})) - f_1(\xi_{n-2}, w_{n-3}(\xi_{n-2}), v_{n-3}(\xi_{n-2}))) d\xi_{n-2} \right| \right. \right. \\
&\quad \left. \left. + \left| \int_{x_0}^{\xi_{n-1}} (f_1(\xi_{n-2}, w_{n-2}(\xi_{n-2}), v_{n-2}(\xi_{n-2})) - f_1(\xi_{n-2}, w_{n-3}(\xi_{n-2}), v_{n-3}(\xi_{n-2}))) d\xi_{n-2} \right| \right) d\xi_{n-1} \right| \\
&\leq K \left| \int_{x_0}^x \left(\int_{x_0}^{\xi_{n-1}} |f_1(\xi_{n-2}, w_{n-2}(\xi_{n-2}), v_{n-2}(\xi_{n-2})) - f_1(\xi_{n-2}, w_{n-3}(\xi_{n-2}), v_{n-3}(\xi_{n-2}))| d\xi_{n-2} \right. \right. \\
&\quad \left. \left. + \int_{x_0}^{\xi_{n-1}} |f_1(\xi_{n-2}, w_{n-2}(\xi_{n-2}), v_{n-2}(\xi_{n-2})) - f_1(\xi_{n-2}, w_{n-3}(\xi_{n-2}), v_{n-3}(\xi_{n-2}))| d\xi_{n-2} \right) d\xi_{n-1} \right| \\
&\leq K^2 \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} (|w_{n-2}(\xi_{n-2}) - w_{n-3}(\xi_{n-2})| + |v_{n-2}(\xi_{n-2}) - v_{n-3}(\xi_{n-2})|) d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq \dots \\
&\leq K^{n-1} \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \dots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} (|w_1(\xi_1) - w_0(\xi_1)| + |v_1(\xi_1) - v_0(\xi_1)|) d\xi_1 d\xi_2 \dots d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq K^{n-1} \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \dots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} \int_{x_0}^{\xi_1} (|f_1(\xi_0, y_0, y'_0)| + |f_2(\xi_0, y_0, y'_0)|) d\xi_0 d\xi_1 d\xi_2 \dots d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq 2K^{n-1} M \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \dots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} \int_{x_0}^{\xi_1} d\xi_0 d\xi_1 d\xi_2 \dots d\xi_{n-2} d\xi_{n-1} \right| \\
&= K^{n-1} \times \frac{2M}{n!} |x - x_0|^n, \tag{6.15}
\end{aligned}$$

$$\begin{aligned}
|v_n(x) - v_{n-1}(x)| &= \left| \int_{x_0}^x f_2(\xi_{n-1}, w_{n-1}(\xi_{n-1}), v_{n-1}(\xi_{n-1})) - f_2(\xi_{n-1}, w_{n-2}(\xi_{n-1}), v_{n-1}(\xi_{n-1})) d\xi_{n-1} \right| \\
&\leq \left| \int_{x_0}^x |f_2(\xi_{n-1}, w_{n-1}(\xi_{n-1}), v_{n-1}(\xi_{n-1})) - f_2(\xi_{n-1}, w_{n-2}(\xi_{n-1}), v_{n-1}(\xi_{n-1}))| d\xi_{n-1} \right| \\
&\leq K \left| \int_{x_0}^x (|w_{n-1}(\xi_{n-1}) - w_{n-2}(\xi_{n-1})| + |v_{n-1}(\xi_{n-1}) - v_{n-2}(\xi_{n-1})|) d\xi_{n-1} \right| \\
&\leq \dots \\
&\leq K^{n-1} \left| \int_{x_0}^x \int_{x_0}^{\xi_{n-1}} \int_{x_0}^{\xi_{n-2}} \dots \int_{x_0}^{\xi_3} \int_{x_0}^{\xi_2} (|w_1(\xi_1) - w_0(\xi_1)| + |v_1(\xi_1) - v_0(\xi_1)|) d\xi_1 d\xi_2 \dots d\xi_{n-2} d\xi_{n-1} \right| \\
&\leq K^{n-1} \times \frac{2M}{n!} |x - x_0|^n. \tag{6.16}
\end{aligned}$$

By means of Eqs.(6.15) and (6.16), the absolutely convergence of the series $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{n=1}^n (w_k(x) - w_{k-1}(x)) \right)$ and $\lim_{n \rightarrow +\infty} \left(y_0 + \sum_{n=1}^n (v_k(x) - v_{k-1}(x)) \right)$ can be obtained as follows:

$$\begin{aligned}
 |w_n(x)| &\leq |y_0| + \sum_{k=1}^n |w_k(x) - w_{k-1}(x)| \\
 &\leq |y_0| + \sum_{k=1}^n K^{k-1} \times \frac{2M}{k!} |x - x_0|^k \\
 &\leq |y_0| + \sum_{k=1}^n K^{k-1} \times \frac{2M}{k!} \alpha^k \\
 &= |y_0| + \frac{2M}{K} (\exp(K\alpha) - 1) \quad (n \rightarrow +\infty, |x - x_0| \leq \alpha). \tag{6.17}
 \end{aligned}$$

$$\begin{aligned}
 |v_n(x)| &\leq |z_0| + \sum_{k=1}^n |v_k(x) - v_{k-1}(x)| \\
 &\leq |z_0| + \sum_{k=1}^n K^{k-1} \times \frac{2M}{k!} |x - x_0|^k \\
 &\leq |z_0| + \sum_{k=1}^n K^{k-1} \times \frac{2M}{k!} \alpha^k \\
 &= |z_0| + \frac{2M}{K} (\exp(K\alpha) - 1) \quad (n \rightarrow +\infty, |x - x_0| \leq \alpha). \tag{6.18}
 \end{aligned}$$

In the final equality, we use the Taylor expansion of exponential function in Eq.(4.15). Then it has been shown that the existence of the first order ordinary differential equation (6.1) with the initial condition (6.2) under boundedness condition (6.3) and the Lipshitz condition (6.6) for the kernels $f_1(x, y, z)$ and $f_2(x, y, z)$. From theorem 3.13, the function sequence $\{w_n(x) | n = 0, 1, 2, \dots\}$ and $\{v_n(x) | n = 0, 1, 2, \dots\}$ defined by the recursion formulas (6.9)-(6.12) under the initial condition (6.2) uniformly converges and theorem 3.14, we can see that the function sequence $\{w_n(x) | n = 0, 1, 2, \dots\}$ always converges to a continuous function $w(x)$ in the bounded and closed interval $|x - x_0| \leq \alpha$ such that

$$\begin{cases} |w(x) - w(\xi)| \leq |w(x) - w_n(x)| + |w_n(x) - w_n(\xi)| + |w_n(\xi) - w(\xi)| \rightarrow 0, \\ |v(x) - v(\xi)| \leq |v(x) - v_n(x)| + |v_n(x) - v_n(\xi)| + |v_n(\xi) - v(\xi)| \rightarrow 0 \end{cases} \quad (|x - x_0| \leq \alpha, \xi \rightarrow x, n \rightarrow +\infty). \tag{6.19}$$

Finally, we shown the uniqueness for solutions in Eq.(6.5). Let us suppose that Eq.(6.5) have two function $\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = \begin{pmatrix} \Phi(x) \\ \Psi(x) \end{pmatrix}$ and $\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = \begin{pmatrix} \tilde{\Phi}(x) \\ \tilde{\Psi}(x) \end{pmatrix}$ as solutions. We consider the differences $|\Phi(x) - \tilde{\Phi}(x)|$ and $|\Psi(x) - \tilde{\Psi}(x)|$ as follows:

$$\begin{aligned}
 |\Phi(x) - \tilde{\Phi}(x)| &= \left| \int_{x_0}^x (f_1(x, \Phi(x), \Psi(x)) - f_1(x, \tilde{\Psi}(x), \tilde{\Phi}(x))) dx \right| \\
 &\leq \left| \int_{x_0}^x |f_1(x, \Phi(x), \Psi(x)) - f_1(x, \tilde{\Phi}(x), \tilde{\Psi}(x))| dx \right| \\
 &\leq \frac{K}{2} \left| \int_{x_0}^x (|\Phi(x) - \tilde{\Phi}(x)| + |\Psi(x) - \tilde{\Psi}(x)|) dx \right| \quad (\forall x \in [x_0 - \alpha, x_0 + \alpha]), \tag{6.20}
 \end{aligned}$$

$$\begin{aligned}
 |\Psi(x) - \tilde{\Psi}(x)| &= \left| \int_{x_0}^x (f_2(x, \Phi(x), \Psi(x)) - f_2(x, \tilde{\Psi}(x), \tilde{\Phi}(x))) dx \right| \\
 &\leq \left| \int_{x_0}^x |f_2(x, \Phi(x), \Psi(x)) - f_2(x, \tilde{\Phi}(x), \tilde{\Psi}(x))| dx \right| \\
 &\leq \frac{K}{2} \left| \int_{x_0}^x (|\Phi(x) - \tilde{\Phi}(x)| + |\Psi(x) - \tilde{\Psi}(x)|) dx \right| \quad (\forall x \in [x_0 - \alpha, x_0 + \alpha]). \tag{6.21}
 \end{aligned}$$

By using the above inequalities (6.20) and (6.21), we obtain

$$\begin{aligned}
|\Phi(x) - \tilde{\Phi}(x)| &\leq K \left| \int_{x_0}^x \left(|\Phi(\xi_1) - \tilde{\Phi}(\xi_1)| + |\Psi(\xi_1) - \tilde{\Psi}(\xi_1)| \right) d\xi_1 \right| \\
&\leq K^2 \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \left(|\Phi(\xi_2) - \tilde{\Phi}(\xi_2)| + |\Psi(\xi_2) - \tilde{\Psi}(\xi_2)| \right) d\xi_2 d\xi_1 \right| \\
&\leq \dots \leq K^n \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \dots \int_{x_0}^{\xi_{n-1}} \left(|\Phi(\xi_n) - \tilde{\Phi}(\xi_n)| + |\Psi(\xi_n) - \tilde{\Psi}(\xi_n)| \right) d\xi_n \dots d\xi_2 d\xi_1 \right| \\
&\leq \left(\max_{|\xi - x_0| \leq \alpha} |\Phi(\xi) - \tilde{\Phi}(\xi)| + \max_{|\xi - x_0| \leq \alpha} |\Psi(\xi) - \tilde{\Psi}(\xi)| \right) K^n \left| \int_{x_0}^x \int_{x_0}^{\xi_1} \dots \int_{x_0}^{\xi_{n-1}} |\xi_n - x_0| d\xi_n \dots d\xi_2 d\xi_1 \right| \\
&= \left(\max_{|\xi - x_0| \leq \alpha} |\Phi(\xi) - \tilde{\Phi}(\xi)| + \max_{|\xi - x_0| \leq \alpha} |\Psi(\xi) - \tilde{\Psi}(\xi)| \right) K^n \times \frac{1}{n!} |x - x_0|^n \\
&\leq \left(\max_{|\xi - x_0| \leq \alpha} |\Phi(\xi) - \tilde{\Phi}(\xi)| + \max_{|\xi - x_0| \leq \alpha} |\Psi(\xi) - \tilde{\Psi}(\xi)| \right) K^n \times \frac{1}{n!} \alpha^n \\
&\rightarrow 0 \quad (n \rightarrow +\infty, \forall x \in [x_0 - \alpha, x_0 + \alpha]).
\end{aligned} \tag{6.22}$$

From Eq.(6.22), $(\Phi(x), \Psi(x))$ and $(\tilde{\Phi}(x), \tilde{\Psi}(x))$ are equivalent of each other and the uniqueness for solutions in Eq.(6.5) has been confirmed.

6.2 Simultaneous Exact Differential Equation

In Eq.(6.1), let us suppose that $v(x)$ can be eliminated. Because it is valid that

$$\begin{aligned}
\frac{d^2}{dx^2} w(x) &= \left[\frac{\partial}{\partial x} f(x, y, z) \right]_{y=w(x), z=v(x)} + \left[\frac{\partial}{\partial y} f(x, y, z) \right]_{y=w(x), z=v(x)} \left(\frac{d}{dx} w(x) \right) \\
&\quad + \left[\frac{\partial}{\partial z} f(x, y, z) \right]_{y=w(x), z=v(x)} \left(\frac{d}{dx} v(x) \right)
\end{aligned} \tag{6.23}$$

Eq.(6.1) can be reduced to a second order ordinary differential equation

$$F\left(x, w(x), \frac{d}{dx} w(x), \frac{d^2}{dx^2} w(x)\right) = 0 \tag{6.24}$$

for a function $F(x, y, z)$. From Eq.(6.24) and the first equation of Eq.(6.1), the general solution of Eq.(6.1) can be expressed as follows:

$$\begin{cases} w(x) = \Phi(x, c_1, c_2), \\ v(x) = \Psi(x, c_1, c_2), \end{cases} \quad |x - x_0| \leq \alpha, \tag{6.25}$$

for functions $\Phi(x, y, z)$ and $\Psi(x, y, z)$ where c_1 and c_2 are constants which are determined so as to satisfy Eq.(6.2).

Two functions $\phi(x, y, z)$ and $\psi(x, y, z)$ are independent of each other in the bounded and closed interval $|x - x_0| \leq \alpha$, if there exists a function $\Theta(y, z)$ which satisfies

$$\Theta(\phi(x, y, z), \psi(x, y, z)) = 0. \tag{6.26}$$

We take partial derivatives for both sides of Eq.(6.26) with respect to x, y and z , respectively, as follows:

$$\begin{cases} \left[\frac{\partial}{\partial \eta} \Theta(\eta, \zeta) \right]_{\eta=\phi(x, y, z), \zeta=\psi(x, y, z)} \left(\frac{\partial}{\partial x} \phi(x, y, z) \right) + \left[\frac{\partial}{\partial \zeta} \Theta(\eta, \zeta) \right]_{\eta=\phi(x, y, z), \zeta=\psi(x, y, z)} \left(\frac{\partial}{\partial x} \psi(x, y, z) \right) = 0, \\ \left[\frac{\partial}{\partial \eta} \Theta(\eta, \zeta) \right]_{\eta=\phi(x, y, z), \zeta=\psi(x, y, z)} \left(\frac{\partial}{\partial y} \phi(x, y, z) \right) + \left[\frac{\partial}{\partial \zeta} \Theta(\eta, \zeta) \right]_{\eta=\phi(x, y, z), \zeta=\psi(x, y, z)} \left(\frac{\partial}{\partial y} \psi(x, y, z) \right) = 0, \\ \left[\frac{\partial}{\partial \eta} \Theta(\eta, \zeta) \right]_{\eta=\phi(x, y, z), \zeta=\psi(x, y, z)} \left(\frac{\partial}{\partial z} \phi(x, y, z) \right) + \left[\frac{\partial}{\partial \zeta} \Theta(\eta, \zeta) \right]_{\eta=\phi(x, y, z), \zeta=\psi(x, y, z)} \left(\frac{\partial}{\partial z} \psi(x, y, z) \right) = 0. \end{cases} \tag{6.27}$$

Eq.(6.27) can be reduced to

$$\frac{\left(\frac{\partial}{\partial x} \psi(x, y, z) \right)}{\left(\frac{\partial}{\partial x} \phi(x, y, z) \right)} = \frac{\left(\frac{\partial}{\partial y} \psi(x, y, z) \right)}{\left(\frac{\partial}{\partial y} \phi(x, y, z) \right)} = \frac{\left(\frac{\partial}{\partial z} \psi(x, y, z) \right)}{\left(\frac{\partial}{\partial z} \phi(x, y, z) \right)}. \tag{6.28}$$

On the other hand, let us suppose that Eq.(6.28) is valid. We introduce a new variable ξ as follows:

$$\xi \equiv \phi(x, y, z). \quad (6.29)$$

If we assume that $\frac{\partial}{\partial x}\phi(x, y, z) \neq 0$, there exists a function $\chi(\xi, y, z)$ uniquely and it satisfies

$$x = \chi(\xi, y, z), \quad (6.30)$$

and

$$\begin{cases} \frac{\partial}{\partial y}\chi(\xi, y, z) = \frac{\left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{x=\chi(\xi, y, z)}}{\left[\frac{\partial}{\partial x}\phi(x, y, z)\right]_{x=\chi(\xi, y, z)}}, \\ \frac{\partial}{\partial z}\chi(\xi, y, z) = \frac{\left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{x=\chi(\xi, y, z)}}{\left[\frac{\partial}{\partial x}\phi(x, y, z)\right]_{x=\chi(\xi, y, z)}}, \end{cases} \quad (6.31)$$

from the implicit function theorem in the calculus. By taking the partial derivatives of $\psi(\chi(\xi, y, z), y, z)$ with respect to y and z , respectively, and by applying Eq.(6.28) and (6.31), we obtain the following equalities:

$$\begin{cases} \frac{\partial}{\partial y}\psi(\chi(\xi, y, z), y, z) = \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{x=\chi(\xi, y, z)} + \left(\frac{\partial}{\partial y}\chi(\xi, y, z)\right)\left[\frac{\partial}{\partial x}\psi(x, y, z)\right]_{x=\chi(\xi, y, z)} = 0, \\ \frac{\partial}{\partial z}\psi(\chi(\xi, y, z), y, z) = \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{x=\chi(\xi, y, z)} + \left(\frac{\partial}{\partial z}\chi(\xi, y, z)\right)\left[\frac{\partial}{\partial x}\psi(x, y, z)\right]_{x=\chi(\xi, y, z)} = 0. \end{cases} \quad (6.32)$$

Eq.(6.30) means that $\psi(\chi(\xi, y, z), y, z)$ depends on only ξ and does not depend on y and z , such that

$$\psi(x, y, z) = \psi(\chi(\xi, y = c_1, z = c_2), y = c_1, z = c_2) = \psi(\chi(\xi = \phi(x, y, z), y = c_1, z = c_2), y = c_1, z = c_2), \quad (6.33)$$

where c_1 and c_2 are constants. Eq.(6.34) means that there exists a function $\Theta(y, z)$ in Eq.(6.26) and then functions $\phi(x, y, z)$ and $\psi(x, y, z)$ are dependent of each other. It is concluded that Eq.(6.28) are the necessary and sufficient condition that functions $\phi(x, y, z)$ and $\psi(x, y, z)$ are dependent of each other. Conversely, if Eq.(6.28) does not hold, such that

$$\frac{\frac{\partial}{\partial x}\psi(x, y, z)}{\frac{\partial}{\partial x}\phi(x, y, z)} = \frac{\frac{\partial}{\partial y}\psi(x, y, z)}{\frac{\partial}{\partial y}\phi(x, y, z)}, \quad \frac{\frac{\partial}{\partial y}\psi(x, y, z)}{\frac{\partial}{\partial y}\phi(x, y, z)} = \frac{\frac{\partial}{\partial z}\psi(x, y, z)}{\frac{\partial}{\partial z}\phi(x, y, z)} \quad \text{or} \quad \frac{\frac{\partial}{\partial x}\psi(x, y, z)}{\frac{\partial}{\partial x}\phi(x, y, z)} \neq \frac{\frac{\partial}{\partial z}\psi(x, y, z)}{\frac{\partial}{\partial z}\phi(x, y, z)}, \quad (6.34)$$

then $\phi(x, y, z)$ and $\psi(x, y, z)$ are independent of each other.

If Eq.(6.25) can be solved with respect to c_1 and c_2 as follows:

$$\begin{cases} c_1 = \phi(x, w(x), v(x)), \\ c_2 = \psi(x, w(x), v(x)), \end{cases} \quad (6.35)$$

the set of the two functions $\phi(x, y, z)$ and $\psi(x, y, z)$ is referred to as the **first integral** of the simultaneous ordinary differential equations (6.1). The solution of the simultaneous ordinary differential equations (6.1) can be expressed in terms of an intersection of two surfaces

$$\begin{cases} \phi(x, y, z) = c_1, \\ \psi(x, y, z) = c_2, \end{cases} \quad (6.36)$$

in the three dimensional xyz -place. Let us suppose that the **first integral** of the simultaneous ordinary differential equations (6.1) are expressed as Eq.(6.35) in terms of the set of the two mutually independent functions $\phi(x, y, z)$ and $\psi(x, y, z)$. By differentiating both sides of Eq.(6.36) with $y = w(x)$ and $z = v(x)$, such that

$$\begin{cases} \left[\frac{\partial}{\partial x}\phi(x, y, z)\right]_{y=w(x), z=v(x)} + \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=w(x), z=v(x)}\left(\frac{\partial}{\partial x}w(x)\right) + \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=w(x), z=v(x)}\left(\frac{\partial}{\partial x}v(x)\right) = 0, \\ \left[\frac{\partial}{\partial x}\psi(x, y, z)\right]_{y=w(x), z=v(x)} + \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=w(x), z=v(x)}\left(\frac{\partial}{\partial x}w(x)\right) + \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=w(x), z=v(x)}\left(\frac{\partial}{\partial x}v(x)\right) = 0, \end{cases} \quad (6.37)$$

and by applying Eq.(6.1) to Eq.(6.38), we have

$$\begin{cases} \left[\frac{\partial}{\partial x}\phi(x, y, z)\right]_{y=w(x), z=v(x)} + \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=w(x), z=v(x)}f_1(x, w(x), v(x)) \\ \quad + \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=w(x), z=v(x)}f_2(x, w(x), v(x)) = 0, \\ \left[\frac{\partial}{\partial x}\psi(x, y, z)\right]_{y=w(x), z=v(x)} + \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=w(x), z=v(x)}f_1(x, w(x), v(x)) \\ \quad + \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=w(x), z=v(x)}f_2(x, w(x), v(x)) = 0, \end{cases} \quad (6.38)$$

Because of the mutually independence of $\phi(x, y, z)$ and $\psi(x, y, z)$, we have Eq.(6.34). If it satisfies that

$$\left(\frac{\partial}{\partial x}\phi(x, y, z)\right)\left(\frac{\partial}{\partial y}\phi(x, y, z)\right) - \left(\frac{\partial}{\partial y}\phi(x, y, z)\right)\left(\frac{\partial}{\partial x}\phi(x, y, z)\right) \neq 0, \quad (6.39)$$

or

$$\left(\frac{\partial}{\partial x}\phi(x, y, z)\right)\left(\frac{\partial}{\partial z}\phi(x, y, z)\right) - \left(\frac{\partial}{\partial z}\phi(x, y, z)\right)\left(\frac{\partial}{\partial x}\phi(x, y, z)\right) \neq 0, \quad (6.40)$$

we derive

$$\left(\frac{\partial}{\partial y}\phi(x, y, z)\right)\left(\frac{\partial}{\partial z}\phi(x, y, z)\right) - \left(\frac{\partial}{\partial z}\phi(x, y, z)\right)\left(\frac{\partial}{\partial y}\phi(x, y, z)\right) \neq 0, \quad (6.41)$$

by means of Eq.(6.38). It means that Eq.(6.41) is always valid for two mutually independent functions $\phi(x, y, z)$ and $\psi(x, y, z)$ in Eq.(6.36). From the implicit function theorem in the calculus, there exists the set of the solutions for Eq.(6.36) with respect to y and z as follows:

$$\begin{cases} y = \tilde{\Phi}(x, c_1, c_2), \\ z = \tilde{\Psi}(x, c_1, c_2), \end{cases} \quad (6.42)$$

in terms of functions $\tilde{\Phi}(x, \eta, \zeta)$ and $\tilde{\Psi}(x, \eta, \zeta)$. From (6.41), we have

$$\begin{cases} \left[\frac{\partial}{\partial x}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} + \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} f_1(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Phi}(x, \eta, \zeta)) \\ \quad + \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} f_2(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Phi}(x, \eta, \zeta)) = 0, \\ \left[\frac{\partial}{\partial x}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} + \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} f_1(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Phi}(x, \eta, \zeta)) \\ \quad + \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} f_2(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Phi}(x, \eta, \zeta)) = 0, \end{cases} \quad (6.43)$$

By differentiating both sides of Eq.(6.36) with $y = w(x)$ and $z = v(x)$, we can derive

$$\begin{cases} \left[\frac{\partial}{\partial x}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} + \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\frac{\partial}{\partial x}\tilde{\Phi}(x, \eta, \zeta)\right) \\ \quad + \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\frac{\partial}{\partial x}\tilde{\Psi}(x, \eta, \zeta)\right) = 0, \\ \left[\frac{\partial}{\partial x}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} + \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\frac{\partial}{\partial x}\tilde{\Phi}(x, \eta, \zeta)\right) \\ \quad + \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\frac{\partial}{\partial x}\tilde{\Psi}(x, \eta, \zeta)\right) = 0, \end{cases} \quad (6.44)$$

and Eqs.(6.43) and (6.44) can be reduced to

$$\begin{cases} \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\left(\frac{\partial}{\partial x}\tilde{\Phi}(x, \eta, \zeta)\right) - f_1(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta))\right) \\ \quad + \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\left(\frac{\partial}{\partial x}\tilde{\Psi}(x, \eta, \zeta)\right) - f_2(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta))\right) = 0, \\ \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\left(\frac{\partial}{\partial x}\tilde{\Phi}(x, \eta, \zeta)\right) - f_1(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta))\right) \\ \quad + \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \left(\left(\frac{\partial}{\partial x}\tilde{\Psi}(x, \eta, \zeta)\right) - f_2(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta))\right) = 0, \end{cases} \quad (6.45)$$

such that

$$\begin{pmatrix} \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} & \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \\ \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} & \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \end{pmatrix} \begin{pmatrix} \frac{\partial}{\partial x}\tilde{\Phi}(x, \eta, \zeta) - f_1(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta)) \\ \frac{\partial}{\partial x}\tilde{\Psi}(x, \eta, \zeta) - f_2(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta)) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (6.46)$$

From Eq.(6.41), we have

$$\det \begin{pmatrix} \left[\frac{\partial}{\partial y}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} & \left[\frac{\partial}{\partial z}\phi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \\ \left[\frac{\partial}{\partial y}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} & \left[\frac{\partial}{\partial z}\psi(x, y, z)\right]_{y=\tilde{\Phi}(x, \eta, \zeta), z=\tilde{\Phi}(x, \eta, \zeta)} \end{pmatrix} \neq 0, \quad (6.47)$$

and then it should be valid that

$$\begin{pmatrix} \frac{\partial}{\partial x} \tilde{\Phi}(x, \eta, \zeta) - f_1(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta)) \\ \frac{\partial}{\partial x} \tilde{\Psi}(x, \eta, \zeta) - f_2(x, \tilde{\Phi}(x, \eta, \zeta), \tilde{\Psi}(x, \eta, \zeta)) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (6.48)$$

Eq.(6.48) means that $\tilde{\Phi}(x, \eta, \zeta)$ and $\tilde{\Psi}(x, \eta, \zeta)$ in Eq.(6.42) are the solutions of the simultaneous ordinary differential equations in Eq.(6.1).

Finally we consider the following simultaneous differential equations:

$$\begin{cases} P_1(x, w(x), v(x)) + Q_1(x, w(x), v(x)) \frac{d}{dx} w(x) + R_1(x, w(x), v(x)) \frac{d}{dx} v(x) = 0, \\ P_2(x, w(x), v(x)) + Q_2(x, w(x), v(x)) \frac{d}{dx} w(x) + R_2(x, w(x), v(x)) \frac{d}{dx} v(x) = 0. \end{cases} \quad (6.49)$$

This equation can be reduced to

$$\begin{aligned} \det \begin{pmatrix} Q_1(x, w(x), v(x)) & R_1(x, w(x), v(x)) \\ Q_2(x, w(x), v(x)) & R_2(x, w(x), v(x)) \end{pmatrix}^{-1} &= \det \begin{pmatrix} R_1(x, w(x), v(x)) & P_1(x, w(x), v(x)) \\ R_2(x, w(x), v(x)) & P_2(x, w(x), v(x)) \end{pmatrix}^{-1} \frac{d}{dx} w(x) \\ &= \det \begin{pmatrix} P_1(x, w(x), v(x)) & Q_1(x, w(x), v(x)) \\ P_2(x, w(x), v(x)) & Q_2(x, w(x), v(x)) \end{pmatrix}^{-1} \frac{d}{dx} v(x). \end{aligned} \quad (6.50)$$

Eqs.(6.49) and (6.50) are expressed as

$$\begin{cases} P_1(x, y, z) dx + Q_1(x, y, z) dy + R_1(x, y, z) dz = 0, \\ P_2(x, y, z) dx + Q_2(x, y, z) dy + R_2(x, y, z) dz = 0. \end{cases} \quad (6.51)$$

$$\begin{aligned} \det \begin{pmatrix} Q_1(x, y, z) & R_1(x, y, z) \\ Q_2(x, y, z) & R_2(x, y, z) \end{pmatrix}^{-1} dx &= \det \begin{pmatrix} R_1(x, y, z) & P_1(x, y, z) \\ R_2(x, y, z) & P_2(x, y, z) \end{pmatrix}^{-1} dy \\ &= \det \begin{pmatrix} P_1(x, y, z) & Q_1(x, y, z) \\ P_2(x, y, z) & Q_2(x, y, z) \end{pmatrix}^{-1} dz, \end{aligned} \quad (6.52)$$

in terms of representations of exact equations, respectively.

6.3 Simultaneous Linear Ordinary Differential Equations

Let us consider the following simultaneous linear ordinary differential equations:

$$\begin{cases} \frac{d}{dx} w(x) = p_{11}(x)w(x) + p_{12}(x)v(x), \\ \frac{d}{dx} v(x) = p_{21}(x)w(x) + p_{22}(x)v(x), \end{cases} \quad (|x - x_0| \leq \alpha), \quad (6.53)$$

where $p_{11}(x)$, $p_{12}(x)$, $p_{21}(x)$ and $p_{22}(x)$ are bounded and continuous functions in $|x - x_0| \leq \alpha$. The present equation corresponds to the case of

$$\begin{cases} f_1(x, y, z) = p_{11}(x)y + p_{12}(x)z, \\ f_2(x, y, z) = p_{21}(x)y + p_{22}(x)z, \end{cases} \quad (|x - x_0| \leq \alpha), \quad (6.54)$$

in Eq.(6.1). It is easy to confirm that $f_1(x, y, z)$ and $f_2(x, y, z)$ in Eq.(6.54) satisfy the boundedness (6.3) and the Lipschitz condition (6.6). Hence the differential equation (6.53) with Eq.(6.2) has the continuousness, the existence and the uniqueness of the solution.

Now we consider two solutions $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ as solutions in Eq.(6.54) as follows:

$$\begin{cases} \frac{d}{dx} w^{(1)}(x) = p_{11}(x)w^{(1)}(x) + p_{12}(x)v^{(1)}(x), \\ \frac{d}{dx} v^{(1)}(x) = p_{21}(x)w^{(1)}(x) + p_{22}(x)v^{(1)}(x), \end{cases} \quad (|x - x_0| \leq \alpha), \quad (6.55)$$

$$\begin{cases} \frac{d}{dx} w^{(2)}(x) = p_{11}(x)w^{(2)}(x) + p_{12}(x)v^{(2)}(x), \\ \frac{d}{dx} v^{(2)}(x) = p_{21}(x)w^{(2)}(x) + p_{22}(x)v^{(2)}(x), \end{cases} \quad (|x - x_0| \leq \alpha). \quad (6.56)$$

The necessary and sufficient condition that $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ are independent of each other is that the simultaneous linear equation for $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$:

$$\begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (6.57)$$

always have non-trivial solutions in $|x - x_0| \leq \alpha$, such that

$$R(x) \equiv \det \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix} = \det \begin{pmatrix} w^{(1)}(x) & v^{(1)}(x) \\ w^{(2)}(x) & v^{(2)}(x) \end{pmatrix} \neq 0 \quad (|x - x_0| \leq \alpha). \quad (6.58)$$

We consider the derivative of the above determinant $R(x)$ with respect to x

$$\begin{aligned} \frac{d}{dx} R(x) &= \det \begin{pmatrix} \frac{d}{dx} w^{(1)}(x) & v^{(1)}(x) \\ \frac{d}{dx} w^{(2)}(x) & v^{(2)}(x) \end{pmatrix} + \det \begin{pmatrix} w^{(1)}(x) & \frac{d}{dx} v^{(1)}(x) \\ w^{(2)}(x) & \frac{d}{dx} v^{(2)}(x) \end{pmatrix} \\ &= \det \begin{pmatrix} p_{11}(x)w^{(1)}(x) + p_{12}(x)v^{(1)}(x) & v^{(1)}(x) \\ p_{11}(x)w^{(2)}(x) + p_{12}(x)v^{(2)}(x) & v^{(2)}(x) \end{pmatrix} + \det \begin{pmatrix} w^{(1)}(x) & p_{21}(x)w^{(1)}(x) + p_{22}(x)v^{(1)}(x) \\ w^{(2)}(x) & p_{21}(x)w^{(2)}(x) + p_{22}(x)v^{(2)}(x) \end{pmatrix} \\ &= \det \begin{pmatrix} p_{11}(x)w^{(1)}(x) & v^{(1)}(x) \\ p_{11}(x)w^{(2)}(x) & v^{(2)}(x) \end{pmatrix} + \det \begin{pmatrix} w^{(1)}(x) & p_{22}(x)v^{(1)}(x) \\ w^{(2)}(x) & p_{22}(x)v^{(2)}(x) \end{pmatrix} \\ &= (p_{11}(x) + p_{22}(x)) \det \begin{pmatrix} w^{(1)}(x) & v^{(1)}(x) \\ w^{(2)}(x) & v^{(2)}(x) \end{pmatrix} \quad (|x - x_0| \leq \alpha), \end{aligned} \quad (6.59)$$

such that we have

$$\frac{d}{dx} R(x) = (p_{11}(x) + p_{22}(x)) R(x) \quad (|x - x_0| \leq \alpha), \quad (6.60)$$

By solving Eq.(6.60), $R(x)$ can be reduced to the following closed expression un to a constant factor:

$$R(x) = \text{Constant} \times \int_{x_0}^x \exp((p_{11}(x) + p_{22}(x))) \quad (|x - x_0| \leq \alpha), \quad (6.61)$$

$\exp\left(-\int_{x_0}^x (p_{11}(\xi) + p_{22}(\xi)) d\xi\right)$ is always nonzero for any real number x in $|x - x_0| \leq \alpha$ because $p_{11}(x) + p_{22}(x)$ is a continuous function in the present section. We conclude the following two statements:

1. If $\det \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix} = 0$ at a point $x = x_0$ in $|x - x_0| \leq \alpha$, then the constant factor in the right side of Eq.(6.61) should be zero and the determinant $\det \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix}$ of the solutions $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ in Eq.(6.53) is always zero at every point x in $|x - x_0| \leq \alpha$, such that $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ are linearly dependent on each other in $|x - x_0| \leq \alpha$.
2. If $\det \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix} \neq 0$ at a point $x = x_0$ in $|x - x_0| \leq \alpha$, then the constant factor in the right side of Eq.(6.61) should be nonzero and the determinant $\det \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix}$ of the solutions $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ in Eq.(6.53) is always nonzero at every point x in $|x - x_0| \leq \alpha$, such that $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ are linearly independent of each other in $|x - x_0| \leq \alpha$.

For the second order linear ordinary differential equation in Eq.(6.53), if two solutions $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$ are linearly independent of each other in $|x - x_0| \leq \alpha$, then the set of the solutions are a **fundamental system of solutions** of Eq.(6.53). Moreover, every solution $\begin{pmatrix} w(x) \\ v(x) \end{pmatrix}$ for Eq.(6.53) is expressed as

$$\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = c_1 \begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix} + c_2 \begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix} \quad (|x - x_0| \leq \alpha), \quad (6.62)$$

in terms of the fundamental system of solutions, $\begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix}$ and $\begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix}$. And by specifying value of $\begin{pmatrix} w(x) \\ v(x) \end{pmatrix}$ at a point $x = x_0$ in $|x - x_0| \leq \alpha$ as shown in Eq.(6.2), the solution $\begin{pmatrix} w(x) \\ v(x) \end{pmatrix}$ in Eq.(6.53) are uniquely expressed because

$$\det \begin{pmatrix} w^{(1)}(x_0) & v^{(1)}(x_0) \\ w^{(2)}(x_0) & v^{(2)}(x_0) \end{pmatrix} \neq 0, \quad (6.63)$$

and the solutions c_1 and c_2 of the following simultaneous linear equations:

$$\begin{cases} c_1 w^{(1)}(x_0) + c_2 v^{(1)}(x_0) = y_0, \\ c_1 w^{(2)}(x_0) + c_2 v^{(2)}(x_0) = z_0, \end{cases} \quad (6.64)$$

are uniquely determined.

For Eq.(6.53), we introduce the following inhomogeneous simultaneous linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx} w(x) = p_{11}(x)w(x) + p_{12}(x)v(x) + h_1(x), \\ \frac{d}{dx} v(x) = p_{21}(x)w(x) + p_{22}(x)v(x) + h_2(x), \end{cases} \quad (|x - x_0| \leq \alpha). \quad (6.65)$$

The inhomogeneous simultaneous linear equations can be solved by means of the variation of constants. The general solution of Eq.(6.65) can be expressed in terms of the general solution (6.62) of the homogeneous equation (6.53) and a special solution $\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}$ of Eq.(6.65) as follows:

$$\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = c_1 \begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix} + c_2 \begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix} + \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} \quad (6.66)$$

$$= \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} \quad (|x - x_0| \leq \alpha). \quad (6.67)$$

In the variations of constants for the present inhomogeneous case, the special solution $\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}$ is introduced by replacing $u_1(x)$ and $u_2(x)$ with arbitrary constants c_1 and c_2 in Eq.(6.62) as follows:

$$\begin{aligned} \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} &= u_1(x) \begin{pmatrix} w^{(1)}(x) \\ v^{(1)}(x) \end{pmatrix} + u_2(x) \begin{pmatrix} w^{(2)}(x) \\ v^{(2)}(x) \end{pmatrix} \\ &= \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ v^{(1)}(x) & v^{(2)}(x) \end{pmatrix} \begin{pmatrix} u_1(x) \\ u_2(x) \end{pmatrix} \quad (|x - x_0| \leq \alpha). \end{aligned} \quad (6.68)$$

By substituting Eq.(6.68) to Eq.(6.65) as $\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix}$, we have the following simultaneous equations to determine $u_1(x)$ and $u_2(x)$:

$$\begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ w^{(1)}(x) & w^{(2)}(x) \end{pmatrix} \begin{pmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \end{pmatrix} = \begin{pmatrix} h_1(x) \\ h_2(x) \end{pmatrix} \quad (|x - x_0| \leq \alpha). \quad (6.69)$$

From Eq.(6.69), $u_1(x)$ and $u_2(x)$ can be obtained as follows:

$$\begin{pmatrix} \frac{d}{dx} u_1(x) \\ \frac{d}{dx} u_2(x) \end{pmatrix} = \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ w^{(1)}(x) & w^{(2)}(x) \end{pmatrix}^{-1} \begin{pmatrix} h_1(x) \\ h_2(x) \end{pmatrix} \quad (|x - x_0| \leq \alpha), \quad (6.70)$$

such that

$$\begin{pmatrix} u_1(x) \\ u_2(x) \end{pmatrix} = \int \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ w^{(1)}(x) & w^{(2)}(x) \end{pmatrix}^{-1} \begin{pmatrix} h_1(x) \\ h_2(x) \end{pmatrix} dx \quad (|x - x_0| \leq \alpha), \quad (6.71)$$

By substituting Eq.(6.72) to Eq.(6.68), the special solution of Eq.(6.65) is derived to

$$\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \end{pmatrix} = \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ w^{(1)}(x) & w^{(2)}(x) \end{pmatrix} \int \begin{pmatrix} w^{(1)}(x) & w^{(2)}(x) \\ w^{(1)}(x) & w^{(2)}(x) \end{pmatrix}^{-1} \begin{pmatrix} h_1(x) \\ h_2(x) \end{pmatrix} dx \quad (|x - x_0| \leq \alpha). \quad (6.72)$$

6.4 Simultaneous Constant Coefficient Linear Ordinary Differential Equations

We consider the simultaneous homogeneous linear constant coefficient ordinary differential equations as follows:

$$\begin{cases} \frac{d}{dx}w(x) = a_{11}w(x) + a_{12}v(x), \\ \frac{d}{dx}v(x) = a_{21}w(x) + a_{22}v(x), \end{cases} \quad (6.73)$$

Eq.(6.73) is replaced to

$$\begin{cases} v(x) = \frac{1}{a_{21}} \left(\frac{d}{dx}v(x) - a_{22}v(x) \right), \\ w(x) = \frac{1}{a_{21}} \left(\frac{d}{dx}v(x) - a_{22}v(x) \right), \end{cases} \quad (6.74)$$

and by substituting the first and the second equation in Eq.(6.74) to the second and the first equation in Eq.(6.73), we can confirm that the simultaneous homogeneous first order linear constant coefficient equations (6.73) can be reduced to the separated forms of the homogeneous second order linear coefficient equations as follows:

$$\begin{cases} \frac{d}{dx} \left\{ \frac{1}{a_{21}} \left(\frac{d}{dx}v(x) - a_{22}v(x) \right) \right\} = a_{11} \left\{ \frac{1}{a_{21}} \left(\frac{d}{dx}v(x) - a_{22}v(x) \right) \right\} + a_{12}v(x), \\ \frac{d}{dx} \left\{ \frac{1}{a_{12}} \left(\frac{d}{dx}w(x) - a_{11}w(x) \right) \right\} = a_{21}w(x) + a_{22} \left\{ \frac{1}{a_{12}} \left(\frac{d}{dx}w(x) - a_{11}w(x) \right) \right\}, \end{cases} \quad (6.75)$$

$$\begin{cases} \frac{d^2}{dx^2}v(x) - (a_{22} + a_{11})\frac{d}{dx}v(x) + (a_{11}a_{22} - a_{21}a_{12})v(x), \\ \frac{d^2}{dx^2}w(x) - (a_{11} + a_{22})\frac{d}{dx}w(x) + (a_{22}a_{11} - a_{12}a_{21})w(x), \end{cases} \quad (6.76)$$

Eq.(6.76) means that Eq.(6.76) can be solved by setting

$$\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = \begin{pmatrix} C_1(\lambda)e^{\lambda x} \\ C_2(\lambda)e^{\lambda x} \end{pmatrix}, \quad (6.77)$$

such that Eq.(6.76) can be reduced to

$$e^{\lambda x} \begin{pmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{pmatrix} \begin{pmatrix} C_1(\lambda) \\ C_2(\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \quad (6.78)$$

The necessary and sufficient condition that the simultaneous linear equation (6.78) have non-trivial solutions with being $\begin{pmatrix} C_1(\lambda) \\ C_2(\lambda) \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ is as follows:

$$\det \begin{pmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{pmatrix} = 0. \quad (6.79)$$

It can be confirmed that Eq.(6.79) is equivalent to the characteristic equation of the second order linear constant coefficient ordinary differential equations in Eq.(6.76). If Eq.(6.79) have non-degenerate solutions λ_1 and λ_2 ($\lambda_1 \neq \lambda_2$),

we have the fundamental systems of solutions as $\left\{ \begin{pmatrix} C_1(\lambda_1)e^{\lambda_1 x} \\ C_2(\lambda_1)e^{\lambda_1 x} \end{pmatrix}, \begin{pmatrix} C_1(\lambda_2)e^{\lambda_2 x} \\ C_2(\lambda_2)e^{\lambda_2 x} \end{pmatrix} \right\}$ and the general solution is given by

$$\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = \begin{pmatrix} C_1(\lambda_1)e^{\lambda_1 x} \\ C_2(\lambda_1)e^{\lambda_1 x} \end{pmatrix} + \begin{pmatrix} C_1(\lambda_2)e^{\lambda_2 x} \\ C_2(\lambda_2)e^{\lambda_2 x} \end{pmatrix}. \quad (6.80)$$

If λ_1 and λ_2 are given as the complex numbers $\kappa + i\omega$ and $\kappa - i\omega$, respectively, where κ and ω are real numbers, the fundamental systems of solutions and the general equation are expressed in terms of $e^{\kappa x} \cos(\omega x)$ and $e^{\kappa x} \sin(\omega x)$ by using $e^{(\kappa + i\omega)x} = e^{\kappa x} (\cos(\omega x) + i \sin(\omega x))$. If Eq.(6.79) have a degenerate solution λ_1 ($\lambda_1 = \lambda_2$), we have not only

$\begin{pmatrix} C_1(\lambda_1)e^{\lambda_1 x} \\ C_2(\lambda_1)e^{\lambda_1 x} \end{pmatrix}$ but also $\begin{pmatrix} \left[\frac{\partial}{\partial \lambda} (C_1(\lambda)e^{\lambda x}) \right]_{\lambda=\lambda_1} \\ \left[\frac{\partial}{\partial \lambda} (C_2(\lambda)e^{\lambda x}) \right]_{\lambda=\lambda_1} \end{pmatrix} = \begin{pmatrix} \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} + C_1(\lambda_1) x e^{\lambda_1 x} \\ \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} + C_2(\lambda_1) x e^{\lambda_1 x} \end{pmatrix}$ as solutions of

Eq.(6.79), because it can be confirm, for any real number x , that

$$\left[\frac{\partial}{\partial x} \left(\frac{\partial}{\partial \lambda} \begin{pmatrix} C_1(\lambda)e^{\lambda x} \\ C_2(\lambda)e^{\lambda x} \end{pmatrix} \right) - \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \frac{\partial}{\partial \lambda} \begin{pmatrix} C_1(\lambda)e^{\lambda x} \\ C_2(\lambda)e^{\lambda x} \end{pmatrix} \right]_{\lambda=\lambda_1}$$

$$\begin{aligned}
&= \left[\frac{\partial}{\partial x} \left(e^{\lambda x} \frac{\partial}{\partial \lambda} \begin{pmatrix} C_1(\lambda) \\ C_2(\lambda) \end{pmatrix} \right) + x e^{\lambda x} \begin{pmatrix} C_1(\lambda) \\ C_2(\lambda) \end{pmatrix} \right] \\
&\quad - \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \left(e^{\lambda x} \frac{\partial}{\partial \lambda} \begin{pmatrix} C_1(\lambda) \\ C_2(\lambda) \end{pmatrix} + x e^{\lambda x} \begin{pmatrix} C_1(\lambda) \\ C_2(\lambda) \end{pmatrix} \right) \Big|_{\lambda=\lambda_1} \\
&= \begin{pmatrix} \left[\frac{d}{d\lambda} C_1(\lambda) \right]_{\lambda=\lambda_1} & 0 \\ 0 & \left[\frac{d}{d\lambda} C_2(\lambda) \right]_{\lambda=\lambda_1} \end{pmatrix} \begin{pmatrix} \frac{d}{dx} e^{\lambda_1 x} \\ \frac{d}{dx} e^{\lambda_1 x} \end{pmatrix} + \lambda_1 x e^{\lambda_1 x} \begin{pmatrix} C_1(\lambda_1) \\ C_2(\lambda_1) \end{pmatrix} \\
&\quad - \begin{pmatrix} \left[\frac{d}{d\lambda} C_1(\lambda) \right]_{\lambda=\lambda_1} & 0 \\ 0 & \left[\frac{d}{d\lambda} C_2(\lambda) \right]_{\lambda=\lambda_1} \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} e^{\lambda_1 x} \\ e^{\lambda_1 x} \end{pmatrix} - x e^{\lambda_1 x} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} C_1(\lambda_1) \\ C_2(\lambda_1) \end{pmatrix} \\
&= \begin{pmatrix} \left[\frac{d}{d\lambda} C_1(\lambda) \right]_{\lambda=\lambda_1} & 0 \\ 0 & \left[\frac{d}{d\lambda} C_2(\lambda) \right]_{\lambda=\lambda_1} \end{pmatrix} \begin{pmatrix} \frac{d}{dx} \begin{pmatrix} e^{\lambda_1 x} \\ e^{\lambda_1 x} \end{pmatrix} - \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} e^{\lambda_1 x} \\ e^{\lambda_1 x} \end{pmatrix} \end{pmatrix} \\
&\quad - x e^{\lambda_1 x} \begin{pmatrix} a_{11} - \lambda_1 & a_{12} \\ a_{21} & a_{22} - \lambda_1 \end{pmatrix} \begin{pmatrix} C_1(\lambda_1) \\ C_2(\lambda_1) \end{pmatrix} = 0. \tag{6.81}
\end{aligned}$$

We have

$$\begin{aligned}
&\det \begin{pmatrix} C_1(\lambda_1) e^{\lambda_1 x} & \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} + C_1(\lambda_1) x e^{\lambda_1 x} \\ C_2(\lambda_1) e^{\lambda_1 x} & \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} + C_2(\lambda_1) x e^{\lambda_1 x} \end{pmatrix} \\
&= \det \begin{pmatrix} C_1(\lambda_1) e^{\lambda_1 x} & \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} \\ C_2(\lambda_1) e^{\lambda_1 x} & \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} \end{pmatrix} + \det \begin{pmatrix} C_1(\lambda_1) e^{\lambda_1 x} & C_1(\lambda_1) x e^{\lambda_1 x} \\ C_2(\lambda_1) e^{\lambda_1 x} & C_2(\lambda_1) x e^{\lambda_1 x} \end{pmatrix} \\
&= e^{2\lambda_1 x} \det \begin{pmatrix} C_1(\lambda_1) & \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} \\ C_2(\lambda_1) & \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} \end{pmatrix}. \tag{6.82}
\end{aligned}$$

If it is valid that $\det \begin{pmatrix} C_1(\lambda_1) & \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} \\ C_2(\lambda_1) & \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} \end{pmatrix} \neq 0$, then the set

$$\left\{ \begin{pmatrix} C_1(\lambda_1) e^{\lambda_1 x} \\ C_2(\lambda_1) e^{\lambda_1 x} \end{pmatrix}, \begin{pmatrix} \left[\frac{\partial}{\partial \lambda} (C_1(\lambda) e^{\lambda x}) \right]_{\lambda=\lambda_1} \\ \left[\frac{\partial}{\partial \lambda} (C_2(\lambda) e^{\lambda x}) \right]_{\lambda=\lambda_1} \end{pmatrix} = \begin{pmatrix} \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} + C_1(\lambda_1) x e^{\lambda_1 x} \\ \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda_1 x} + C_2(\lambda_1) x e^{\lambda_1 x} \end{pmatrix} \right\} \text{ is the fundamental}$$

system of solutions and the general solution for $\lambda_1 = \lambda_2$ is given as follows:

$$\begin{pmatrix} w(x) \\ v(x) \end{pmatrix} = A \begin{pmatrix} C_1(\lambda_1) e^{\lambda_1 x} \\ C_2(\lambda_1) e^{\lambda_1 x} \end{pmatrix} + B \begin{pmatrix} \left[\frac{d}{d\lambda} (C_1(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda x} + C_1(\lambda_1) x e^{\lambda_1 x} \\ \left[\frac{d}{d\lambda} (C_2(\lambda)) \right]_{\lambda=\lambda_1} e^{\lambda x} + C_2(\lambda_1) x e^{\lambda_1 x} \end{pmatrix}, \tag{6.83}$$

where A and B are arbitrary constant.

6.5 Some Practical Examples of Simultaneous Linear Constant Coefficient Ordinary Differential Equations

In the present subsection, some practical examples of simultaneous linear constant coefficient ordinary differential equations for both homogeneous case and inhomogeneous case. The below examples 6.1, 6.2, 6.3, 6.4 and 6.5 have been introduced in Chapter 6 of Ref. [6] as some examples and practices.

Example 6.1. Find each solution of the homogeneous second order linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx} w_1(x) = 12w_1(x) - 17w_2(x) + w_3(x), \\ \frac{d}{dx} w_2(x) = 7w_1(x) - 10w_2(x) + w_3(x), \\ \frac{d}{dx} w_3(x) = 5w_1(x) - 9w_2(x) + 4w_3(x). \end{cases} \tag{6.84}$$

(Solution) First we set the solutions of the present simultaneous linear differential equations by

$$\begin{cases} w_1(x) = A_1(\lambda) e^{\lambda x}, \\ w_2(x) = A_2(\lambda) e^{\lambda x}, \\ w_3(x) = A_3(\lambda) e^{\lambda x}, \end{cases} \tag{6.85}$$

and substitute them to Eq.(6.84) as follows:

$$\begin{cases} A_1(\lambda)\lambda e^{\lambda x} = 12A_1(\lambda)e^{\lambda x} - 17A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x}, \\ A_2(\lambda)\lambda e^{\lambda x} = 7A_1(\lambda)e^{\lambda x} - 10A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x}, \\ A_3(\lambda)\lambda e^{\lambda x} = 5A_1(\lambda)e^{\lambda x} - 9A_2(\lambda)e^{\lambda x} + 4A_3(\lambda)e^{\lambda x}, \end{cases} \quad (6.86)$$

$$\begin{cases} (12 - \lambda)A_1(\lambda)e^{\lambda x} - 17A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x} = 0, \\ 7A_1(\lambda)e^{\lambda x} + (-10 - \lambda)A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x} = 0, \\ 5A_1(\lambda)e^{\lambda x} - 9A_2(\lambda)e^{\lambda x} + (4 - \lambda)A_3(\lambda)e^{\lambda x} = 0. \end{cases} \quad (6.87)$$

$$\begin{pmatrix} 12 - \lambda & -17 & 1 \\ 7 & -10 - \lambda & 1 \\ 5 & -9 & 4 - \lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda)e^{\lambda x} \\ A_2(\lambda)e^{\lambda x} \\ A_3(\lambda)e^{\lambda x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad (6.88)$$

$$\begin{pmatrix} 12 - \lambda & -17 & 1 \\ 7 & -10 - \lambda & 1 \\ 5 & -9 & 4 - \lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (6.89)$$

Here we remark that $e^{\lambda x}$ is always non zero for any real number x in the interval $(-\infty, +\infty)$ The condition that the homogeneous simultaneous linear equations (6.89) for $\begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix}$ has non-trivial solutions is given as follows:

$$\det \begin{pmatrix} 12 - \lambda & -17 & 1 \\ 7 & -10 - \lambda & 1 \\ 5 & -9 & 4 - \lambda \end{pmatrix} = 0, \quad (6.90)$$

$$\det \begin{pmatrix} 12 - \lambda & -17 & 1 \\ 7 - (12 - \lambda) & -10 - \lambda - (-17) & 1 - 1 \\ 5 - (12 - \lambda)(4 - \lambda) & -9 - (-17)(4 - \lambda) & 4 - \lambda - (4 - \lambda) \end{pmatrix} = 0, \quad (6.91)$$

$$\det \begin{pmatrix} 12 - \lambda & -17 & 1 \\ -5 + \lambda & 7 - \lambda & 0 \\ -43 + 16\lambda - \lambda^2 & 59 - 17\lambda & 0 \end{pmatrix} = 0, \quad (6.92)$$

$$\det \begin{pmatrix} -5 + \lambda & 7 - \lambda \\ -43 + 16\lambda - \lambda^2 & 59 - 17\lambda \end{pmatrix} = 0, \quad (6.93)$$

$$(-5 + \lambda)(59 - 17\lambda) - (7 - \lambda)(-43 + 16\lambda - \lambda^2) = 0, \quad (6.94)$$

$$-295 + 144\lambda - 17\lambda^2 + 301 - 112\lambda + 7\lambda^2 - 43\lambda + 16\lambda^2 - \lambda^3 = 0, \quad (6.95)$$

$$6 - 11\lambda + 6\lambda^2 - \lambda^3 = -(\lambda - 1)(\lambda^2 - 6\lambda + 5) = -(\lambda - 1)(\lambda - 2)(\lambda - 3) = 0, \quad (6.96)$$

$$\lambda = 1, 2, 3. \quad (6.97)$$

For each solution, $\lambda = 1$, $\lambda = 2$ and $\lambda = 3$, the simultaneous linear equation (6.89) can be written down to

$$\begin{cases} 11A_1(1) - 17A_2(1) + A_3(1) = 0 \\ 7A_1(1) - 11A_2(1) + A_3(1) = 0 \\ 5A_1(1) - 9A_2(1) + 3A_3(1) = 0 \end{cases} \rightarrow \begin{cases} A_2(1) = \frac{2}{3}A_1(1), \\ A_3(1) = \frac{1}{3}A_1(1), \end{cases} \quad (6.98)$$

$$\begin{cases} 10A_1(2) - 17A_2(2) + A_3(2) = 0, \\ 7A_1(2) - 12A_2(2) + A_3(2) = 0, \\ 5A_1(2) - 9A_2(2) + 2A_3(2) = 0, \end{cases} \rightarrow \begin{cases} A_2(2) = \frac{3}{5}A_1(2), \\ A_3(2) = \frac{1}{5}A_1(2), \end{cases} \quad (6.99)$$

$$\begin{cases} 9A_1(3) - 17A_2(3) + A_3(3) = 0 \\ 7A_1(3) - 13A_2(3) + A_3(3) = 0 \\ 5A_1(3) - 9A_2(3) + A_3(3) = 0 \end{cases} \rightarrow \begin{cases} A_2(3) = \frac{1}{2}A_1(3), \\ A_3(3) = -\frac{1}{2}A_1(3), \end{cases} \quad (6.100)$$

and then their corresponding solutions are derived as

$$\begin{pmatrix} w_1^{(1)}(x) \\ w_2^{(1)}(x) \\ w_3^{(1)}(x) \end{pmatrix} = \begin{pmatrix} A_1(1)e^x \\ \frac{2}{3}A_1(1)e^x \\ \frac{1}{3}A_1(1)e^x \end{pmatrix} = \frac{1}{3}A^{(1)}(1) \begin{pmatrix} 3e^x \\ 2e^x \\ e^x \end{pmatrix}, \quad (6.101)$$

$$\begin{pmatrix} w_1^{(2)}(x) \\ w_2^{(2)}(x) \\ w_3^{(2)}(x) \end{pmatrix} = \begin{pmatrix} A_1(2)e^{2x} \\ \frac{3}{5}A_1(2)e^{2x} \\ -\frac{1}{5}A_1(2)e^{2x} \end{pmatrix} = \frac{1}{5}A^{(1)}(2) \begin{pmatrix} 5e^{2x} \\ 3e^{2x} \\ -e^{2x} \end{pmatrix}, \quad (6.102)$$

$$\begin{pmatrix} w_1^{(3)}(x) \\ w_2^{(3)}(x) \\ w_3^{(3)}(x) \end{pmatrix} = \begin{pmatrix} A_1(3)e^{3x} \\ \frac{1}{2}A_1(3)e^{3x} \\ -\frac{1}{2}A_1(3)e^{3x} \end{pmatrix} = \frac{1}{2}A^{(1)}(3) \begin{pmatrix} 2e^{3x} \\ e^{3x} \\ -e^{3x} \end{pmatrix}, \quad (6.103)$$

respectively. We have

$$\begin{aligned} \det \begin{pmatrix} 3e^x & 5e^{2x} & 2e^{3x} \\ 2e^x & 3e^{2x} & e^{3x} \\ e^x & -e^{2x} & -e^{3x} \end{pmatrix} &= \det \begin{pmatrix} 5e^x & 3e^{2x} & 0 \\ 3e^x & 2e^{2x} & 0 \\ e^x & -e^{2x} & -e^{3x} \end{pmatrix} \\ &= -e^{3x} \det \begin{pmatrix} 5e^x & 3e^{2x} \\ 3e^x & 2e^{2x} \end{pmatrix} = -e^{3x}(10e^{3x} - 9e^{3x}) = -e^{6x} \neq 0, \end{aligned} \quad (6.104)$$

and then the above three solutions $\begin{pmatrix} 3e^x \\ 2e^x \\ e^x \end{pmatrix}$, $\begin{pmatrix} 5e^{2x} \\ 3e^{2x} \\ -e^{2x} \end{pmatrix}$ and $\begin{pmatrix} 2e^{3x} \\ e^{3x} \\ -e^{3x} \end{pmatrix}$ are linearly independent of each other and are a fundamental system of solutions. The general solution of the present homogeneous simultaneous linear differential equations is given as follows:

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = C_1 \begin{pmatrix} 3e^x \\ 2e^x \\ e^x \end{pmatrix} + C_2 \begin{pmatrix} 5e^{2x} \\ 3e^{2x} \\ -e^{2x} \end{pmatrix} + C_3 \begin{pmatrix} 2e^{3x} \\ e^{3x} \\ -e^{3x} \end{pmatrix} \quad (\forall x \in (-\infty, +\infty)). \quad (6.105)$$

Example 6.2. Find each solution of the homogeneous second order linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx} w_1(x) = 7w_1(x) - 8w_2(x) + 2w_3(x), \\ \frac{d}{dx} w_2(x) = 3w_1(x) - 4w_2(x), \\ \frac{d}{dx} w_3(x) = -4w_1(x) + 5w_2(x) - w_3(x). \end{cases} \quad (6.106)$$

(Solution) First we set the solutions of the present simultaneous linear differential equations by

$$\begin{cases} w_1(x) = A_1(\lambda)e^{\lambda x}, \\ w_2(x) = A_2(\lambda)e^{\lambda x}, \\ w_3(x) = A_3(\lambda)e^{\lambda x}, \end{cases} \quad (6.107)$$

and substitute them to Eq.(6.106) as follows:

$$\begin{cases} A_1(\lambda)\lambda e^{\lambda x} = 7A_1(\lambda)e^{\lambda x} - 8A_2(\lambda)e^{\lambda x} + 2A_3(\lambda)e^{\lambda x}, \\ A_2(\lambda)\lambda e^{\lambda x} = 3A_1(\lambda)e^{\lambda x} - 4A_2(\lambda)e^{\lambda x}, \\ A_3(\lambda)\lambda e^{\lambda x} = -4A_1(\lambda)e^{\lambda x} + 5A_2(\lambda)e^{\lambda x} - A_3(\lambda)e^{\lambda x}, \end{cases} \quad (6.108)$$

$$\begin{cases} (7 - \lambda)A_1(\lambda)e^{\lambda x} - 8A_2(\lambda)e^{\lambda x} + 2A_3(\lambda)e^{\lambda x} = 0, \\ 3A_1(\lambda)e^{\lambda x} + (-4 - \lambda)A_2(\lambda)e^{\lambda x} = 0, \\ -4A_1(\lambda)e^{\lambda x} + 5A_2(\lambda)e^{\lambda x} + (-1 - \lambda)A_3(\lambda)e^{\lambda x} = 0. \end{cases} \quad (6.109)$$

$$\begin{pmatrix} 7-\lambda & -8 & 2 \\ 3 & -4-\lambda & 0 \\ -4 & 5 & -1-\lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda)e^{\lambda x} \\ A_2(\lambda)e^{\lambda x} \\ A_3(\lambda)e^{\lambda x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad (6.110)$$

$$\begin{pmatrix} 7-\lambda & -8 & 2 \\ 3 & -4-\lambda & 0 \\ -4 & 5 & -1-\lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (6.111)$$

Here we remark that $e^{\lambda x}$ is always non zero for any real number x in the interval $(-\infty, +\infty)$ The condition that the homogeneous simultaneous linear equations (6.111) for $\begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix}$ has non-trivial solutions is given as follows:

$$\det \begin{pmatrix} 7-\lambda & -8 & 2 \\ 3 & -4-\lambda & 0 \\ -4 & 5 & -1-\lambda \end{pmatrix} = 0, \quad (6.112)$$

$$\det \begin{pmatrix} 7-\lambda & -8 & 2 \\ 3 & -4-\lambda & 0 \\ -8 & 10 & 2(-1-\lambda) \end{pmatrix} = 0, \quad (6.113)$$

$$\det \begin{pmatrix} 7-\lambda & -8 & 2 \\ 3 & -4-\lambda & 0 \\ -8-(7-\lambda)(-1-\lambda) & 10+8(-1-\lambda) & 0 \end{pmatrix} = 0, \quad (6.114)$$

$$\det \begin{pmatrix} 7-\lambda & -8 & 2 \\ 3 & -4-\lambda & 0 \\ -1+6\lambda+\lambda^2 & 2-8\lambda & 0 \end{pmatrix} = 0, \quad (6.115)$$

$$2\det \begin{pmatrix} 3 & -4-\lambda \\ -1+6\lambda+\lambda^2 & 2-8\lambda \end{pmatrix} = 0, \quad (6.116)$$

$$2(3(2-8\lambda) + (4+\lambda)(-1+6\lambda-\lambda^2)) = 0, \quad (6.117)$$

$$2(6-24\lambda-4+24\lambda-4\lambda^2-\lambda+6\lambda^2-\lambda^3) = 0, \quad (6.118)$$

$$2(2-\lambda+2\lambda^2-\lambda^3) = 0, \quad (6.119)$$

$$-2(\lambda-2)(\lambda^2+1) = 0, \quad (6.120)$$

$$\lambda = 2, \pm i. \quad (6.121)$$

For each solution, $\lambda = +2$, $\lambda = +i$ and $\lambda = -i$, the simultaneous linear equation (6.111) can be written down to

$$\begin{cases} 5A_1(+2) - 8A_2(+2) + 2A_3(+2) = 0, \\ 3A_1(+2) - 6A_2(+2) = 0, \\ -4A_1(+2) + 5A_2(+2) - 3A_3(+2) = 0. \end{cases} \rightarrow \begin{cases} A_2(+2) = \frac{1}{2}A_1(+2), \\ A_3(+2) = -\frac{1}{2}A_1(+2), \end{cases} \quad (6.122)$$

$$\begin{cases} (7-i)A_1(+i) - 8A_2(+i) + 2A_3(+i) = 0, \\ 3A_1(+i) + (-4-i)A_2(+i) = 0, \\ -4A_1(+i) + 5A_2(+i) + (-1-i)A_3(+i) = 0. \end{cases} \rightarrow \begin{cases} A_2(+i) = \frac{3(4-i)}{17}A_1(+i), \\ A_3(+i) = \frac{-23-7i}{34}A_1(+i), \end{cases} \quad (6.123)$$

$$\begin{cases} (7+i)A_1(-i) - 8A_2(-i) + 2A_3(-i) = 0, \\ 3A_1(-i) + (-4+i)A_2(-i) = 0, \\ -4A_1(-i) + 5A_2(-i) + (-1+i)A_3(-i) = 0. \end{cases} \rightarrow \begin{cases} A_2(3) = \frac{3(4+i)}{17}A_1(3), \\ A_3(3) = \frac{-23+7i}{34}A_1(3), \end{cases} \quad (6.124)$$

and then their corresponding solutions are derived as

$$\begin{pmatrix} w_1^{(+2)}(x) \\ w_2^{(+2)}(x) \\ w_3^{(+2)}(x) \end{pmatrix} = \begin{pmatrix} A_1(1)e^{2x} \\ \frac{1}{2}A_1(1)e^{2x} \\ -\frac{1}{2}A_1(1)e^{2x} \end{pmatrix} = \frac{1}{2}A^{(1)}(1) \begin{pmatrix} 2e^{+2x} \\ e^{+2x} \\ -e^{+2x} \end{pmatrix}, \quad (6.125)$$

$$\begin{aligned} \begin{pmatrix} w_1^{(+i)}(x) \\ w_2^{(+i)}(x) \\ w_3^{(+i)}(x) \end{pmatrix} &= \begin{pmatrix} A_1(+i)e^{+ix} \\ \frac{3(4+i)}{17}A_1(+i)e^{+ix} \\ \frac{-23-7i}{34}A_1(+i)e^{+ix} \end{pmatrix} = \frac{1}{34}A^{(1)}(+i) \begin{pmatrix} 34e^{+ix} \\ 6(4-i)e^{+ix} \\ (-23-7i)e^{+ix} \end{pmatrix} \\ &= \frac{1}{34}A^{(1)}(+i) \begin{pmatrix} 34(\cos(x) + i\sin(x)) \\ 6(4-i)(\cos(x) + i\sin(x)) \\ (-23-7i)(\cos(x) + i\sin(x)) \end{pmatrix} \\ &= \frac{1}{34}A^{(1)}(+i) \begin{pmatrix} 34(\cos(x) + i\sin(x)) \\ 6(4-i)(\cos(x) + i\sin(x)) \\ (-23-7i)(\cos(x) + i\sin(x)) \end{pmatrix}. \end{aligned} \quad (6.126)$$

$$\begin{aligned} \begin{pmatrix} w_1^{(-i)}(x) \\ w_2^{(-i)}(x) \\ w_3^{(-i)}(x) \end{pmatrix} &= \begin{pmatrix} A_1(-i)e^{-ix} \\ \frac{3(4+i)}{17}A_1(-i)e^{-ix} \\ \frac{-23+7i}{34}A_1(-i)e^{-ix} \end{pmatrix} = \frac{1}{34}A^{(1)}(-i) \begin{pmatrix} 34e^{-ix} \\ 6(4+i)e^{-ix} \\ (-23+7i)e^{-ix} \end{pmatrix} \\ &= \frac{1}{34}A^{(1)}(-i) \begin{pmatrix} 34(\cos(x) - i\sin(x)) \\ 6(4+i)(\cos(x) - i\sin(x)) \\ (-23+7i)(\cos(x) - i\sin(x)) \end{pmatrix}. \end{aligned} \quad (6.127)$$

respectively. From Eqs.(6.126) and (6.127), we see that the following two functions are also solutions of Eq.(6.106)

$$\begin{aligned} \frac{1}{2} \begin{pmatrix} 34(\cos(x) + i\sin(x)) \\ 6(4-i)(\cos(x) + i\sin(x)) \\ (-23-7i)(\cos(x) + i\sin(x)) \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 34(\cos(x) - i\sin(x)) \\ 6(4+i)(\cos(x) - i\sin(x)) \\ (-23+7i)(\cos(x) - i\sin(x)) \end{pmatrix} \\ = \begin{pmatrix} 34\cos(x) \\ 24\cos(x) + 6\sin(x) \\ -23\cos(x) + 7\sin(x) \end{pmatrix} \end{aligned} \quad (6.128)$$

$$\begin{aligned} \frac{1}{2}i \begin{pmatrix} 34(\cos(x) + i\sin(x)) \\ 6(4-i)(\cos(x) + i\sin(x)) \\ (-23-7i)(\cos(x) + i\sin(x)) \end{pmatrix} - \frac{1}{2}i \begin{pmatrix} 34(\cos(x) - i\sin(x)) \\ 6(4+i)(\cos(x) - i\sin(x)) \\ (-23+7i)(\cos(x) - i\sin(x)) \end{pmatrix} \\ = \begin{pmatrix} -34\sin(x) \\ 6\cos(x) - 24\sin(x) \\ 7\cos(x) + 23\sin(x) \end{pmatrix} \end{aligned} \quad (6.129)$$

We have

$$\begin{aligned} &\det \begin{pmatrix} 2e^{2x} & 34\cos(x) & -34\sin(x) \\ e^{2x} & 24\cos(x) + 6\sin(x) & 6\cos(x) - 24\sin(x) \\ -e^{2x} & -23\cos(x) + 7\sin(x) & 7\cos(x) + 23\sin(x) \end{pmatrix} \\ &= 2\det \begin{pmatrix} e^{2x} & 17\cos(x) & -17\sin(x) \\ e^{2x} & 24\cos(x) + 6\sin(x) & 6\cos(x) - 24\sin(x) \\ -e^{2x} & -23\cos(x) + 7\sin(x) & 7\cos(x) + 23\sin(x) \end{pmatrix} \\ &= 2\det \begin{pmatrix} e^{2x} & 17\cos(x) & -17\cos(x) \\ 0 & 7\cos(x) + 6\sin(x) & 6\cos(x) - 7\sin(x) \\ 0 & -6\cos(x) + 7\sin(x) & 7\cos(x) + 6\sin(x) \end{pmatrix} \\ &= 2e^{2x}\det \begin{pmatrix} 7\cos(x) + 6\sin(x) & 6\cos(x) - 7\sin(x) \\ -6\cos(x) + 7\sin(x) & 7\cos(x) + 6\sin(x) \end{pmatrix} \\ &= 2e^{2x}((7\cos(x) + 6\sin(x))(7\cos(x) + 6\sin(x))) \end{aligned}$$

$$\begin{aligned}
& - (6\cos(x) - 7\sin(x))(-6\cos(x) + 7\sin(x)) \\
& = 2e^{2x} \times 85(\cos^2(x) + \sin^2(x)) = 170e^{2x} \neq 0,
\end{aligned} \tag{6.130}$$

and then the above three solutions $\begin{pmatrix} 2e^{2x} \\ e^{2x} \\ -e^{2x} \end{pmatrix}$, $\begin{pmatrix} 34\cos(x) \\ 24\cos(x) + 6\sin(x) \\ -23\cos(x) + 7\sin(x) \end{pmatrix}$ and $\begin{pmatrix} -34\sin(x) \\ 6\cos(x) - 24\sin(x) \\ 7\cos(x) + 23\sin(x) \end{pmatrix}$ are linearly independent of each other and are a fundamental system of solutions. The general solution of the present homogeneous simultaneous linear differential equations is given as follows:

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = C_1 \begin{pmatrix} 2e^{2x} \\ e^{2x} \\ -e^{2x} \end{pmatrix} + C_2 \begin{pmatrix} 34\cos(x) \\ 24\cos(x) + 6\sin(x) \\ -23\cos(x) + 7\sin(x) \end{pmatrix} + C_3 \begin{pmatrix} -34\sin(x) \\ 6\cos(x) - 24\sin(x) \\ 7\cos(x) + 23\sin(x) \end{pmatrix} \quad (\forall x \in (-\infty, +\infty)), \tag{6.131}$$

where C_1, C_2 and C_3 are arbitrary constants.

Example 6.3. Find each solution of the homogeneous second order linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx}w_1(x) = 3w_1(x) - w_2(x) + w_3(x), \\ \frac{d}{dx}w_2(x) = w_1(x) + w_2(x) + w_3(x), \\ \frac{d}{dx}w_3(x) = -w_1(x) + w_2(x) + 2w_3(x). \end{cases} \tag{6.132}$$

(Solution) First we set the solutions of the present simultaneous linear differential equations by

$$\begin{cases} w_1(x) = A_1(\lambda)e^{\lambda x}, \\ w_2(x) = A_2(\lambda)e^{\lambda x}, \\ w_3(x) = A_3(\lambda)e^{\lambda x}, \end{cases} \tag{6.133}$$

and substitute them to Eq.(6.132) as follows:

$$\begin{cases} A_1(\lambda)\lambda e^{\lambda x} = 3A_1(\lambda)e^{\lambda x} - A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x}, \\ A_2(\lambda)\lambda e^{\lambda x} = A_1(\lambda)e^{\lambda x} + A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x}, \\ A_3(\lambda)\lambda e^{\lambda x} = -A_1(\lambda)e^{\lambda x} + A_2(\lambda)e^{\lambda x} + 2A_3(\lambda)e^{\lambda x}, \end{cases} \tag{6.134}$$

$$\begin{cases} (3 - \lambda)A_1(\lambda)e^{\lambda x} - A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x} = 0, \\ A_1(\lambda)e^{\lambda x} + (1 - \lambda)A_2(\lambda)e^{\lambda x} + A_3(\lambda)e^{\lambda x} = 0, \\ -A_1(\lambda)e^{\lambda x} + A_2(\lambda)e^{\lambda x} + (2 - \lambda)A_3(\lambda)e^{\lambda x} = 0. \end{cases} \tag{6.135}$$

$$\begin{pmatrix} 3 - \lambda & -1 & 1 \\ 1 & 1 - \lambda & 1 \\ -1 & 1 & 2 - \lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda)e^{\lambda x} \\ A_2(\lambda)e^{\lambda x} \\ A_3(\lambda)e^{\lambda x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \tag{6.136}$$

$$\begin{pmatrix} 3 - \lambda & -1 & 1 \\ 1 & 1 - \lambda & 1 \\ -1 & 1 & 2 - \lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \tag{6.137}$$

Here we remark that $e^{\lambda x}$ is always non zero for any real number x in the interval $(-\infty, +\infty)$. The condition that the homogeneous simultaneous linear equations (6.137) for $\begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix}$ has non-trivial solutions is given as follows:

$$\det \begin{pmatrix} 3 - \lambda & -1 & 1 \\ 1 & 1 - \lambda & 1 \\ -1 & 1 & 2 - \lambda \end{pmatrix} = 0, \tag{6.138}$$

$$\det \begin{pmatrix} 0 & -(2 - \lambda)^2 & -2 + \lambda \\ 1 & 1 - \lambda & 1 \\ 0 & 2 - \lambda & 3 - \lambda \end{pmatrix} = 0, \tag{6.139}$$

$$-\det \begin{pmatrix} -(2-\lambda)^2 & -2+\lambda \\ 2-\lambda & 3-\lambda \end{pmatrix} = 0, \quad (6.140)$$

$$-(2-\lambda)^2(3-\lambda) + (2-\lambda)^2 = 0, \quad (6.141)$$

$$(2-\lambda)^2(-(3-\lambda)+1) = 0, \quad (6.142)$$

$$-(2-\lambda)^3 = 0, \quad (6.143)$$

$$\lambda = 2 \text{ (TripleDegeneracy)}. \quad (6.144)$$

In this case, we see that the solution of Eq.(6.138) have a triple degeneracy. In such cases, it is known that we can obtain the general solution of Eq.(6.132) by setting the solution as

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = \begin{pmatrix} \Phi_1(x; \lambda = 2) \\ \Phi_2(x; \lambda = 2) \\ \Phi_3(x; \lambda = 2) \end{pmatrix}, \quad (6.145)$$

where

$$\begin{aligned} \begin{pmatrix} \Phi_1(x; \lambda) \\ \Phi_2(x; \lambda) \\ \Phi_3(x; \lambda) \end{pmatrix} &\equiv \frac{\partial^2}{\partial \lambda^2} \begin{pmatrix} A_1(\lambda) e^{\lambda x} \\ A_2(\lambda) e^{\lambda x} \\ A_3(\lambda) e^{\lambda x} \end{pmatrix} \\ &= \begin{pmatrix} \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) x e^{\lambda x} + A_1(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) x e^{\lambda x} + A_2(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) x e^{\lambda x} + A_3(\lambda) x^2 e^{\lambda x} \end{pmatrix}, \end{aligned} \quad (6.146)$$

and by determining the coefficients $\left[\frac{d^2}{d\lambda^2} A_i(\lambda) \right]_{\lambda=2}$, $\left[\frac{d}{d\lambda} A_i(\lambda) \right]_{\lambda=2}$ and $A_i(\lambda = 2)$ ($i = 1, 2, 3$) so as to satisfy Eq.(6.132) as follows:

$$\left[\frac{\partial}{\partial x} \begin{pmatrix} \Phi_1(x; \lambda) \\ \Phi_2(x; \lambda) \\ \Phi_3(x; \lambda) \end{pmatrix} - \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} \Phi_1(x; \lambda) \\ \Phi_2(x; \lambda) \\ \Phi_3(x; \lambda) \end{pmatrix} \right]_{\lambda=2} = 0 \quad (6.147)$$

$$\begin{aligned} &\left[\frac{\partial}{\partial x} \begin{pmatrix} \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) x e^{\lambda x} + A_1(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) x e^{\lambda x} + A_2(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) x e^{\lambda x} + A_3(\lambda) x^2 e^{\lambda x} \end{pmatrix} \right. \\ &\quad \left. - \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) x e^{\lambda x} + A_1(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) x e^{\lambda x} + A_2(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) x e^{\lambda x} + A_3(\lambda) x^2 e^{\lambda x} \end{pmatrix} \right]_{\lambda=2} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned} \quad (6.148)$$

$$\begin{aligned} &\left[\begin{pmatrix} \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) \lambda e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) \lambda x e^{\lambda x} + 2 A_1(\lambda) x e^{\lambda x} + A_1(\lambda) \lambda x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) \lambda e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) \lambda x e^{\lambda x} + 2 A_2(\lambda) x e^{\lambda x} + A_2(\lambda) \lambda x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) \lambda e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) \lambda x e^{\lambda x} + 2 A_3(\lambda) x e^{\lambda x} + A_3(\lambda) \lambda x^2 e^{\lambda x} \end{pmatrix} \right. \\ &\quad \left. - \begin{pmatrix} 3 & -1 & 1 \\ 1 & 1 & 1 \\ -1 & 1 & 2 \end{pmatrix} \begin{pmatrix} \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) x e^{\lambda x} + A_1(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) x e^{\lambda x} + A_2(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) x e^{\lambda x} + A_3(\lambda) x^2 e^{\lambda x} \end{pmatrix} \right]_{\lambda=2} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \end{aligned} \quad (6.149)$$

$$\left\{ \begin{array}{l} \left[\left(\left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) (\lambda - 3) + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) + \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) - \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) \right) e^{\lambda x} \right. \\ \quad + \left(2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) (\lambda - 3) + 2 A_1(\lambda) + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) - 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) \right) x e^{\lambda x} \\ \quad \left. + (A_1(\lambda)(\lambda - 3) + A_2(\lambda) - A_3(\lambda)) x^2 e^{\lambda x} \right]_{\lambda=2} = 0, \\ \left[\left(- \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) + \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) (\lambda - 1) - \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) \right) e^{\lambda x} \right. \\ \quad + \left(-2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) (\lambda - 1) + 2 A_2(\lambda) - 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) \right) x e^{\lambda x} \\ \quad \left. + (-A_1(\lambda) + A_2(\lambda)(\lambda - 1) - A_3(\lambda)) x^2 e^{\lambda x} \right]_{\lambda=2} = 0, \\ \left[\left(2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) + \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) - \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) + \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) (\lambda - 2) \right) e^{\lambda x} \right. \\ \quad + \left(2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) - 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) (\lambda - 2) + 2 A_3(\lambda) \right) x e^{\lambda x} \\ \quad \left. + (A_1(\lambda) - A_2(\lambda) + A_3(\lambda)(\lambda - 2)) x^2 e^{\lambda x} \right]_{\lambda=2} = 0, \end{array} \right. \quad (6.150)$$

so that we have

$$\left\{ \begin{array}{l} (-\alpha_1 + \alpha_2 - \alpha_3 + \beta_1) e^{2x} + (-\beta_1 + \beta_2 - \beta_3 + 2\gamma_1) x e^{2x} + (-\gamma_1 + \gamma_2 - \gamma_3) x^2 e^{2x} = 0, \\ (-\alpha_1 + \alpha_2 - \alpha_3 + \beta_2) e^{2x} + (-\beta_1 + \beta_2 - \beta_3 + 2\gamma_2) x e^{2x} + (-\gamma_1 + \gamma_2 - \gamma_3) x^2 e^{2x} = 0, \\ (\alpha_1 - \alpha_2 + \beta_3) e^{2x} + (\beta_1 - \beta_2 + 2\gamma_3) x e^{2x} + (\gamma_1 - \gamma_2) x^2 e^{2x} = 0, \end{array} \right. \quad (6.151)$$

where

$$\left\{ \begin{array}{l} \alpha_1 \equiv \left[\frac{d^2}{d\lambda^2} A_1(\lambda) \right]_{\lambda=2}, \quad \beta_1 \equiv \left[\frac{d}{d\lambda} A_1(\lambda) \right]_{\lambda=2}, \quad \gamma_1 \equiv A_1(2), \\ \alpha_2 \equiv \left[\frac{d^2}{d\lambda^2} A_2(\lambda) \right]_{\lambda=2}, \quad \beta_2 \equiv \left[\frac{d}{d\lambda} A_2(\lambda) \right]_{\lambda=2}, \quad \gamma_2 \equiv A_2(2), \\ \alpha_3 \equiv \left[\frac{d^2}{d\lambda^2} A_3(\lambda) \right]_{\lambda=2}, \quad \beta_3 \equiv \left[\frac{d}{d\lambda} A_3(\lambda) \right]_{\lambda=2}, \quad \gamma_3 \equiv A_3(2). \end{array} \right. \quad (6.152)$$

Here we remark that e^x , $x e^x$ and $x^2 e^x$ are independent of each other for any real number x in the interval $(-\infty, +\infty)$, because of

$$\begin{aligned} W[e^x, x e^x, x^2 e^x] &= \det \begin{pmatrix} e^{2x} & \frac{d}{dx} e^{2x} & \frac{d^2}{dx^2} e^{2x} \\ x e^{2x} & \frac{d}{dx} (x e^{2x}) & \frac{d^2}{dx^2} (x e^{2x}) \\ x^2 e^{2x} & \frac{d}{dx} (x^2 e^{2x}) & \frac{d^2}{dx^2} (x^2 e^{2x}) \end{pmatrix} \\ &= \det \begin{pmatrix} e^{2x} & 2e^{2x} & 4e^{2x} \\ x e^{2x} & (2x+1)e^{2x} & (4x+4)e^{2x} \\ x^2 e^{2x} & (2x^2+2x)e^{2x} & (4x^2+8x+2)e^{2x} \end{pmatrix} \\ &= \det \begin{pmatrix} e^{2x} & 0 & 0 \\ x e^{2x} & e^{2x} & 4e^{2x} \\ x^2 e^{2x} & 2x e^{2x} & (8x+2)e^{2x} \end{pmatrix} \\ &= \det \begin{pmatrix} e^{2x} & 0 & 0 \\ x e^{2x} & e^{2x} & 0 \\ x^2 e^{2x} & 2x e^{2x} & 2e^{2x} \end{pmatrix} = 2e^{6x} \neq 0 \quad (\forall x \in (-\infty, +\infty)). \end{aligned} \quad (6.153)$$

Sufficient conditions for identity (6.151) can be given as

$$\left\{ \begin{array}{l} -\alpha_1 + \alpha_2 - \alpha_3 + \beta_1 = 0, \quad -\beta_1 + \beta_2 - \beta_3 + 2\gamma_1 = 0, \quad -\gamma_1 + \gamma_2 - \gamma_3 = 0, \\ -\alpha_1 + \alpha_2 - \alpha_3 + \beta_2 = 0, \quad -\beta_1 + \beta_2 - \beta_3 + 2\gamma_2 = 0, \quad -\gamma_1 + \gamma_2 - \gamma_3 = 0, \\ \alpha_1 - \alpha_2 + \beta_3 = 0, \quad \beta_1 - \beta_2 + 2\gamma_3 = 0, \quad \gamma_1 - \gamma_2 = 0, \end{array} \right. \quad (6.154)$$

and then

$$\left\{ \begin{array}{l} \alpha_2 = c_1 + 2c_3, \quad \beta_2 = c_2, \quad \gamma_2 = c_3, \\ \alpha_3 = c_2 + 2c_3, \quad \beta_3 = 2c_3, \quad \gamma_3 = 0, \end{array} \right. \quad (6.155)$$

where

$$c_1 \equiv \alpha_1, \quad c_2 \equiv \beta_1, \quad c_3 \equiv \gamma_1. \quad (6.156)$$

Here c_1 , c_2 and c_3 are arbitrary constants. The general solution is given by

$$\begin{aligned}
 \begin{pmatrix} \Phi_1(x; \lambda = 2) \\ \Phi_2(x; \lambda = 2) \\ \Phi_3(x; \lambda = 2) \end{pmatrix} &= \left[\begin{pmatrix} \left(\frac{d^2}{d\lambda^2} A_1(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) x e^{\lambda x} + A_1(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_2(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_2(\lambda) \right) x e^{\lambda x} + A_2(\lambda) x^2 e^{\lambda x} \\ \left(\frac{d^2}{d\lambda^2} A_3(\lambda) \right) e^{\lambda x} + 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) x e^{\lambda x} + A_3(\lambda) x^2 e^{\lambda x} \end{pmatrix} \right]_{\lambda=2} \\
 &= \begin{pmatrix} \left[\frac{d^2}{d\lambda^2} A_1(\lambda) \right]_{\lambda=2} e^{2x} + 2 \left[\frac{d}{d\lambda} A_1(\lambda) \right]_{\lambda=2} x e^{2x} + A_1(2) x^2 e^{2x} \\ \left[\frac{d^2}{d\lambda^2} A_2(\lambda) \right]_{\lambda=2} e^{2x} + 2 \left[\frac{d}{d\lambda} A_2(\lambda) \right]_{\lambda=2} x e^{2x} + A_2(2) x^2 e^{2x} \\ \left[\frac{d^2}{d\lambda^2} A_3(\lambda) \right]_{\lambda=2} e^{2x} + 2 \left[\frac{d}{d\lambda} A_3(\lambda) \right]_{\lambda=2} x e^{2x} + A_3(2) x^2 e^{2x} \end{pmatrix} \\
 &= \begin{pmatrix} \alpha_1 e^{2x} + \beta_1 x e^{2x} + \gamma_1 x^2 e^{2x} \\ \alpha_2 e^{2x} + \beta_2 x e^{2x} + \gamma_2 x^2 e^{2x} \\ \alpha_3 e^{2x} + \beta_3 x e^{2x} + \gamma_3 x^2 e^{2x} \end{pmatrix} \\
 &= \begin{pmatrix} c_1 e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x} \\ (c_1 + 2c_3) e^{2x} + c_2 x e^{2x} + c_3 x^2 e^{2x} \\ (c_2 + 2c_3) e^{2x} + 2c_3 x e^{2x} \end{pmatrix} \\
 &= c_1 \begin{pmatrix} e^{2x} \\ e^{2x} \\ 0 \end{pmatrix} + c_2 \begin{pmatrix} x e^{2x} \\ x e^{2x} \\ e^{2x} \end{pmatrix} + c_3 \begin{pmatrix} x^2 e^{2x} \\ 2e^{2x} + x^2 e^{2x} \\ 2e^{2x} + 2x e^{2x} \end{pmatrix}. \tag{6.157}
 \end{aligned}$$

Because we can confirm that

$$\det \begin{pmatrix} e^{2x} & x e^{2x} & x^2 e^{2x} \\ e^{2x} & x e^{2x} & 2e^{2x} + x^2 e^{2x} \\ 0 & e^{2x} & 2e^{2x} + 2x e^{2x} \end{pmatrix} = \det \begin{pmatrix} e^{2x} & x e^{2x} & x^2 e^{2x} \\ 0 & 0 & 2e^{2x} \\ 0 & e^{2x} & 2e^{2x} + 2x e^{2x} \end{pmatrix} = -2e^{6x} \neq 0 \quad (\forall x \in (-\infty, +\infty)), \tag{6.158}$$

the solutions $\begin{pmatrix} e^{2x} \\ e^{2x} \\ 0 \end{pmatrix}$, $\begin{pmatrix} x e^{2x} \\ x e^{2x} \\ e^{2x} \end{pmatrix}$ and $\begin{pmatrix} x^2 e^{2x} \\ 2e^{2x} + x^2 e^{2x} \\ 2e^{2x} + 2x e^{2x} \end{pmatrix}$ are independent of each other and their set

$$\left\{ \begin{pmatrix} e^{2x} \\ e^{2x} \\ 0 \end{pmatrix}, \begin{pmatrix} x e^{2x} \\ x e^{2x} \\ e^{2x} \end{pmatrix}, \begin{pmatrix} x^2 e^{2x} \\ 2e^{2x} + x^2 e^{2x} \\ 2e^{2x} + 2x e^{2x} \end{pmatrix} \right\}, \tag{6.159}$$

is the fundamental systems of solutions in Eq.(6.132). We can see that Eq.(6.153) is the general solution of Eq.(6.132).

Example 6.4. Find each solution of the homogeneous second order linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx} w_1(x) = 2w_1(x) + w_2(x) + w_3(x), \\ \frac{d}{dx} w_2(x) = 2w_1(x) + 3w_2(x) + 2w_3(x), \\ \frac{d}{dx} w_3(x) = w_1(x) + w_2(x) + 2w_3(x). \end{cases} \tag{6.160}$$

(Solution) First we set the solutions of the present simultaneous linear differential equations by

$$\begin{cases} w_1(x) = A_1(\lambda) e^{\lambda x}, \\ w_2(x) = A_2(\lambda) e^{\lambda x}, \\ w_3(x) = A_3(\lambda) e^{\lambda x}, \end{cases} \tag{6.161}$$

and substitute them to Eq.(6.160) as follows:

$$\begin{cases} A_1(\lambda) \lambda e^{\lambda x} = A_1(\lambda) e^{\lambda x} + A_2(\lambda) e^{\lambda x} + 2A_3(\lambda) e^{\lambda x}, \\ A_2(\lambda) \lambda e^{\lambda x} = 2A_1(\lambda) e^{\lambda x} + 3A_2(\lambda) e^{\lambda x} + 2A_3(\lambda) e^{\lambda x}, \\ A_3(\lambda) \lambda e^{\lambda x} = A_1(\lambda) e^{\lambda x} + A_2(\lambda) e^{\lambda x} + 2A_3(\lambda) e^{\lambda x}, \end{cases} \tag{6.162}$$

$$\begin{cases} (1 - \lambda) A_1(\lambda) e^{\lambda x} + A_2(\lambda) e^{\lambda x} + A_3(\lambda) e^{\lambda x} = 0, \\ 2A_1(\lambda) e^{\lambda x} + (3 - \lambda) A_2(\lambda) e^{\lambda x} + 2A_3(\lambda) e^{\lambda x} = 0, \\ A_1(\lambda) e^{\lambda x} + A_2(\lambda) e^{\lambda x} + (2 - \lambda) A_3(\lambda) e^{\lambda x} = 0. \end{cases} \tag{6.163}$$

$$\begin{pmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda)e^{\lambda x} \\ A_2(\lambda)e^{\lambda x} \\ A_3(\lambda)e^{\lambda x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad (6.164)$$

$$\begin{pmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (6.165)$$

Here we remark that $e^{\lambda x}$ is always non zero for any real number x in the interval $(-\infty, +\infty)$. The condition that the homogeneous simultaneous linear equations (6.165) for $\begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix}$ has non-trivial solutions is given as follows:

$$\det \begin{pmatrix} 2-\lambda & 1 & 1 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{pmatrix} = 0, \quad (6.166)$$

$$\det \begin{pmatrix} 0 & 1-(2-\lambda) & 1-(2-\lambda)^2 \\ 2 & 3-\lambda & 2 \\ 1 & 1 & 2-\lambda \end{pmatrix} = 0, \quad (6.167)$$

$$\det \begin{pmatrix} 0 & 1-(2-\lambda) & 1-(2-\lambda)^2 \\ 0 & 3-\lambda-2 & 2-2(2-\lambda) \\ 1 & 1 & 2-\lambda \end{pmatrix} = 0, \quad (6.168)$$

$$\det \begin{pmatrix} 0 & -1+\lambda & (-1+\lambda)(3-\lambda) \\ 0 & 1-\lambda & 2(-1+\lambda) \\ 1 & 1 & 2-\lambda \end{pmatrix} = 0, \quad (6.169)$$

$$(\lambda-1)\det \begin{pmatrix} 0 & 1 & (-1+\lambda)(3-\lambda) \\ 0 & -1 & 2(-1+\lambda) \\ 1 & 1 & 2-\lambda \end{pmatrix} = 0, \quad (6.170)$$

$$(\lambda-1)(-1)^{3+1}\det \begin{pmatrix} 1 & (-1+\lambda)(3-\lambda) \\ -1 & 2(-1+\lambda) \end{pmatrix} = 0, \quad (6.171)$$

$$(\lambda-1)(2(-1+\lambda) + (\lambda-1)(3-\lambda)) = 0, \quad (6.172)$$

$$(\lambda-1)(-2+2\lambda-\lambda^2+4\lambda-3) = 0, \quad (6.173)$$

$$-(\lambda-1)(\lambda^2-6\lambda+5) = 0, \quad (6.174)$$

$$-(\lambda-1)^2(\lambda-5) = 0, \quad (6.175)$$

$$\begin{cases} \lambda = 5 \text{ (Non Degeneracy),} \\ \lambda = 1 \text{ (Double Degeneracy).} \end{cases} \quad (6.176)$$

For $\lambda = 5$, Eq.(6.163) can be written down as follows:

$$\begin{cases} -4A_1(+5)e^{5x} + A_2(+5)e^{5x} + A_3(+5)e^{5x} = 0, \\ 2A_1(+5)e^{5x} - 2A_2(+5)e^{5x} + 2A_3(+5)e^{5x} = 0, \\ A_1(+5)e^{5x} + A_2(+5)e^{5x} - 3A_3(+5)e^{5x} = 0, \end{cases} \rightarrow \begin{cases} A_2(+5) = 2A_1(+5), \\ A_3(+5) = A_1(+5). \end{cases} \quad (6.177)$$

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = A_1(+5) \begin{pmatrix} e^{5x} \\ 2e^{5x} \\ e^{5x} \end{pmatrix}, \quad (6.178)$$

In this case, we see that the solution of Eq.(6.166) have a triple degeneracy. In such cases, it is known that we can

obtain the general solution of Eq.(6.160) by setting the solution as

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = \begin{pmatrix} \Phi_1(x; \lambda = 1) \\ \Phi_2(x; \lambda = 1) \\ \Phi_3(x; \lambda = 1) \end{pmatrix}, \quad (6.179)$$

where

$$\begin{aligned} \begin{pmatrix} \Phi_1(x; \lambda) \\ \Phi_2(x; \lambda) \\ \Phi_3(x; \lambda) \end{pmatrix} &\equiv \frac{\partial}{\partial \lambda} \begin{pmatrix} A_1(\lambda) e^{\lambda x} \\ A_2(\lambda) e^{\lambda x} \\ A_3(\lambda) e^{\lambda x} \end{pmatrix} \\ &= \begin{pmatrix} \left(\frac{d}{d\lambda} A_1(\lambda) \right) e^{\lambda x} + A_1(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_2(\lambda) \right) e^{\lambda x} + A_2(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_3(\lambda) \right) e^{\lambda x} + A_3(\lambda) x e^{\lambda x} \end{pmatrix}, \end{aligned} \quad (6.180)$$

and by determining the coefficients $\left[\frac{d}{d\lambda} A_i(\lambda) \right]_{\lambda=1}$ and $A_i(\lambda = 1)$ ($i = 1, 2, 3$) so as to satisfy Eq.(6.160) as follows:

$$\left[\frac{\partial}{\partial x} \begin{pmatrix} \Phi_1(x; \lambda) \\ \Phi_2(x; \lambda) \\ \Phi_3(x; \lambda) \end{pmatrix} - \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} \Phi_1(x; \lambda) \\ \Phi_2(x; \lambda) \\ \Phi_3(x; \lambda) \end{pmatrix} \right]_{\lambda=1} = 0 \quad (6.181)$$

$$\begin{aligned} &\left[\frac{\partial}{\partial x} \begin{pmatrix} \left(\frac{d}{d\lambda} A_1(\lambda) \right) e^{\lambda x} + A_1(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_2(\lambda) \right) e^{\lambda x} + A_2(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_3(\lambda) \right) e^{\lambda x} + A_3(\lambda) x e^{\lambda x} \end{pmatrix} \right. \\ &\quad \left. - \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} \left(\frac{d}{d\lambda} A_1(\lambda) \right) e^{\lambda x} + A_1(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_2(\lambda) \right) e^{\lambda x} + A_2(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_3(\lambda) \right) e^{\lambda x} + A_3(\lambda) x e^{\lambda x} \end{pmatrix} \right]_{\lambda=1} = 0, \end{aligned} \quad (6.182)$$

$$\begin{aligned} &\left[\begin{pmatrix} \left(\frac{d}{d\lambda} A_1(\lambda) \right) \lambda e^{\lambda x} + A_1(\lambda) e^{\lambda x} + A_1(\lambda) \lambda x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_2(\lambda) \right) \lambda e^{\lambda x} + A_2(\lambda) e^{\lambda x} + A_2(\lambda) \lambda x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_3(\lambda) \right) \lambda e^{\lambda x} + A_3(\lambda) e^{\lambda x} + A_3(\lambda) \lambda x e^{\lambda x} \end{pmatrix} \right. \\ &\quad \left. - \begin{pmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{pmatrix} \begin{pmatrix} \left(\frac{d}{d\lambda} A_1(\lambda) \right) e^{\lambda x} + A_1(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_2(\lambda) \right) e^{\lambda x} + A_2(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_3(\lambda) \right) e^{\lambda x} + A_3(\lambda) x e^{\lambda x} \end{pmatrix} \right]_{\lambda=1} = 0, \end{aligned} \quad (6.183)$$

$$\left\{ \begin{aligned} &\left[\left(\left(\frac{d}{d\lambda} A_1(\lambda) \right) (\lambda - 2) + A_1(\lambda) - \left(\frac{d}{d\lambda} A_2(\lambda) \right) - \left(\frac{d}{d\lambda} A_3(\lambda) \right) \right) e^{\lambda x} \right. \\ &\quad \left. + (A_1(\lambda) (\lambda - 2) - A_2(\lambda) - A_3(\lambda)) x e^{\lambda x} \right]_{\lambda=1} = 0, \\ &\left[\left(-2 \left(\frac{d}{d\lambda} A_1(\lambda) \right) + \left(\frac{d}{d\lambda} A_2(\lambda) \right) (\lambda - 3) + A_2(\lambda) - 2 \left(\frac{d}{d\lambda} A_3(\lambda) \right) \right) e^{\lambda x} \right. \\ &\quad \left. + (-2A_1(\lambda) + A_2(\lambda) (\lambda - 3) - 2A_3(\lambda)) x e^{\lambda x} \right]_{\lambda=1} = 0, \\ &\left[\left(- \left(\frac{d}{d\lambda} A_1(\lambda) \right) - \left(\frac{d}{d\lambda} A_2(\lambda) \right) + \left(\frac{d}{d\lambda} A_3(\lambda) \right) (\lambda - 2) + A_3(\lambda) \right) e^{\lambda x} \right. \\ &\quad \left. + (-A_1(\lambda) - A_2(\lambda) + A_3(\lambda) (\lambda - 2)) x e^{\lambda x} \right]_{\lambda=1} = 0, \end{aligned} \right. \quad (6.184)$$

so that we have

$$\begin{cases} (-\alpha_1 + \beta_1 - \alpha_2 - \alpha_3) e^x + (-\beta_1 - \beta_2 - \beta_3) x e^x = 0, \\ (-2\alpha_1 - 2\alpha_2 + \beta_2 - 2\alpha_3) e^x + (-2\beta_1 - 2\beta_2 - 2\beta_3) x e^x = 0, \\ (-\alpha_1 - \alpha_2 - \alpha_3 + \beta_3) e^x + (-\beta_1 - \beta_2 - \beta_3) x e^x = 0, \end{cases} \quad (6.185)$$

where

$$\begin{cases} \alpha_1 \equiv \left[\frac{d}{d\lambda} A_1(\lambda) \right]_{\lambda=+1}, & \beta_1 \equiv A_1(+1), \\ \alpha_2 \equiv \left[\frac{d}{d\lambda} A_2(\lambda) \right]_{\lambda=+1}, & \beta_2 \equiv A_2(+1), \\ \alpha_3 \equiv \left[\frac{d}{d\lambda} A_3(\lambda) \right]_{\lambda=+1}, & \beta_3 \equiv A_3(+1). \end{cases} \quad (6.186)$$

Here we remark that e^x and $x e^x$ are independent of each other for any real number x in the interval $(-\infty, +\infty)$,

because of

$$\begin{aligned}
 W[e^x, xe^x] &= \det \begin{pmatrix} e^x & \frac{d}{dx}e^x \\ xe^x & \frac{d}{dx}(xe^x) \end{pmatrix} \\
 &= \det \begin{pmatrix} e^x & e^{2x} \\ xe^x & (x+1)e^x \end{pmatrix} \\
 &= e^{2x} \neq 0 \quad (\forall x \in (-\infty, +\infty)).
 \end{aligned} \tag{6.187}$$

Sufficient conditions for identity (6.186) can be given as

$$\begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = \beta_1, \\ 2\alpha_1 + 2\alpha_2 + 2\alpha_3 = \beta_2, \quad \beta_1 + \beta_2 + \beta_3 = 0, \\ \alpha_1 + \alpha_2 + \alpha_3 = \beta_3, \end{cases} \tag{6.188}$$

and then

$$\begin{cases} \alpha_1 = c_1 & \beta_1 = 0, \\ \alpha_2 = c_2 & \beta_2 = 0, \\ \alpha_3 = -c_1 - c_2, & \beta_3 = 0, \end{cases} \tag{6.189}$$

where

$$c_1 \equiv \alpha_1, \quad c_2 \equiv \beta_1. \tag{6.190}$$

Here c_1 and c_2 are arbitrary constants.

$$\begin{aligned}
 \begin{pmatrix} \Phi_1(x; \lambda = 1) \\ \Phi_2(x; \lambda = 1) \\ \Phi_3(x; \lambda = 1) \end{pmatrix} &= \left[\begin{pmatrix} \left(\frac{d}{d\lambda} A_1(\lambda) \right) e^{\lambda x} + A_1(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_2(\lambda) \right) e^{\lambda x} + A_2(\lambda) x e^{\lambda x} \\ \left(\frac{d}{d\lambda} A_3(\lambda) \right) e^{\lambda x} + A_3(\lambda) x e^{\lambda x} \end{pmatrix} \right]_{\lambda=1} \\
 &= \begin{pmatrix} \left[\frac{d}{d\lambda} A_1(\lambda) \right]_{\lambda=+1} x e^x + A_1(+1) x e^x \\ \left[\frac{d}{d\lambda} A_2(\lambda) \right]_{\lambda=+1} x e^x + A_2(+1) x e^x \\ \left[\frac{d}{d\lambda} A_3(\lambda) \right]_{\lambda=+1} x e^x + A_3(+1) x e^x \end{pmatrix} \\
 &= \begin{pmatrix} \alpha_1 e^x + \beta_1 x e^x \\ \alpha_2 e^x + \beta_2 x e^x \\ \alpha_3 e^x + \beta_3 x e^x \end{pmatrix} \\
 &= \begin{pmatrix} c_1 e^x \\ c_2 e^x \\ (c_1 + c_2) e^x \end{pmatrix} \\
 &= c_1 \begin{pmatrix} e^x \\ 0 \\ e^x \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^x \\ e^x \end{pmatrix}.
 \end{aligned} \tag{6.191}$$

From Eqs.(6.192) and (6.192), the general solution of Eq.(6.160) is given as

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = c_1 \begin{pmatrix} e^x \\ 0 \\ e^x \end{pmatrix} + c_2 \begin{pmatrix} 0 \\ e^x \\ e^x \end{pmatrix} + c_3 \begin{pmatrix} e^{5x} \\ 2e^{5x} \\ e^{5x} \end{pmatrix}. \tag{6.192}$$

Because we can confirm that

$$\det \begin{pmatrix} e^x & e^x & e^{5x} \\ 0 & e^x & 2e^{5x} \\ e^x & 0 & e^{5x} \end{pmatrix} = \det \begin{pmatrix} e^x & e^x & e^{5x} \\ 0 & e^x & 2e^{5x} \\ 0 & -e^x & 0 \end{pmatrix} = e^x \det \begin{pmatrix} e^x & 2e^{5x} \\ -e^x & 0 \end{pmatrix} = 2e^{6x} \neq 0 \quad (\forall x \in (-\infty, +\infty)) \tag{6.193}$$

the solutions $\begin{pmatrix} e^x \\ 0 \\ e^x \end{pmatrix}$, $\begin{pmatrix} e^x \\ e^x \\ 0 \end{pmatrix}$ and $\begin{pmatrix} e^{5x} \\ 2e^{5x} \\ e^{5x} \end{pmatrix}$ are independent of each other and their set

$$\left\{ \begin{pmatrix} e^x \\ 0 \\ e^x \end{pmatrix}, \begin{pmatrix} e^x \\ e^x \\ 0 \end{pmatrix}, \begin{pmatrix} e^{5x} \\ 2e^{5x} \\ e^{5x} \end{pmatrix} \right\}, \tag{6.194}$$

is the fundamental systems of solutions in Eq.(6.160). We can see that Eq.(6.192) is the general solution of Eq.(6.160).

Example 6.5. Find each solution of the inhomogeneous second order linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx}w_1(x) = 16w_1(x) + 14w_2(x) + 38w_3(x) - 2e^{-x}, \\ \frac{d}{dx}w_2(x) = -9w_1(x) - 7w_2(x) - 18w_3(x) - 3e^{-x}, \\ \frac{d}{dx}w_3(x) = -4w_1(x) - 4w_2(x) - 11w_3(x) + 2e^{-x}. \end{cases} \quad (6.195)$$

(Solution) First we consider its corresponding homogeneous second order linear ordinary differential equation:

$$\begin{cases} \frac{d}{dx}w_1(x) = 16w_1(x) + 14w_2(x) + 38w_3(x), \\ \frac{d}{dx}w_2(x) = -9w_1(x) - 7w_2(x) - 18w_3(x), \\ \frac{d}{dx}w_3(x) = -4w_1(x) - 4w_2(x) - 11w_3(x). \end{cases} \quad (6.196)$$

We set the solutions of the present simultaneous linear differential equations by

$$\begin{cases} w_1(x) = A_1(\lambda)e^{\lambda x}, \\ w_2(x) = A_2(\lambda)e^{\lambda x}, \\ w_3(x) = A_3(\lambda)e^{\lambda x}, \end{cases} \quad (6.197)$$

and substitute them to Eq.(6.195) as follows:

$$\begin{cases} A_1(\lambda)\lambda e^{\lambda x} = 16A_1(\lambda)e^{\lambda x} + 14A_2(\lambda)e^{\lambda x} + 38A_3(\lambda)e^{\lambda x}, \\ A_2(\lambda)\lambda e^{\lambda x} = -9A_1(\lambda)e^{\lambda x} - 7A_2(\lambda)e^{\lambda x} - 18A_3(\lambda)e^{\lambda x}, \\ A_3(\lambda)\lambda e^{\lambda x} = -4A_1(\lambda)e^{\lambda x} - 4A_2(\lambda)e^{\lambda x} - 11A_3(\lambda)e^{\lambda x}, \end{cases} \quad (6.198)$$

$$\begin{cases} (16 - \lambda)A_1(\lambda)e^{\lambda x} + 14A_2(\lambda)e^{\lambda x} + 38A_3(\lambda)e^{\lambda x} = 0, \\ -9A_1(\lambda)e^{\lambda x} + (-7 - \lambda)A_2(\lambda)e^{\lambda x} - 18A_3(\lambda)e^{\lambda x} = 0, \\ -4A_1(\lambda)e^{\lambda x} - 4A_2(\lambda)e^{\lambda x} + (-11 - \lambda)A_3(\lambda)e^{\lambda x} = 0. \end{cases} \quad (6.199)$$

$$\begin{pmatrix} 16 - \lambda & 14 & 38 \\ -9 & -7 - \lambda & -18 \\ -4 & -4 & -11 - \lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda)e^{\lambda x} \\ A_2(\lambda)e^{\lambda x} \\ A_3(\lambda)e^{\lambda x} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \quad (6.200)$$

$$\begin{pmatrix} 16 - \lambda & 14 & 38 \\ -9 & -7 - \lambda & -18 \\ -4 & -4 & -11 - \lambda \end{pmatrix} \begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}. \quad (6.201)$$

Here we remark that $e^{\lambda x}$ is always non zero for any real number x in the interval $(-\infty, +\infty)$ The condition that the

homogeneous simultaneous linear equations (6.201) for $\begin{pmatrix} A_1(\lambda) \\ A_2(\lambda) \\ A_3(\lambda) \end{pmatrix}$ has non-trivial solutions is given as follows:

$$\det \begin{pmatrix} 16 - \lambda & 14 & 38 \\ -9 & -7 - \lambda & -18 \\ -4 & -4 & -11 - \lambda \end{pmatrix} = 0, \quad (6.202)$$

$$\det \begin{pmatrix} 2 - \lambda & 14 & 38 \\ -2 + \lambda & -7 - \lambda & -18 \\ 0 & -4 & -11 - \lambda \end{pmatrix} = 0, \quad (6.203)$$

$$\det \begin{pmatrix} 2 - \lambda & 14 & 38 \\ 0 & 7 - \lambda & 20 \\ 0 & -4 & -11 - \lambda \end{pmatrix} = 0, \quad (6.204)$$

$$(2 - \lambda)\det \begin{pmatrix} 7 - \lambda & 20 \\ -4 & -11 - \lambda \end{pmatrix} = 0, \quad (6.205)$$

$$(2 - \lambda)((7 - \lambda)(-11 - \lambda) + 80) = 0, \quad (6.206)$$

$$(2 - \lambda)(-77 + 4\lambda + \lambda^2 + 80) = 0, \quad (6.207)$$

$$(2 - \lambda)(3 + 4\lambda + \lambda^2) = 0, \quad (6.208)$$

$$-(\lambda - 2)(\lambda + 1)(\lambda + 3) = 0, \quad (6.209)$$

$$\lambda = 2, -1, -3. \quad (6.210)$$

For each solution, $\lambda = 2$, $\lambda = -1$ and $\lambda = -3$, the simultaneous linear equation (6.201) can be written down to

$$\begin{cases} 19A_1(-3) + 14A_2(-3) + 38A_3(-3) = 0 \\ -9A_1(-3) - 4A_2(-3) - 18A_3(-3) = 0 \\ -4A_1(-3) - 4A_2(-3) - 8A_3(-3) = 0 \end{cases} \rightarrow \begin{cases} A_2(-3) = 0, \\ A_3(-3) = -\frac{1}{2}A_1(-3), \end{cases} \quad (6.211)$$

$$\begin{cases} 17A_1(-1) + 14A_2(-1) + 38A_3(-1) = 0, \\ -9A_1(-1) - 6A_2(-1) - 18A_3(-1) = 0, \\ -4A_1(-1) - 4A_2(-1) - 10A_3(-1) = 0, \end{cases} \rightarrow \begin{cases} A_2(-1) = \frac{3}{2}A_1(-1), \\ A_3(-1) = -A_1(-1), \end{cases} \quad (6.212)$$

$$\begin{cases} 14A_1(2) + 14A_2(2) + 38A_3(2) = 0 \\ -9A_1(2) - 9A_2(2) - 18A_3(2) = 0 \\ -4A_1(2) - 4A_2(2) - 13A_3(2) = 0 \end{cases} \rightarrow \begin{cases} A_2(2) = -A_1(2), \\ A_3(2) = 0, \end{cases} \quad (6.213)$$

and then their corresponding solutions are derived as

$$\begin{pmatrix} w_1^{(-3)}(x) \\ w_2^{(-3)}(x) \\ w_3^{(-3)}(x) \end{pmatrix} = \begin{pmatrix} A_1(-3)e^{-3x} \\ 0 \\ -\frac{1}{2}A_1(-3)e^{-3x} \end{pmatrix} = \frac{1}{2}A^{(1)}(-3) \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix}, \quad (6.214)$$

$$\begin{pmatrix} w_1^{(-1)}(x) \\ w_2^{(-1)}(x) \\ w_3^{(-1)}(x) \end{pmatrix} = \begin{pmatrix} A_1(-1)e^{-x} \\ \frac{3}{2}A_1(-1)e^{-x} \\ -A_1(-1)e^{-x} \end{pmatrix} = \frac{1}{2}A^{(1)}(-1) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix}, \quad (6.215)$$

$$\begin{pmatrix} w_1^{(+2)}(x) \\ w_2^{(+2)}(x) \\ w_3^{(+2)}(x) \end{pmatrix} = \begin{pmatrix} A_1(+2)e^{2x} \\ -A_1(+2)e^{2x} \\ 0 \end{pmatrix} = A^{(1)}(+2) \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix}, \quad (6.216)$$

respectively. We have

$$\begin{aligned} \det \begin{pmatrix} 2e^{-3x} & 2e^{-x} & e^{2x} \\ 0 & 3e^{-x} & -e^{2x} \\ -e^{-3x} & -2e^{-x} & 0 \end{pmatrix} &= \det \begin{pmatrix} 2e^{-3x} & 2e^{-x} & e^{2x} \\ 2e^{-3x} & 5e^{-x} & 0 \\ -e^{-3x} & -2e^{-x} & 0 \end{pmatrix} \\ &= e^{2x} \det \begin{pmatrix} 2e^{-3x} & 5e^{-x} \\ -e^{-3x} & -2e^{-x} \end{pmatrix} \\ &= e^{2x}(-4e^{-4x} + 5e^{-4x}) = e^{-2x} \neq 0, \end{aligned} \quad (6.217)$$

and then the above three solutions $\begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix}$, $\begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix}$ and $\begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix}$ are linearly independent of each other and are a fundamental system of solutions. The general solution of the present homogeneous simultaneous linear differential equations is given as follows:

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = B_{-3} \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + B_{-1} \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + B_{+2} \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} \quad (\forall x \in (-\infty, +\infty)), \quad (6.218)$$

where B_{-3} , B_{-1} and B_{+2} are arbitrary constants. From the general solution (6.218) for the simultaneous homogeneous

linear differential equations (6.196), the special solution $\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix}$ of the present simultaneous inhomogeneous linear differential equations (6.195) can be obtained by introducing a variation of constants as follows:

$$\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix} = u_{-3}(x) \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + u_{-1}(x) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + u_{+2}(x) \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} \quad (\forall x \in (-\infty, +\infty)), \quad (6.219)$$

$$\frac{d}{dx} \begin{pmatrix} \frac{d}{dx} \psi_1(x) \\ \frac{d}{dx} \psi_2(x) \\ \frac{d}{dx} \psi_3(x) \end{pmatrix} - \begin{pmatrix} 16 & 14 & 38 \\ -9 & -7 & -18 \\ -4 & -4 & -11 \end{pmatrix} \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix} - \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} = 0, \quad (6.220)$$

$$\frac{d}{dx} \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix} - \begin{pmatrix} 16 & 14 & 38 \\ -9 & -7 & -18 \\ -4 & -4 & -11 \end{pmatrix} \begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix} - \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} = 0, \quad (6.221)$$

$$\begin{aligned} & \frac{d}{dx} \left(u_{-3}(x) \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + u_{-1}(x) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + u_{+2}(x) \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} \right) \\ & - \begin{pmatrix} 16 & 14 & 38 \\ -9 & -7 & -18 \\ -4 & -4 & -11 \end{pmatrix} \left(u_{-3}(x) \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + u_{-1}(x) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + u_{+2}(x) \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} \right) \\ & - \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} = 0, \end{aligned} \quad (6.222)$$

$$\begin{aligned} & \left(\frac{d}{dx} u_{-3}(x) \right) \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + \left(\frac{d}{dx} u_{-1}(x) \right) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + \left(\frac{d}{dx} u_{+2}(x) \right) \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} \\ & + u_{-3}(x) \begin{pmatrix} -6e^{-3x} \\ 0 \\ +3e^{-3x} \end{pmatrix} + u_{-1}(x) \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ +2e^{-x} \end{pmatrix} + u_{+2}(x) \begin{pmatrix} 2e^{2x} \\ -2e^{2x} \\ 0 \end{pmatrix} \\ & - u_{-3}(x) \begin{pmatrix} 16 & 14 & 38 \\ -9 & -7 & -18 \\ -4 & -4 & -11 \end{pmatrix} \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + u_{-1}(x) \begin{pmatrix} 16 & 14 & 38 \\ -9 & -7 & -18 \\ -4 & -4 & -11 \end{pmatrix} \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} \\ & + u_{+2}(x) \begin{pmatrix} 16 & 14 & 38 \\ -9 & -7 & -18 \\ -4 & -4 & -11 \end{pmatrix} \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} - \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} = 0, \end{aligned} \quad (6.223)$$

$$\left(\frac{d}{dx} u_{-3}(x) \right) \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + \left(\frac{d}{dx} u_{-1}(x) \right) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + \left(\frac{d}{dx} u_{+2}(x) \right) \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} - \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} = 0, \quad (6.224)$$

$$\begin{pmatrix} 2e^{-3x} & 2e^{-x} & e^{2x} \\ 0 & 3e^{-x} & -e^{2x} \\ -e^{-3x} & -2e^{-x} & 0 \end{pmatrix} \begin{pmatrix} \left(\frac{d}{dx} u_{-3}(x) \right) \\ \left(\frac{d}{dx} u_{-1}(x) \right) \\ \left(\frac{d}{dx} u_{+2}(x) \right) \end{pmatrix} = \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix}, \quad (6.225)$$

$$\begin{pmatrix} \left(\frac{d}{dx} u_{-3}(x) \right) \\ \left(\frac{d}{dx} u_{-1}(x) \right) \\ \left(\frac{d}{dx} u_{+2}(x) \right) \end{pmatrix} = \begin{pmatrix} 2e^{-3x} & 2e^{-x} & e^{2x} \\ 0 & 3e^{-x} & -e^{2x} \\ -e^{-3x} & -2e^{-x} & 0 \end{pmatrix}^{-1} \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix}$$

$$\begin{aligned}
&= \left(\det \begin{pmatrix} 2e^{-3x} & 2e^{-x} & e^{2x} \\ 0 & 3e^{-x} & -e^{2x} \\ -e^{-3x} & -2e^{-x} & 0 \end{pmatrix} \right)^{-1} \\
&\times \begin{pmatrix} (-1)^{1+1} \det \begin{pmatrix} 3e^{-x} & -e^{2x} \\ -2e^{-x} & 0 \end{pmatrix} & (-1)^{2+1} \det \begin{pmatrix} 2e^{-x} & e^{2x} \\ -2e^{-x} & 0 \end{pmatrix} & (-1)^{3+1} \det \begin{pmatrix} 2e^{-x} & e^{2x} \\ 3e^{-x} & -e^{2x} \end{pmatrix} \\ (-1)^{1+2} \det \begin{pmatrix} 0 & -e^{2x} \\ -e^{-3x} & 0 \end{pmatrix} & (-1)^{2+2} \det \begin{pmatrix} 2e^{-3x} & e^{2x} \\ -e^{-3x} & 0 \end{pmatrix} & (-1)^{3+2} \det \begin{pmatrix} 2e^{-3x} & e^{2x} \\ 0 & -e^{2x} \end{pmatrix} \\ (-1)^{1+3} \det \begin{pmatrix} 0 & 3e^{-x} \\ -e^{-3x} & -2e^{-x} \end{pmatrix} & (-1)^{2+3} \det \begin{pmatrix} 2e^{-3x} & 2e^{-x} \\ -e^{-3x} & -2e^{-x} \end{pmatrix} & (-1)^{3+3} \det \begin{pmatrix} 2e^{-3x} & 2e^{-x} \\ 0 & 3e^{-x} \end{pmatrix} \end{pmatrix} \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} \\
&= \frac{1}{e^{-2x}} \begin{pmatrix} -2e^x & -2e^x & -5e^x \\ e^{-x} & e^{-x} & 2e^{-x} \\ 3e^{-4x} & 2e^{-4x} & 6e^{-4x} \end{pmatrix} \begin{pmatrix} -2e^{-x} \\ -3e^{-x} \\ 2e^{-x} \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}, \tag{6.226}
\end{aligned}$$

$$\begin{pmatrix} u_{-3}(x) \\ u_{-1}(x) \\ u_{+2}(x) \end{pmatrix} = \begin{pmatrix} \text{Constant} \\ -x + \text{Constant} \\ \text{Constant} \end{pmatrix}. \tag{6.227}$$

By substituting Eq.(6.227) to Eq.(6.219), the special solutions are derived as follows:

$$\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix} = \text{Constant} \times \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + (-x + \text{Constant}) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + \text{Constant} \times \begin{pmatrix} e^{2x} \\ -e^{2x} \\ 0 \end{pmatrix} \quad (\forall x \in (-\infty, +\infty)). \tag{6.228}$$

From the general solution (6.218) for the simultaneous homogeneous linear differential equations (6.196) and the special solution $\begin{pmatrix} \psi_1(x) \\ \psi_2(x) \\ \psi_3(x) \end{pmatrix}$ in Eq.(6.228) of the present simultaneous inhomogeneous linear differential equations (6.195), the general solution of the present simultaneous inhomogeneous linear differential equations (6.195) can be reduced as follows:

$$\begin{pmatrix} w_1(x) \\ w_2(x) \\ w_3(x) \end{pmatrix} = C_{-3} \begin{pmatrix} 2e^{-3x} \\ 0 \\ -e^{-3x} \end{pmatrix} + (-x + C_{-1}) \begin{pmatrix} 2e^{-x} \\ 3e^{-x} \\ -2e^{-x} \end{pmatrix} + C_{+2} \begin{pmatrix} e^{+2x} \\ -e^{+2x} \\ 0 \end{pmatrix} \quad (\forall x \in (-\infty, +\infty)), \tag{6.229}$$

where C_{-3} , C_{-1} and C_{+2} are arbitrary constants.

7 Orthonormal Systems

In Calculus, it is known that any real-valued function $f(x)$ defined in the bounded and closed interval $[\alpha, \beta]$, that is always bounded and has at most a finite number of discontinuities at $x = \gamma_1, x = \gamma_2, \dots, x = \gamma_K$

($\alpha \leq \gamma_1 < \gamma_2 < \dots < \gamma_K \leq \beta$), is integrable in the interval $[\alpha, \beta]$ by defining the integral $\int_{\alpha}^{\beta} f(x)dx$ as follows:

$$\int_{\alpha}^{\beta} f(x)dx \equiv \int_{\alpha}^{\gamma_1-0} f(x)dx + \sum_{k=1}^{K-1} \int_{\gamma_k+0}^{\gamma_{k+1}-0} f(x)dx + \int_{\gamma_K+0}^{\beta} f(x)dx, \tag{7.1}$$

where

$$\begin{cases} \int_{\alpha}^{\gamma_1-0} f(x)dx \equiv \lim_{\varepsilon \rightarrow +0} \int_{\alpha}^{\gamma_1-\varepsilon} f(x)dx, \\ \int_{\gamma_k+0}^{\gamma_{k+1}-0} f(x)dx \equiv \lim_{\varepsilon \rightarrow +0} \lim_{\varepsilon' \rightarrow +0} \int_{\gamma_k+\varepsilon}^{\gamma_{k+1}-\varepsilon'} f(x)dx, \\ \int_{\gamma_K+0}^{\beta} f(x)dx \equiv \lim_{\varepsilon \rightarrow +0} \int_{\gamma_K+\varepsilon}^{\beta} f(x)dx. \end{cases} \tag{7.2}$$

Here we remark that $f(x)$ has to be always bounded in all the intervals $[\alpha, \gamma_1], [\gamma_1, \gamma_2], [\gamma_2, \gamma_3], \dots, [\gamma_{K-1}, \gamma_K]$ and $[\gamma_K, \beta]$. and also should be always continuous in all the intervals $[\alpha, \gamma_1), (\gamma_1, \gamma_2), (\gamma_2, \gamma_3), \dots, (\gamma_{K-1}, \gamma_K)$ and $(\gamma_K, \beta]$. The integral defined by Eq.(7.1) is referred to as an **improper integral** in the bounded interval. This fact has been established the following theorems in Calculus:

Theorem 7.1 Any continuous function $f(x)$ in a bounded and closed interval $[\alpha, \beta]$ is always integrable in the interval,

such that there exists $\int_{\alpha}^{\beta} f(x)dx$.

Theorem 7.2 Any function $f(x)$ that is bounded in $[\alpha, \beta]$ and is continuous in (α, β) is always integrable in the interval $[\alpha, \beta]$, such that there exists $\int_{\alpha}^{\beta} f(x)dx$ and we have

$$\int_{\alpha}^{\beta} f(x)dx = \int_{\alpha+0}^{\beta} f(x)dx. \quad (7.3)$$

In the similar arguments, we have

$$\int_{\alpha}^{\beta} f(x)dx = \int_{\alpha}^{\beta-0} f(x)dx, \quad (7.4)$$

for any function $f(x)$ that is bounded in $[\alpha, \beta]$ and is continuous in $[\alpha, \beta)$.

Theorem 7.3 Any function $f(x)$, that is always bounded in $[\alpha, \beta]$ and is always continuous in the interval except a finite number of points, is integrable in the interval, such that there exists $\int_{\alpha}^{\beta} f(x)dx$.

Theorem 7.4 If functions $f(x)$ and $g(x)$, that is always continuous in the interval (α, β) ($[\alpha, \beta)$), satisfy

(i) For any $x \in (\alpha, \beta]$, it is always valid that $|f(x)| \leq g(x)$,

(ii) There exists $\int_{\alpha+0}^{\beta} g(x)dx$,

then there exists $\int_{\alpha+0}^{\beta} f(x)dx$. In the similar arguments, if functions $f(x)$ and $g(x)$, that is always continuous in the interval $[\alpha, \beta)$, satisfy

(i) For any $x \in [\alpha, \beta)$, it is always valid that $|f(x)| \leq g(x)$,

(ii) There exists $\int_{\alpha}^{\beta-0} g(x)dx$,

then there exists $\int_{\alpha}^{\beta-0} f(x)dx$.

Moreover by setting $g(x) = |f(x)|$ in theorem 7.4 and by considering the inequality $\left| \int_{\alpha}^{\beta} f(x)dx \right| \leq \int_{\alpha}^{\beta} |f(x)|dx$, we can see that if $f(x)$ is **absolutely integrable** in $[\alpha, \beta]$, such that $\int_{\alpha}^{\beta} |f(x)|dx < +\infty$, then the function $f(x)$ is integrable in $[\alpha, \beta]$, such that there exists $\int_{\alpha}^{\beta} f(x)dx$.

Let us suppose that we consider a function sequence:

$$\left\{ \phi_n(x) \mid n = 0, 1, 2, \dots, x \in [\alpha, \beta] \right\}, \quad (7.5)$$

with

$$\begin{cases} \int_{\alpha}^{\beta} |\phi_n(x)|dx < +\infty, \\ \int_{\alpha}^{\beta} |\phi_n(x)|^2 dx < +\infty, \end{cases} \quad (n = 0, 1, 2, \dots), \quad (7.6)$$

where all the functions $\phi_n(x)$ in the function sequence are always bounded, continuous and $C^{+\infty}$ -class function in $[\alpha, \beta]$. The function sequence satisfies the following **orthonormal relations**:

$$\int_{\alpha}^{\beta} \phi_m(x)\phi_n(x)dx = \delta_{m,n} \quad (m, n = 0, 1, 2, \dots). \quad (7.7)$$

Moreover, we consider a sequence $\{c_0, c_1, c_2, \dots\}$ and its function series

$$\psi(x) \equiv \sum_{n=0}^{+\infty} c_n \phi_n(x) \quad (\forall x \in ([\alpha, \gamma_1) \cup (\gamma_1, \gamma_2) \cup (\gamma_2, \gamma_3) \cup \dots \cup (\gamma_{K-1}, \gamma_K) \cup (\gamma_K, \beta])), \quad (7.8)$$

that always uniformly converges in each interval of $[\alpha, \gamma_1]$, $[\gamma_1, \gamma_2]$, $[\gamma_2, \gamma_3]$, $\dots$, $[\gamma_{K-1}, \gamma_K]$ and $[\gamma_K, \beta]$ ($\alpha \leq \gamma_1 < \gamma_2 < \dots < \gamma_K \leq \beta$). We integrate the both sides of Eq.(7.8) after multiplying $\phi_m(x)$ and by using the orthonormal relations (7.7), we derive

$$\int_{\alpha}^{\beta} \psi(x)\phi_m(x)dx = \int_{\alpha}^{\beta} \left(\sum_{n=0}^{+\infty} c_n \phi_n(x)\phi_m(x) \right) dx. \quad (7.9)$$

From theorem 3.10, the integral and the summation in the right side of Eq.(7.9) can be exchanged as follows:

$$\int_{\alpha}^{\beta} \psi(x) \phi_m(x) dx = \sum_{n=0}^{+\infty} c_n \int_{\alpha}^{\beta} \phi_n(x) \phi_m(x) dx. \quad (7.10)$$

By applying the orthonormal relations in Eq.(7.7) to the right side of Eq.(7.10), all the coefficients in $\{c_0, c_1, c_2, \dots\}$ are expressed in terms of the function $\phi(x)$ as follows:

$$c_m = \int_{\alpha}^{\beta} \psi(x) \phi_m(x) dx \quad (m = 0, 1, 2, \dots). \quad (7.11)$$

Here we remark that the following equality has been used in the derivation of Eq.(7.11):

$$\sum_{n=0}^{+\infty} c_n \delta_{m,n} = c_m \quad (m = 0, 1, 2, \dots). \quad (7.12)$$

The expression in Eqs.(7.8) and (7.11) is referred to as an **orthonormal expansion** of the function $\psi(x)$ in terms of the **orthonormal function system** $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ in the interval $[\alpha, \beta]$. In order to establish the equality (7.8) with Eq.(7.11) for a given function $\psi(x)$ that is continuous in each interval of $[\alpha, \gamma_1]$, $[\gamma_1, \gamma_2]$, $[\gamma_2, \gamma_3]$, $\dots$, $[\gamma_{K-1}, \gamma_K]$ and $[\gamma_K, \beta]$ ($\alpha \leq \gamma_1 < \gamma_2 < \dots < \gamma_K \leq \beta$), the piecewise uniform convergency for the function series of the right side of Eq.(7.8) should be required in each interval of $[\alpha, \gamma_1]$, $[\gamma_1, \gamma_2]$, $[\gamma_2, \gamma_3]$, $\dots$, $[\gamma_{K-1}, \gamma_K]$ and $[\gamma_K, \beta]$. Eqs.(7.8) and (7.11) are rewritten as follows:

$$\begin{cases} \psi(x) = \sum_{n=0}^{+\infty} c_n \phi_n(x) \quad (\forall x \in [\alpha, \gamma_1] \cup (\gamma_1, \gamma_2) \cup (\gamma_2, \gamma_3) \cup \dots \cup (\gamma_{K-1}, \gamma_K) \cup (\gamma_K, \beta]), \\ c_m \equiv \int_{\alpha}^{\beta} \psi(x) \phi_m(x) dx \quad (m = 0, 1, 2, \dots). \end{cases} \quad (7.13)$$

If it is valid that any function $\psi(x)$ defined with at most a finite number of discontinuities in the interval $[\alpha, \beta]$ satisfying

$$\int_{\alpha}^{\beta} f(x) \phi_n(x) dx = 0 \quad (n = 0, 1, 2, \dots), \quad (7.14)$$

should be identically equal to 0 in the interval $[\alpha, \beta]$, then the function sequence $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is a **complete system** in the interval. Let us suppose that if there exists a function $\psi(x)$ that is not indetically equal to 0 under

satisfying Eq.(7.14), it is valid that $\sum_{n=0}^{+\infty} c_n \phi_n(x) = 0$ although $\sum_{n=0}^{+\infty} c_n \phi_n(x)$ should not identically equal to 0 under the

present assumption. This is a contraction. Then Eq.(7.13) for any function $\psi(x)$ is equivalent to the proposition that the function sequence $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is a complete system in the interval $[\alpha, \beta]$.

Theorem 7.5 Any complete and orthonormal system in an interval $[\alpha, \beta]$ should includes an infinite number of functions defined in the interval.

(Proof) Let us suppose that we can construct a complete system from a finite number of orthonormal functions $\phi_0(x), \phi_1(x), \dots, \phi_n(x)$.

On the other hand, we consider any function $\psi(x)$ that is expressed in terms of a set of $(n+1)$ continuous functions $\{\psi_0(x), \psi_1(x), \dots, \psi_{n+1}(x)\}$ as follows:

$$\psi(x) = \sum_{l=0}^{n+1} c_l \psi_l(x) \quad (\forall x \in [\alpha, \beta]), \quad (7.15)$$

where the function sequence $\{\psi_0(x), \psi_1(x), \psi_2(x), \dots\}$ is linearly independent in $[\alpha, \beta]$ and all of the coefficients $c_0, c_1, c_2, \dots, c_{n+1}$ is not eqaul to 0.

$$\int_{\alpha}^{\beta} \psi(x) \phi_m(x) dx = \int_{\alpha}^{\beta} \left(\sum_{l=0}^{n+1} c_l \psi_l(x) \phi_m(x) \right) dx \quad (m = 0, 1, 2, \dots, n), \quad (7.16)$$

$$\int_{\alpha}^{\beta} \psi(x) \phi_m(x) dx = \sum_{l=0}^{n+1} c_l \int_{\alpha}^{\beta} \psi_l(x) \phi_m(x) dx \quad (m = 0, 1, 2, \dots, n). \quad (7.17)$$

We can choose $\{c_0, c_1, c_2, \dots, c_{n+1}\}$ so as to satisfy the following simultaneous linear equations:

$$\sum_{l=0}^{n+1} c_l \int_{\alpha}^{\beta} \psi_l(x) \phi_m(x) dx = 0 \quad (m = 0, 1, 2, \dots, n), \quad (7.18)$$

under the constraint that all of the coefficients $c_0, c_1, c_2, \dots, c_{n+1}$ is not equal to 0. Eq.(7.18) means that

$$\int_{\alpha}^{\beta} \psi(x) \phi_m(x) dx = 0 \quad (m = 0, 1, 2, \dots, n), \quad (7.19)$$

and then $\psi(x)$ in Eq.(7.15) should be identically equal to 0, such that

$$\psi(x) = 0 \quad (\forall x \in [\alpha, \beta]), \quad (7.20)$$

from the definition of the completeness of the function sequence $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots, \phi_n(x)\}$. This is a contrast. Finally, we consider a mean-square distance

$$\int_{\alpha}^{\beta} (\psi(x) - S_n(x))^2 dx, \quad (7.21)$$

between the orthonormal expansion of $\psi(x)$ in Eq.(7.13) and a function sequence $\{S_0(x), S_1(x), S_2(x), \dots, S_n(x)\}$ defined by

$$S_n(x) \equiv \sum_{l=0}^n c_l \phi_l(x) \quad (\forall x \in [\alpha, \beta]). \quad (7.22)$$

If the orthonormal expansion for $\psi(x)$ in Eq.(7.13) uniformly converges in $[\alpha, \beta]$, we can derive

$$\begin{aligned} \int_{\alpha}^{\beta} (\psi(x) - S_n(x))^2 dx &= \int_{\alpha}^{\beta} \left(\psi(x) - \sum_{l=0}^n c_l \phi_l(x) \right)^2 dx \\ &= \int_{\alpha}^{\beta} \psi(x)^2 dx - 2 \int_{\alpha}^{\beta} \left(\sum_{l=0}^n c_l \psi(x) \phi_l(x) \right) dx + \int_{\alpha}^{\beta} \left(\sum_{l=0}^n c_l \phi_l(x) \right)^2 dx \\ &= \int_{\alpha}^{\beta} \psi(x)^2 dx - 2 \sum_{l=0}^n c_l \int_{\alpha}^{\beta} \psi(x) \phi_l(x) dx + \sum_{l=0}^n c_l^2 \\ &= \int_{\alpha}^{\beta} \psi(x)^2 dx - 2 \sum_{l=0}^n c_l^2 + \sum_{l=0}^n c_l^2 \\ &= \int_{\alpha}^{\beta} \psi(x)^2 dx - \sum_{l=0}^n c_l^2 \\ &\geq \int_{\alpha}^{\beta} \psi(x)^2 dx - \sum_{l=0}^n c_l^2, \end{aligned} \quad (7.23)$$

such that we obtain the following **Bessel inequality**:

$$\int_{\alpha}^{\beta} \psi(x)^2 dx. \quad (7.24)$$

Because $\lim_{n \rightarrow +\infty} S_n(x)$ uniformly converges to $\psi(x)$ in each interval of $[\alpha, \gamma_1]$, $[\gamma_1, \gamma_2]$, $[\gamma_2, \gamma_3]$, $\dots$, $[\gamma_{K-1}, \gamma_K]$ and $[\gamma_K, \beta]$ ($\alpha \leq \gamma_1 < \gamma_2 < \dots < \gamma_K \leq \beta$), we can conform the following **Parseval identity**:

$$\int_{\alpha}^{\beta} \psi(x)^2 dx = \sum_{l=0}^{+\infty} c_l^2. \quad (7.25)$$

8 Sturm-Liouville Equation

8.1 Inhomogeneous Sturm-Liouville Equation

We consider the homogeneous second order ordinary differential equation:

$$-\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) w(x) + q(x) w(x) = 0, \quad (8.1)$$

with the following initial conditions:

$$w(x_0) = y_0, \quad \left[\frac{d}{dx} w(x) \right]_{x=x_0} = y'_0, \quad (8.2)$$

for any real value x in the bounded and closed interval $[\alpha, \beta]$. Here $p(x)$, $q(x)$ and $f(x)$ are continuous functions and

they satisfy $p(x) > 0$ and $q(x) \geq 0$ for any $x \in [\alpha, \beta]$. By replacing $\begin{pmatrix} p(x) \\ q(x) \end{pmatrix}$ with $\begin{pmatrix} \exp\left(\int p(x)dx\right) \\ q(x)\exp\left(\int p(x)dx\right) \end{pmatrix}$, Eq.(8.1)

can be reduced to the homogeneous second order linear differential equation (5.1). If the homogeneous differential equation (8.1) has a fundamental system of solutions, $w^{(1)}(x)$ and $w^{(2)}(x)$, which are linearly independent of each other, the general solution is given by

$$w(x) = c_1 w^{(1)}(x) + c_2 w^{(2)}(x). \quad (8.3)$$

Next we consider the following inhomogeneous differential equation:

$$-\frac{d}{dx}\left(p(x)\frac{d}{dx}\right)w(x) + q(x)w(x) = -f(x) \quad (\forall x \in [\alpha, \beta]), \quad (8.4)$$

under the boundary condition:

$$\begin{cases} a_1 w(\alpha) + a_2 \left[\frac{d}{dx}w(x)\right]_{x=\alpha} = 0, \\ b_1 w(\beta) + b_2 \left[\frac{d}{dx}w(x)\right]_{x=\beta} = 0, \end{cases} \quad (8.5)$$

where $p(x)$, $q(x)$ and $f(x)$ is also a continuous function and it is valid that $p(x) > 0$ and $q(x) \geq 0$ in the bounded and closed interval $[\alpha, \beta]$. Moreover the constants a_1 , b_1 , a_2 and b_2 do not depend on x and we have $a_1^2 + a_2^2 \neq 0$ and $b_1^2 + b_2^2 \neq 0$. The second order linear differential equations (8.1) and (8.4) are referred to as **Sturm-Liouville equations**. We should remark that the homogeneous differential equation (8.1) does not always have two linearly independent solutions satisfying both equalities in the specified boundary condition (8.5). Let us suppose that the homogeneous differential equation (8.1) has the basic solutions, $w^{(1)}(x)$ and $w^{(2)}(x)$, which are linearly independent of each other in the interval $[\alpha, \beta]$, and each solution satisfies only one of two equalities in Eq.(8.5) that is different from each other as follows:

$$\begin{cases} a_1 w^{(1)}(\alpha) + a_2 \left[\frac{d}{dx}w^{(1)}(x)\right]_{x=\alpha} = 0, \\ b_1 w^{(2)}(\beta) + b_2 \left[\frac{d}{dx}w^{(2)}(x)\right]_{x=\beta} = 0, \end{cases} \quad (8.6)$$

and

$$\begin{cases} a_1 w^{(2)}(\alpha) + a_2 \left[\frac{d}{dx}w^{(2)}(x)\right]_{x=\alpha} \neq 0, \\ b_1 w^{(1)}(\beta) + b_2 \left[\frac{d}{dx}w^{(1)}(x)\right]_{x=\beta} \neq 0, \end{cases} \quad (8.7)$$

By applying the fundamental system of solutions, $w^{(1)}(x)$ and $w^{(2)}(x)$, for the homogeneous differential equation (8.1) to the method of variation of constants, we obtain the special solution $w(x)$. In the method of variation of constants, the special solution is introduced in terms of unknown functions $u_1(x)$ and $u_2(x)$ as follows:

$$w(x) = u_1(x)w^{(1)}(x) + u_2(x)w^{(2)}(x), \quad (8.8)$$

and the unknown functions $u_1(x)$ and $u_2(x)$ are determined so as to satisfy the inhomogeneous linear equation (8.4).

First we substitute Eq.(8.8) to Eq.(8.4) as follows:

$$-\frac{d}{dx}\left(p(x)\frac{d}{dx}(u_1(x)w^{(1)}(x) + u_2(x)w^{(2)}(x))\right) + q(x)(u_1(x)w^{(1)}(x) + u_2(x)w^{(2)}(x)) = -f(x). \quad (8.9)$$

$$-\frac{d}{dx}\left(p(x)\frac{d}{dx}(u_1(x)w^{(1)}(x))\right) + q(x)u_1(x)w^{(1)}(x) - \frac{d}{dx}\left(p(x)\frac{d}{dx}(u_2(x)w^{(2)}(x))\right) + q(x)u_2(x)w^{(2)}(x) = -f(x), \quad (8.10)$$

$$\begin{aligned} & -\frac{d}{dx}\left(p(x)\frac{du_1(x)}{dx}f_1(x)\right) - \frac{d}{dx}\left(p(x)u_1(x)\frac{df_1(x)}{dx}\right) + q(x)u_1(x)w^{(1)}(x) \\ & -\frac{d}{dx}\left(p(x)\frac{du_2(x)}{dx}f_2(x)\right) - \frac{d}{dx}\left(p(x)u_2(x)\frac{df_2(x)}{dx}\right) + q(x)u_2(x)w^{(2)}(x) = -f(x), \end{aligned} \quad (8.11)$$

$$\begin{aligned} & -\frac{d}{dx}\left(p(x)\frac{du_1(x)}{dx}f_1(x)\right) - \frac{du_1(x)}{dx}p(x)\frac{df_1(x)}{dx} - u_1(x)\frac{d}{dx}\left(p(x)\frac{df_1(x)}{dx}\right) \\ & + q(x)u_1(x)w^{(1)}(x) \end{aligned}$$

$$\begin{aligned}
& -\frac{d}{dx}\left(p(x)\frac{du_2(x)}{dx}w^{(2)}(x)\right) - \frac{du_2(x)}{dx}p(x)\frac{dw^{(2)}(x)}{dx} - u_2(x)\frac{d}{dx}\left(p(x)\frac{dw^{(2)}(x)}{dx}\right) \\
& + q(x)u_2(x)w^{(2)}(x) = -f(x),
\end{aligned} \tag{8.12}$$

$$\begin{aligned}
& -\frac{d}{dx}\left(p(x)\frac{du_1(x)}{dx}w^{(1)}(x)\right) - p(x)\left(\frac{du_1(x)}{dx}\right)\left(\frac{dw^{(1)}(x)}{dx}\right) - \frac{d}{dx}\left(p(x)\frac{du_2(x)}{dx}w^{(2)}(x)\right) - p(x)\left(\frac{du_2(x)}{dx}\right)\left(\frac{dw^{(2)}(x)}{dx}\right) \\
& + u_1(x)\left\{-\frac{d}{dx}\left(p(x)\frac{d}{dx}\right) + q(x)\right\}w^{(1)}(x) + u_2(x)\left\{-\frac{d}{dx}\left(p(x)\frac{d}{dx}\right) + q(x)\right\}w^{(2)}(x) = -f(x),
\end{aligned} \tag{8.13}$$

$$\begin{aligned}
& -\frac{d}{dx}\left(p(x)\frac{du_1(x)}{dx}w^{(1)}(x)\right) - \frac{du_1(x)}{dx}p(x)\frac{dw^{(1)}(x)}{dx} - \frac{d}{dx}\left(p(x)\frac{du_2(x)}{dx}w^{(2)}(x)\right) - \frac{du_2(x)}{dx}p(x)\frac{dw^{(2)}(x)}{dx} = -f(x),
\end{aligned} \tag{8.14}$$

$$\begin{aligned}
& -\frac{d}{dx}\left(p(x)\frac{du_1(x)}{dx}w^{(1)}(x)\right) + \frac{d}{dx}\left(p(x)\frac{du_2(x)}{dx}w^{(2)}(x)\right) \\
& - p(x)\left\{\left(\frac{du_1(x)}{dx}\right)\left(\frac{dw^{(1)}(x)}{dx}\right) + \left(\frac{du_2(x)}{dx}\right)\left(\frac{dw^{(2)}(x)}{dx}\right)\right\} = -f(x),
\end{aligned} \tag{8.15}$$

$$\begin{aligned}
& \frac{dp(x)}{dx}\left\{\left(\frac{du_1(x)}{dx}\right)w^{(1)}(x) + \left(\frac{du_2(x)}{dx}\right)w^{(2)}(x)\right\} + p(x)\left\{\left(\frac{d^2u_1(x)}{dx^2}\right)w^{(1)}(x) + \left(\frac{d^2u_2(x)}{dx^2}\right)w^{(2)}(x)\right\} \\
& + 2p(x)\left\{\left(\frac{du_1(x)}{dx}\right)\left(\frac{dw^{(1)}(x)}{dx}\right) + \left(\frac{du_2(x)}{dx}\right)\left(\frac{dw^{(2)}(x)}{dx}\right)\right\} = f(x).
\end{aligned} \tag{8.16}$$

By imposing the following constraint condition

$$\left(\frac{du_1(x)}{dx}\right)w^{(1)}(x) + \left(\frac{du_2(x)}{dx}\right)w^{(2)}(x) = 0, \tag{8.17}$$

for the unknown functions $u_1(x)$ and $u_2(x)$, the above equation can be reduced to

$$\begin{aligned}
& \frac{d}{dx}\left\{\left(\frac{du_1(x)}{dx}\right)w^{(1)}(x) + \left(\frac{du_2(x)}{dx}\right)w^{(2)}(x)\right\} \\
& = \left\{\left(\frac{d^2u_1(x)}{dx^2}\right)w^{(1)}(x) + \left(\frac{d^2u_2(x)}{dx^2}\right)w^{(2)}(x)\right\} + \left\{\left(\frac{du_1(x)}{dx}\right)\left(\frac{dw^{(1)}(x)}{dx}\right) + \left(\frac{du_2(x)}{dx}\right)\left(\frac{dw^{(2)}(x)}{dx}\right)\right\} = 0,
\end{aligned} \tag{8.18}$$

and then we have

$$p(x)\left\{\left(\frac{du_1(x)}{dx}\right)\left(\frac{df_1(x)}{dx}\right) + \left(\frac{du_2(x)}{dx}\right)\left(\frac{df_2(x)}{dx}\right)\right\} = f(x). \tag{8.19}$$

By solving Eq.(8.19) and Eq.(8.17), $\frac{du_1(x)}{dx}$ and $\frac{du_2(x)}{dx}$ are expressed as follows:

$$\begin{cases} \frac{du_1(x)}{dx} = -\frac{w^{(2)}(x)f(x)}{W[w^{(1)}(x), w^{(2)}(x)]p(x)}, \\ \frac{du_2(x)}{dx} = \frac{w^{(1)}(x)f(x)}{W[w^{(1)}(x), w^{(2)}(x)]p(x)}. \end{cases} \tag{8.20}$$

By integrating the both sides of Eq.(8.20) with respect to x , $u_1(x)$ and $u_2(x)$ are derived as follows:

$$\begin{cases} u_1(x) = -\int_{\alpha}^x \frac{w^{(2)}(\xi)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} d\xi, \\ u_2(x) = \int_{\alpha}^x \frac{w^{(1)}(\xi)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} d\xi. \end{cases} \tag{8.21}$$

The special solution of the inhomogeneous differential equation (8.4) is derived to

$$w(x) = w^{(2)}(x)\int_{\alpha}^x \frac{w^{(1)}(\xi)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} d\xi, -w^{(1)}(x)\int_{\alpha}^x \frac{w^{(2)}(\xi)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} d\xi$$

$$\begin{aligned}
&= \int_{\alpha}^x \frac{w^{(1)}(\xi)w^{(2)}(x)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} d\xi - \int_{\alpha}^x \frac{w^{(1)}(x)w^{(2)}(\xi)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} d\xi \\
&= \int_{\alpha}^x \left\{ \frac{w^{(1)}(\xi)w^{(2)}(x)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} - \frac{w^{(1)}(x)w^{(2)}(\xi)f(\xi)}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} \right\} d\xi \\
&= \int_{\alpha}^x \frac{1}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} \left\{ w^{(1)}(\xi)w^{(2)}(x) - w^{(1)}(x)w^{(2)}(\xi) \right\} f(\xi) d\xi. \tag{8.22}
\end{aligned}$$

By using the fundamental system of solutions $w^{(1)}(x)$ and $w^{(2)}(x)$ for Eq.(8.1) and the special solution (8.22), the general solution of the inhomogeneous linear differential equation (8.4) is expressed as follows:

$$w(x) = c_1 w^{(1)}(x) + c_2 w^{(2)}(x) + \int_{\alpha}^x \frac{1}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} \left\{ w^{(1)}(\xi)w^{(2)}(x) - w^{(1)}(x)w^{(2)}(\xi) \right\} f(\xi) d\xi. \tag{8.23}$$

By substituting Eq.(8.23) to Eq.(8.5), the simultaneous equations for the constants c_1 and c_2 are reduced to

$$c_1 \left(A_1 w^{(1)}(\alpha) + A_2 \left[\frac{d}{dx} w^{(1)}(x) \right]_{x=\alpha} \right) + c_2 \left(A_1 w^{(2)}(\alpha) + A_2 \left[\frac{d}{dx} w^{(2)}(x) \right]_{x=\alpha} \right) = 0, \tag{8.24}$$

$$\begin{aligned}
&c_1 \left(B_1 w^{(1)}(\beta) + B_2 \left[\frac{d}{dx} w^{(1)}(x) \right]_{x=\beta} \right) + c_2 \left(B_1 w^{(2)}(\beta) + B_2 \left[\frac{d}{dx} w^{(2)}(x) \right]_{x=\beta} \right) \\
&+ \int_{\alpha}^{\beta} \frac{1}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} w^{(1)}(\xi) \left(B_1 w^{(2)}(\beta) + B_2 \left[\frac{d}{dx} w^{(2)}(x) \right]_{x=\beta} \right) f(\xi) d\xi \\
&- \int_{\alpha}^{\beta} \frac{1}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} \left(B_1 w^{(1)}(\beta) + B_2 \left[\frac{d}{dx} w^{(1)}(x) \right]_{x=\beta} \right) w^{(2)}(\xi) f(\xi) d\xi = 0, \tag{8.25}
\end{aligned}$$

By using Eqs.(8.6) and (8.7), we obtain

$$\begin{cases} c_2 = 0, \\ c_1 = - \int_{\alpha}^{\beta} \frac{1}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} w^{(2)}(\xi) f(\xi) d\xi. \end{cases} \tag{8.26}$$

By substituting Eq.(8.26) to Eq.(8.23), the solution of Eq.(8.4) with Eq.(8.5) is expressed as follows:

$$w(x) = \lim_{\varepsilon \rightarrow +0} \left(\int_{\alpha}^{x-\varepsilon} G(x, \xi) f(\xi) d\xi + \int_{x+\varepsilon}^{\beta} G(x, \xi) f(\xi) d\xi \right), \tag{8.27}$$

where

$$G(x, \xi) \equiv \frac{1}{W[w^{(1)}(\xi), w^{(2)}(\xi)]p(\xi)} \left\{ w^{(1)}(\xi)w^{(2)}(x)\theta(x-\xi) + w^{(1)}(x)w^{(2)}(\xi)\theta(\xi-x) \right\}, \tag{8.28}$$

$$\theta(x) \equiv \begin{cases} 1 & (x > 0), \\ 0 & (x < 0). \end{cases} \tag{8.29}$$

8.2 Sturm-Liouville Eigenvalue Problem.

We consider the following eigenvalue problem for the Sturm-Liouville equation:

$$-\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) w(x) + q(x)w(x) = \lambda w(x) \quad (\forall x \in [\alpha, \beta]), \tag{8.30}$$

under the boundary condition (8.5). For any value λ , if we have a pair of linearly independent solution $w_1(x; \lambda)$ and $w_2(x; \lambda)$, the general solution $\phi(x; \lambda)$ is given by

$$\phi(x; \lambda) = c_1 w_1(x; \lambda) + c_2 w_2(x; \lambda) \tag{8.31}$$

where the constants c_1 and c_2 should be determined so as to satisfy the boundary condition (8.5). By substituting $w(x) = \phi(x; \lambda)$ to Eq.(8.5), we have Eq.(8.5) can be reduced to

$$\begin{pmatrix} a_1 w_1(\alpha; \lambda) + a_2 \left[\frac{d}{dx} w_1(x; \lambda) \right]_{x=\alpha} & a_1 w_2(\alpha; \lambda) + a_2 \left[\frac{d}{dx} w_2(x; \lambda) \right]_{x=\alpha} \\ b_1 w_1(\beta; \lambda) + b_2 \left[\frac{d}{dx} w_1(x; \lambda) \right]_{x=\beta} & b_1 w_2(\beta; \lambda) + b_2 \left[\frac{d}{dx} w_2(x; \lambda) \right]_{x=\beta} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{8.32}$$

If Eq.(8.32) has a non-trivial solution $\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, it should be valid that

$$\det \begin{pmatrix} a_1 w_1(\alpha; \lambda) + a_2 \left[\frac{d}{dx} w_1(x; \lambda) \right]_{x=\alpha} & a_1 w_2(\alpha; \lambda) + a_2 \left[\frac{d}{dx} w_2(x; \lambda) \right]_{x=\alpha} \\ b_1 w_1(\beta; \lambda) + b_2 \left[\frac{d}{dx} w_1(x; \lambda) \right]_{x=\beta} & b_1 w_2(\beta; \lambda) + b_2 \left[\frac{d}{dx} w_2(x; \lambda) \right]_{x=\beta} \end{pmatrix} = 0. \quad (8.33)$$

Eq.(8.33) that $w_1(x; \lambda)$ and $w_2(x; \lambda)$ are the fundamental system of solutions in Eq.(8.1) satisfies only a sequence of specific values of λ , such that $\{\lambda_0, \lambda_1, \lambda_2, \dots\}$. It means that Eq.(8.1) with Eq.(8.5) provides a sequence of solutions $\phi(x; \lambda)$ in Eq.(8.31), such that $\phi_0(x) = \phi(x; \lambda_0), \phi_1(x) = \phi(x; \lambda_1), \phi_2(x) = \phi(x; \lambda_2), \dots (\forall x \in [\alpha, \beta])$, for only a sequence of specific values of $\lambda = \lambda_0, \lambda = \lambda_1, \lambda = \lambda_2, \dots$.

The sequence $\{\lambda_0, \lambda_1, \lambda_2, \dots\}$ is a set of eigenvalue and $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ is a set of their corresponding eigenfunctions of the following operator $\mathbf{K}$ defined by

$$\mathbf{K}w(x) \equiv \left\{ -\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x) \right\} w(x). \quad (8.34)$$

Theorem 8.1. Every pair of functions $w_1(x)$ and $w_2(x)$ with the boundary condition (8.5) satisfies the following equality:

$$\int_{\alpha}^{\beta} (\mathbf{K}w_1(x)) w_2(x) dx = \int_{\alpha}^{\beta} w_1(x) (\mathbf{K}w_2(x)) dx. \quad (8.35)$$

(Proof) For the operator $\mathbf{K}$ in Eq.(8.34), we have

$$\begin{cases} \mathbf{K}w_1(x) = -\frac{d}{dx} \left(p(x) \frac{d}{dx} w_1(x) \right) + q(x) w_1(x), \\ \mathbf{K}w_2(x) = -\frac{d}{dx} \left(p(x) \frac{d}{dx} w_2(x) \right) + q(x) w_2(x), \end{cases} \quad (8.36)$$

and we derive the following equality:

$$\begin{aligned} (\mathbf{K}w_1(x)) w_2(x) - w_1(x) (\mathbf{K}w_2(x)) &= -w_2(x) \left(\frac{d}{dx} \left(p(x) \frac{d}{dx} w_1(x) \right) \right) + w_1(x) \left(\frac{d}{dx} \left(p(x) \frac{d}{dx} w_2(x) \right) \right) \\ &= p(x) \left(w_1(x) \frac{d^2}{dx^2} w_2(x) - w_2(x) \frac{d^2}{dx^2} w_1(x) \right) \\ &\quad + \left(\frac{d}{dx} p(x) \right) \left(w_1(x) \frac{d}{dx} w_2(x) - w_2(x) \frac{d}{dx} w_1(x) \right) \\ &= p(x) \frac{d}{dx} \left(w_1(x) \frac{d}{dx} w_2(x) - w_2(x) \frac{d}{dx} w_1(x) \right) \\ &\quad + \left(\frac{d}{dx} p(x) \right) \left(w_1(x) \frac{d}{dx} w_2(x) - w_2(x) \frac{d}{dx} w_1(x) \right) \\ &= \frac{d}{dx} \left\{ p(x) \left(w_1(x) \frac{d}{dx} w_2(x) - w_2(x) \frac{d}{dx} w_1(x) \right) \right\}. \end{aligned} \quad (8.37)$$

In terms of the Wronski determinant of $w_1(x)$ and $w_2(x)$, we remark that

$$w_1(x) \frac{d}{dx} w_2(x) - w_2(x) \frac{d}{dx} w_1(x) = \det \begin{pmatrix} w_1(x) & w_2(x) \\ \frac{dw_1}{dx}(x) & \frac{dw_2}{dx}(x) \end{pmatrix} = W[w_1(x), w_2(x)]. \quad (8.38)$$

By integrating Eq.(8.37) with respect to x in the interval $[\alpha, \beta]$ and applying Eq.(8.38), we have

$$\begin{aligned} &\int_{\alpha}^{\beta} \left\{ w_2(x) (\mathbf{K}w_1(x)) - w_1(x) (\mathbf{K}w_2(x)) \right\} dx \\ &= \left[p(x) \left(w_1(x) \frac{d}{dx} w_2(x) - w_2(x) \frac{d}{dx} w_1(x) \right) \right]_{\alpha}^{\beta} \\ &= \left[p(x) W[w_1(x), w_2(x)] \right]_{\alpha}^{\beta} \\ &= p(\beta) [W[w_1(x), w_2(x)]]_{x=\beta} - p(\alpha) [W[w_1(x), w_2(x)]]_{x=\alpha} \end{aligned} \quad (8.39)$$

From the boundary condition (8.5), we have

$$\begin{cases} a_1 w_1(\alpha) + a_2 \left[\frac{dw_1(x)}{dx} \right]_{x=\alpha} = 0 \\ a_1 w_2(\alpha) + a_2 \left[\frac{dw_2(x)}{dx} \right]_{x=\alpha} = 0 \end{cases}, \quad \begin{cases} b_1 w_1(\beta) + b_2 \left[\frac{dw_1(x)}{dx} \right]_{x=\beta} = 0 \\ b_1 w_2(\beta) + b_2 \left[\frac{dw_2(x)}{dx} \right]_{x=\beta} = 0 \end{cases} \quad (8.40)$$

and $a_1^2 + a_2^2 \neq 0$, $b_1^2 + b_2^2 \neq 0$. We can confirm that

$$\left[W[w_1(x), w_2(x)] \right]_{x=\alpha} = \det \begin{pmatrix} w_1(\alpha) & w_2(\alpha) \\ \left[\frac{dw_1(x)}{dx} \right]_{x=\alpha} & \left[\frac{dw_2(x)}{dx} \right]_{x=\alpha} \end{pmatrix} = 0, \quad (8.41)$$

$$\left[W[w_1(x), w_2(x)] \right]_{x=\beta} = \det \begin{pmatrix} w_1(\beta) & w_2(\beta) \\ \left[\frac{dw_1(x)}{dx} \right]_{x=\beta} & \left[\frac{dw_2(x)}{dx} \right]_{x=\beta} \end{pmatrix} = 0, \quad (8.42)$$

and then it is valid that

$$\begin{aligned} I &= \int_{\alpha}^{\beta} \left\{ w_2(x) (\mathbf{K} w_1(x)) - w_1(x) (\mathbf{K} w_2(x)) \right\} dx \\ &= p(\beta) \left[W[w_1(x), w_2(x)] \right]_{x=\beta} - p(\alpha) \left[W[w_1(x), w_2(x)] \right]_{x=\alpha} = 0. \end{aligned} \quad (8.43)$$

Theorem 8.2. For any pair of functions $w_1(x)$ and $w_2(x)$ which satisfy

$$\mathbf{K} w_1(x) = \lambda_1 \rho(x) w_1(x), \quad \mathbf{K} w_2(x) = \lambda_2 \rho(x) w_2(x) \quad (\forall x \in [\alpha, \beta]) \quad (8.44)$$

with

$$\begin{cases} a_1 w_1(\alpha) + a_2 \left[\frac{dw_1(x)}{dx} \right]_{x=\alpha} = 0, & \begin{cases} b_1 w_1(\beta) + b_2 \left[\frac{dw_1(x)}{dx} \right]_{x=\beta} = 0, \\ b_1 w_2(\beta) + b_2 \left[\frac{dw_2(x)}{dx} \right]_{x=\beta} = 0, \end{cases} \\ a_1 w_2(\alpha) + a_2 \left[\frac{dw_2(x)}{dx} \right]_{x=\alpha} = 0, & \end{cases} \quad (8.45)$$

it is valid that the following two items:

(i) For the case of $\lambda_1 = \lambda_2$, One of $w_1(x)$ and $w_2(x)$ are equal to each other up to a constant multiple, such that

$$w_1(x) = \text{Constant} \times w_2(x). \quad (8.46)$$

(ii) For the case of $\lambda_1 \neq \lambda_2$, we have

$$\int_{\alpha}^{\beta} \rho(x) w_1(x) w_2(x) dx = 0. \quad (8.47)$$

(Proof) (i) In setting $\lambda_1 = \lambda_2 = \lambda$, $w_1(x)$ and $w_2(x)$ are solutions of the following homogeneous Sturm-Liouville equation:

$$\left\{ -\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) + q(x) \right\} w(x) = \lambda \rho(x) w(x) \quad (8.48)$$

and they satisfy the following boundary condition:

$$\begin{cases} a_1 w_1(\alpha) + a_2 \left[\frac{dw_1(x)}{dx} \right]_{x=\alpha} = 0, \\ a_1 w_2(\alpha) + a_2 \left[\frac{dw_2(x)}{dx} \right]_{x=\alpha} = 0. \end{cases} \quad (8.49)$$

Because of $a_1^2 + a_2^2 \neq 0$, it is valid that

$$\left[W[w_1(x), w_2(x)] \right]_{x=\alpha} = \det \begin{pmatrix} w_1(\alpha) & \left[\frac{dw_1(x)}{dx} \right]_{x=\alpha} \\ w_2(\alpha) & \left[\frac{dw_2(x)}{dx} \right]_{x=\alpha} \end{pmatrix} = 0, \quad (8.50)$$

and then $w_1(x)$ and $w_2(x)$ are dependent on each other and one of them is a constant multiple of each other.

(ii) From theorem 8.1, we have

$$\int_{\alpha}^{\beta} (\mathbf{K} w_1(x)) w_2(x) - w_1(x) (\mathbf{K} w_2(x)) dx = 0. \quad (8.51)$$

By substituting Eq.(8.44) to Eq.(8.51), we can confirm that

$$(\lambda_1 - \lambda_2) \int_{\alpha}^{\beta} \lambda \rho(x) w_1(x) w_2(x) dx = 0. \quad (8.52)$$

Because of $\lambda_1 - \lambda_2 \neq 0$, we obtain

$$\int_{\alpha}^{\beta} \lambda \rho(x) w_1(x) w_2(x) dx = 0. \quad (8.53)$$

From theorem 8.2, The sequence of the eigenfunctions of the operator $\mathbf{K}$, $\{\phi_n(x)|n=0,1,2,\dots\}$, that $\phi_n(x)$ is the eigenfunction corresponding to the eigenvalue λ_n of the operator $\mathbf{K}$, satisfies the following orthonormal relations:

$$\int_{\alpha}^{\beta} \rho(x) \phi_m(x) \phi_n(x) dx = \delta_{m,n} \quad (m, n = 0, 1, 2, \dots). \quad (8.54)$$

Every C^2 -class functions $w(x)$ satisfying the boundary condition (8.5) can be expanded in terms of the orthonormal systems $\{\phi_0(x), \phi_1(x), \phi_2(x), \dots\}$ as follows:

$$w(x) = \sum_{l=0}^{+\infty} c_l \phi_l(x), \quad (8.55)$$

$$c_n \equiv \int_{\alpha}^{\beta} \phi_n(x) w(x) \rho(x) dx \quad (n = 0, 1, 2, \dots). \quad (8.56)$$

Now we consider the following inhomogeneous Sturm-Liouville equation:

$$-\frac{d}{dx} \left(p(x) \frac{d}{dx} \right) w(x) + q(x) w(x) - \lambda w(x) = -f(x) \quad (\forall x \in [\alpha, \beta]) \quad (\forall x \in [\alpha, \beta]), \quad (8.57)$$

such that

$$\mathbf{K}w(x) - \lambda \rho(x) w(x) = -f(x) \quad (\forall x \in [\alpha, \beta]), \quad (8.58)$$

under the boundary condition (8.5) and $\lambda \neq \lambda_0, \lambda_1, \lambda_2, \dots$. By substituting Eq.(8.55) to Eq.(8.58), we have

$$\mathbf{K} \sum_{l=0}^{+\infty} c_l \phi_l(x) - \lambda \rho(x) \sum_{l=0}^{+\infty} c_l \phi_l(x) = -f(x). \quad (8.59)$$

If $\phi_n(x)$ is always a C^2 -class function and the function sequence $\left\{ \sum_{l=0}^n c_l \phi_l(x) \middle| n = 0, 1, 2, \dots \right\}$ uniformly converges in the interval $[\alpha, \beta]$, we can derive

$$\sum_{l=0}^{+\infty} c_l \mathbf{K} \phi_l(x) - \lambda \rho(x) \sum_{l=0}^{+\infty} c_l \phi_l(x) = -f(x), \quad (8.60)$$

$$-\sum_{l=0}^{+\infty} c_l (\lambda - \lambda_l) \rho(x) \phi_l(x) = -f(x). \quad (8.61)$$

By multiplying $\phi_m(x)$ by the both sides of Eq.(8.61) and integrating them with respect to x in the interval $[\alpha, \beta]$, we derive the expression of c_m as follows:

$$-\int_{\alpha}^{\beta} \left(\sum_{l=0}^{+\infty} c_l (\lambda - \lambda_l) \rho(x) \phi_l(x) \phi_n(x) \right) dx = -\int_{\alpha}^{\beta} f(x) \phi_n(x) dx, \quad (8.62)$$

$$-\sum_{l=0}^{+\infty} c_l (\lambda - \lambda_l) \int_{\alpha}^{\beta} \rho(x) \phi_l(x) \phi_n(x) dx = -\int_{\alpha}^{\beta} f(x) \phi_n(x) dx. \quad (8.63)$$

$$-\sum_{l=0}^{+\infty} c_l (\lambda - \lambda_l) \delta_{l,n} = -\int_{\alpha}^{\beta} f(x) \phi_n(x) dx, \quad (8.64)$$

$$c_n (\lambda - \lambda_n) = \int_{\alpha}^{\beta} f(x) \phi_n(x) dx, \quad (8.65)$$

$$c_n = \frac{1}{\lambda - \lambda_n} \int_{\alpha}^{\beta} f(x) \phi_n(x) dx. \quad (8.66)$$

By substituting Eq.(8.66) to Eq.(8.55), the solution $w(x)$ for Eq.(8.30) under the boundary condition (8.5) can be expressed as follows:

$$w(x) = \sum_{l=0}^{+\infty} \left(\frac{1}{\lambda - \lambda_l} \int_{\alpha}^{\beta} f(\xi) \phi_l(\xi) d\xi \right) \phi_l(x). \quad (8.67)$$

If the function sequence $\left\{ \sum_{l=0}^n \frac{\phi_l(x)\phi_l(\xi)}{\lambda - \lambda_l} \middle| n = 0, 1, 2, \dots \right\}$ uniformly converges for any $\xi \in [\alpha, \beta]$, Eq.(8.67) can be reduced to

$$w(x) = \int_{\alpha}^{\beta} \left(\sum_{l=0}^{+\infty} \frac{\phi_l(x)\phi_l(\xi)}{\lambda - \lambda_l} \right) f(\xi) d\xi, \quad (8.68)$$

such that

$$w(x) = \int_{\alpha}^{\beta} G(x, \xi) f(\xi) d\xi \quad (\forall x \in [\alpha, \beta]), \quad (8.69)$$

$$G(x, \xi) \equiv \sum_{l=0}^{+\infty} \frac{\phi_l(x)\phi_l(\xi)}{\lambda - \lambda_l}. \quad (8.70)$$

9 Concluding Remarks

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